

Public Procurement, the Nature of Innovation, and Growth: Firm-Level Evidence from Germany*

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Abstract

German firms invest in R&D, yet fewer introduce products new to the market. Public procurement may contribute to the pattern: contracts that reward specification-compliant delivery over frontier novelty may reduce the payoff to frontier innovation and shift the composition of innovation toward existing products. Linking firm-level innovation surveys to public contract award records, we find that a one-standard-deviation increase in sector procurement exposure is associated with a 0.70 percentage-point lower probability of market-novel product introduction and a 1.3 percentage-point higher share of revenue from existing products. Whether such composition shifts slow growth depends on general-equilibrium forces. We build and estimate a quality-ladder endogenous-growth model incorporating this mechanism. Redirecting procurement funds to an R&D tax credit raises growth by 10.18 basis points per year and delivers a welfare gain of 1.55 percent of permanent consumption.

JEL codes: D22, H57, O31, O38, O41.

Keywords: public procurement, innovation composition, quality ladders, market novelty, demand-side policy.

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1 Introduction

Public procurement channels government demand for goods and services through contracts that specify technical standards, certification requirements, and delivery schedules. Meeting those specifications is more profitable for existing products than for frontier novelty. In this paper, we study public procurement as a demand-side channel that may redirect the composition of private innovation and quantify the aggregate growth consequences of procurement design.

Germany provides our empirical laboratory. Between 2007 and 2023, public procurement rose from 6.6 to 9.2 percent of GDP (Eurostat, 2026) while the market-novelty rate fell from 20 to 8 percent of firms (Figure 1). The same firms that produce for private markets also compete for public contracts, suggesting a link between contract design and economy-wide innovation composition.

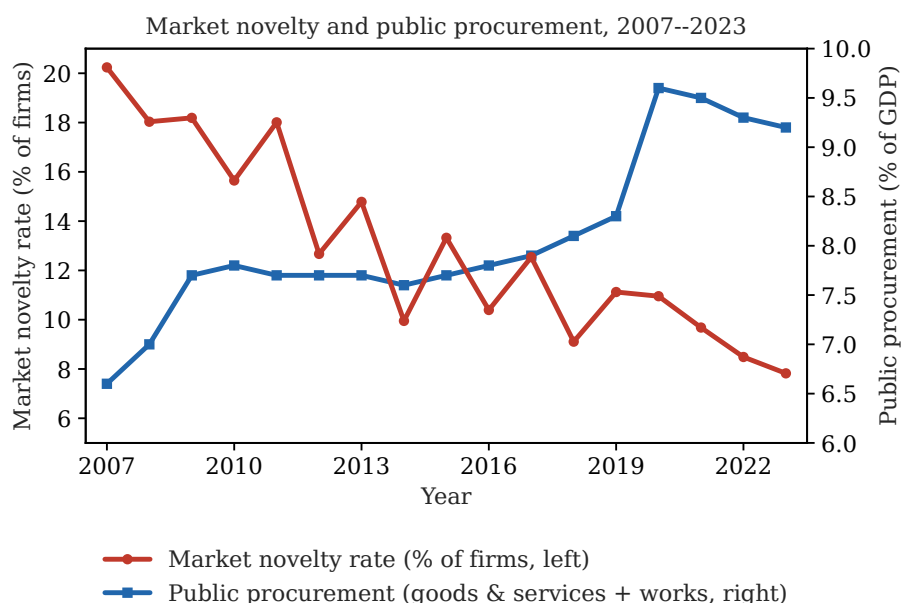


Figure 1: Public procurement rose as market novelty fell

Notes: The red line plots the market-novelty rate, defined as the share of firms introducing products new to the market, on the left axis. The blue line plots public procurement on the right axis: the sum of intermediate consumption (government purchases of goods and services) and public investment (works), the two national-accounts components that map to competitively tendered contracts.

Source: Mannheim Innovation Panel, 2007–2023; Eurostat (2026), Germany, 2007–2023.

To test whether procurement design shapes the composition of innovation, we link German procurement contract records to firm-level innovation surveys and financial accounts. The Mannheim Innovation Panel (MIP), administered by ZEW Mannheim on behalf of the German Federal Ministry of Education and Research, provides firm-year measures of market novelty, R&D expenditure, process innovation, and new-product revenue for a representative sample of firms from 2009 to

2023. We combine the survey with Tenders Electronic Daily (TED) award records, which report publicly disclosed German procurement awards at the contract level: product category, award value, buyer identity, buyer category, and technical-requirement fields. The sample covers a large population of contract awards, one per contract and winning firm, over 2009–2023. We also merge with ORBIS, a firm-level financial database covering European companies, for roughly 15,000 matched contract winners, providing employment, turnover, productivity, and profitability over the same period. The MIP captures innovation composition, the contract records identify sector-level exposure and contract design, and ORBIS describes how winning firms scale and perform.

Empirically, we estimate panel regressions relating sector-level procurement exposure to firm-year innovation outcomes, with firm and year fixed effects. Identifying variation comes from within-sector changes in procurement activity over time. Three facts emerge.

First, procurement exposure is associated with lower frontier product innovation and higher revenue from existing products. Firms in more exposed sectors introduce market-novel products less often; their share of turnover from existing products is correspondingly higher. The pattern extends to cost-reducing innovations: the revenue share from such products is also lower in more exposed sectors, while the rate of cost-reducing activity is unchanged. The results document a composition shift in commercial orientation, not a contraction of innovative effort.

Second, the pattern concentrates in supplies-and-services contracts. Works contracts deliver a physical structure rather than a reproducible product and show no significant association on either margin. The split favors specification requirements over public demand generally as the driver. By category, the composition association concentrates in central-government procurement; regional and local government procurement and utilities show no significant association on either margin.

Third, contract winners expand in scale without productivity or profitability gains: employment and turnover rise with contract volume, while revenue per worker and return on capital show no statistically significant change.

Contract design offers an interpretation. Procurement specifications that require certification, compatibility standards, and delivery schedules favor existing products over frontier novelty. In software procurement, integration and compatibility standards are associated with lower rates of new-architecture introduction; in public R&D service contracts, tightly specified deliverables are associated with incremental over exploratory innovation; in medical-device procurement, regulatory certification requirements may reward known-products over diagnostic innovators. The composition association concentrates in central-government procurement, where awards cluster in

the specification-intensive supply-and-service categories that show the largest wedge, and is most pronounced in industries where the same firms serve both public and private buyers.

The within-firm estimates document the composition shift but do not reveal its aggregate growth implications or the welfare consequences of alternative procurement designs. To formalize the mechanism, we build on the [Klette and Kortum \(2004\)](#) endogenous-growth framework, in which firms hold portfolios of product lines and compete to improve product quality step by step.¹ We add one novel margin: each period, firms decide not only how much to innovate but also what kind of innovation to pursue. Procurement enters as a shift in the relative payoff: frontier product innovation becomes less profitable than specification-compliant delivery; firms shift effort toward incremental improvements. The model rationalizes the patterns in the German innovation and procurement microdata: procurement raises equilibrium innovation effort while narrowing the frontier share, consistent with the observed decline in market novelty and the rise in existing-product revenue in more exposed sectors.

After building the model, we turn to quantification using the same microdata. We estimate four structural parameters by generalized method of moments (GMM), targeting five empirical moments: the frontier share of innovation, the frontier revenue share, firm size, creative destruction, and the R&D personnel share. Estimated quality steps and direction costs fall within the ranges reported in [García-Macià et al. \(2019\)](#) and [Akcigit and Kerr \(2018\)](#). At the estimated parameters, the procurement wedge accounts for the observed composition redirection toward incremental lines.

We conduct three counterfactual exercises to quantify the growth and welfare consequences of procurement design. The primary exercise redirects procurement funds to an R&D tax credit modeled on Germany's *Forschungszulage* (a rule-based credit on eligible R&D labor costs introduced in January 2020). Growth rises by 10.18 basis points per year, and welfare rises by 1.55 percent of permanent consumption. Second, a budget-neutral specification reform replaces certification-based contract requirements with open performance criteria, allowing frontier products to compete for public contracts on equal terms; growth rises by 0.38 basis points. Third, a historically scaled reduction in procurement demand to Germany's 2008 level restores part of the frontier premium, with a growth gain of 0.07 basis points.

Against a German TFP growth baseline of roughly 50 basis points per year, the R&D-credit growth gain represents about one-fifth of trend growth; the 1.55 percent welfare gain lies within the range of trade-liberalization gains for small open economies and exceeds standard inflation-cost estimates. Across exercises, procurement design shapes both the scale and the direction of innovative activity.

¹[Akcigit and Kerr \(2018\)](#), [Akcigit and Ates \(2023\)](#), and [Acemoglu et al. \(2018\)](#) extend the framework.

The paper connects to four strands of the literature. First, we contribute to the endogenous-growth literature on innovation composition (Aghion and Howitt, 1992; Klette and Kortum, 2004; Akcigit and Kerr, 2018; García-Macià et al., 2019; Acemoglu et al., 2022; Aghion et al., 2023; Akcigit and Ates, 2023; Juhász et al., 2024). García-Macià et al. (2019) document that aggregate growth comes from incumbents improving existing products rather than from creative destruction. Acemoglu et al. (2022) establish that the frontier-versus-incremental margin is central for long-run growth when incremental innovations face diminishing returns within a technology cluster. We add a procurement-demand margin to this mechanism: government contracts shift the relative payoff between frontier and incremental activity, redirecting innovation composition without changing its level. The model complements supply-side accounts of composition in Bloom et al. (2020).

Second, we contribute to the empirical literature on demand-side innovation policy (Griffith et al., 2010; Slavtchev and Wiederhold, 2016; Czarnitzki et al., 2020; Chen et al., 2021; Krieger and Zipperer, 2022; Chen and Xu, 2023; Dechezleprêtre et al., 2023; Krieger et al., 2024; Moretti et al., 2025; Chiappinelli et al., 2025; Lerche, 2025; Takalo et al., 2026). Krieger et al. (2024) document that price-criterion tenders reduce product innovation among German firms; Czarnitzki et al. (2020) show that even innovation-directed contracts tilt activity toward incremental rather than frontier novelty. We extend both findings from individual contract variation to sector-level exposure and connect the evidence to quantified design margins. On the subsidies side, Takalo et al. (2026) establish that rule-based tax credits reach a broader firm base than discretionary grants, motivating our counterfactual of redirecting procurement funds to Germany’s *Forschungszulage*; Lerche (2025) shows that investment tax incentives generate local spillovers as large as the direct effect. For German firms, procurement and subsidies operate on distinct margins: procurement redirects innovation composition while subsidies raise its level.

Third, we contribute to the literature on procurement contract design and innovation (Howell et al., 2025; Clemens and Rogers, 2026). Howell et al. (2025) document that open-format defense awards raise patenting and commercial adoption while specification-based awards generate weaker frontier commercialization; Clemens and Rogers (2026) demonstrate historically that fixed-price contracts redirect invention toward cost reduction while quality-focused contracts shift patenting toward buyer-preferred attributes. We extend both mechanisms from individual contract variation to aggregate sector-level exposure, measuring the economy-wide composition association.

Finally, we contribute a demand-side channel to the misallocation literature (Acemoglu et al., 2018; Terry, 2023; Lehr, 2024, 2025; Bachmann et al., 2026). Government procurement changes not only the level of innovative effort but also its direction, adding a public-demand mechanism to accounts of how frictions in factor markets, R&D allocation, and market power slow long-run growth.

The paper proceeds as follows. Section 2 presents the data and empirical evidence. Section 3 develops the model. Section 4 reports the quantitative analysis. Section 5 presents the counterfactuals and robustness exercises. Section 6 concludes.

2 Data and Empirical Evidence

The empirical strategy links public procurement records to firm-level innovation surveys and financial accounts. First, the Mannheim Innovation Panel (MIP) tracks how innovation composition co-varies with sector-level procurement exposure over time. Second, Tenders Electronic Daily (TED) supplies the contract records from which we construct the exposure measure. Third, ORBIS matches procurement award records to firm-level financial accounts, documenting how winners differ in scale, productivity, and profitability. Fourth, the IAB Establishment History Panel provides entry and exit rates for the structural model’s creative-destruction calibration. The MIP and ORBIS evidence serve distinct roles: the MIP captures average patterns across all firms in a sector (winners, unsuccessful bidders, and non-participants alike), while ORBIS describes winner outcomes directly.

2.1 Data Sources

2.1.1 Mannheim Innovation Panel

The Mannheim Innovation Panel is an annual firm-level survey administered by ZEW Mannheim on behalf of the German Federal Ministry of Education and Research.² We use the 2009–2023 releases, which yield 80,028 firm-year observations across 23,547 firms. The panel covers manufacturing and business-service industries; construction, agriculture, and the public sector are excluded. Main outcomes cover market-novel product innovation, unchanged-product revenue share, the market-novel share of new-product revenue, process innovation, and cost-reduction activity and revenue share. Process-innovation regressions use the full 2009–2023 sample.³ Appendix Table A1 reports summary statistics for all variables, including world-first product innovation, R&D intensity, R&D personnel share, new-product revenue share, and public R&D subsidies used in robustness analyses.

Our primary outcome, market novelty, is a binary indicator equal to one if the firm reports having introduced at least one product or service to the market that was new to the market (as opposed to merely new to the firm) over the preceding three-year reference period, following the Community Innovation Survey / Oslo Manual convention (OECD and Eurostat, 2018). Since the classification rests on the respondent’s own assessment of what "new to the market" means for its industry, market

²We access the MIP through a research data use agreement with ZEW Mannheim. The panel uses a stratified random sampling frame, refreshed annually to maintain representativeness across firm size classes, industries, and regions.

³In 2019, the MIP adopted the Oslo-IV definition of process innovation, which expanded the concept and raised the reported rate from 28 to 53 percent. Year fixed effects absorb this level shift in our analysis below.

novelty is a self-reported and inherently subjective measure. Appendix [A.1](#) discusses the measure alongside the survey question wording.

2.1.2 Public Procurement Contracts

Tenders Electronic Daily (TED), the European Commission’s official procurement journal, publishes award notices from German public buyers (federal, state, and municipal) that EU transparency rules require to be disclosed above specified value thresholds.⁴ Because reporting is required only above specified value thresholds, the contracts we observe skew toward larger, more complex procurement: Appendix Figure [A2](#) shows that project and investment contracts account for 69 percent of contracts and 80 percent of value in our sample, against 24 percent and 16 percent for staple and routine purchases. Our exposure measure therefore captures project-based demand rather than routine commodity buying. The main sample contains 519,990 contract-award observations, one per contract and winning firm, covering 508,695 unique contracts from 2009 to 2023; 316,642 map to MIP-covered industries. Each observation reports the product code, award year, buyer region, buyer category, and number of submitted bids. The records also retain contract-design fields such as procedure type and technical-requirement text length. For the pre-period, we use TED award notices from 2006–2008 to construct pre-period exposure.

2.1.3 ORBIS

ORBIS is a firm-level financial database covering European companies, maintained by Bureau van Dijk. It contains annual balance sheet and profit-and-loss accounts, employment, and ownership structure for several firms, with broad coverage of German companies across size classes and industries. We use it in two ways. First, we construct a German firm panel over 2007–2023 with employment, turnover, revenue per employee, profitability, age, foreign ownership, and multinational status. Second, we match procurement participants to firm identifiers using direct identifiers, administrative identifiers, and fuzzy company names. The crosswalk matches 41,801 unique procurement winners to the identifiers; the resulting financial panel contains 124,391 firm-year observations across 15,463 firms and supports the winner scale and selection analyses. Appendix [B](#) reports winner-profile and event-study validation exercises.

⁴Reporting compliance improved markedly after April 2016, when Germany transposed the 2014 EU Procurement Directives; the transposition modernized existing award-notice requirements and coincided with a step change in compliance rates; pre-2016 contract counts therefore understate true sector-level activity. Appendix [A.6](#) details the coverage gap; Appendix Table [A5](#) shows the baseline signs survive with sector-specific trends, and Appendix Table [C2](#) restricts to investment-category CPV contracts.

2.1.4 IAB Establishment History Panel

The IAB Establishment History Panel (BHP) is an annual administrative panel covering the near-universe of German establishments with at least one employee subject to social security contributions.⁵ The annual exit hazard averages 5.07 percent over 2009–2023; we use this rate to calibrate creative destruction in the quality-ladder model (Section 3).

2.2 Empirical Strategy

Procurement markets differ in the degree to which they draw firms that also serve private buyers. Contracts for defense platforms or classified services draw on a different firm base than contracts for engineering services, software, or industrial equipment, industries where the same firms serve both government and private buyers. In such sectors, procurement exposure is associated with lower frontier innovation (Krieger et al., 2024; Chiappinelli et al., 2025). The analysis uses variation in sector-level procurement exposure across these contestable industries and over time within sectors.

We assign each contract to one of the twenty-one MIP industries by mapping its EU product code to the corresponding two-digit German industry classification. Procurement exposure is $\text{arcsinh}(\text{contracts}_{s,t})$, where $\text{contracts}_{s,t}$ is the count of awarded contracts reported in sector s and year t ; the distribution is right-skewed, and the arcsinh transformation behaves similarly to a logarithm for large values. Current MIP releases do not link survey firms to individual contract awards; hence we aggregate contracts to sector-year counts and assign exposure at the industry level.

Figure 2 documents concentration in manufacturing and professional services and substantial within-sector time variation; across 315 sector-years (21 industries $\times$ 15 years, 2009–2023) the measure has a mean of 3.74 and a standard deviation of 3.58. The post-2015 rise in contract counts visible in Panel B reflects the expansion of TED reporting coverage rather than proportional growth in actual procurement: German government procurement grew moderately as a share of GDP, while TED counts more than doubled, with the composition by contract type and buyer stable across years. Year fixed effects absorb common changes in reporting coverage; sector-specific trends absorb smooth differences in reporting growth across industries. Restricting to the post-2016 sample or adding sector-specific trends leaves the coefficient close to baseline (Appendix A.6).

⁵We obtain the data through the IAB Research Data Centre under a restricted data use agreement. The BHP uses establishment-type classification codes to separate genuine new foundations and full closures from administrative re-identifications, spin-offs, and corporate restructurings. We use the entry and exit event files for 2009–2023, covering 22.5 million establishment-years. The annual exit hazard is the number of establishments classified as genuine exits in year t divided by the stock active in $t - 1$.

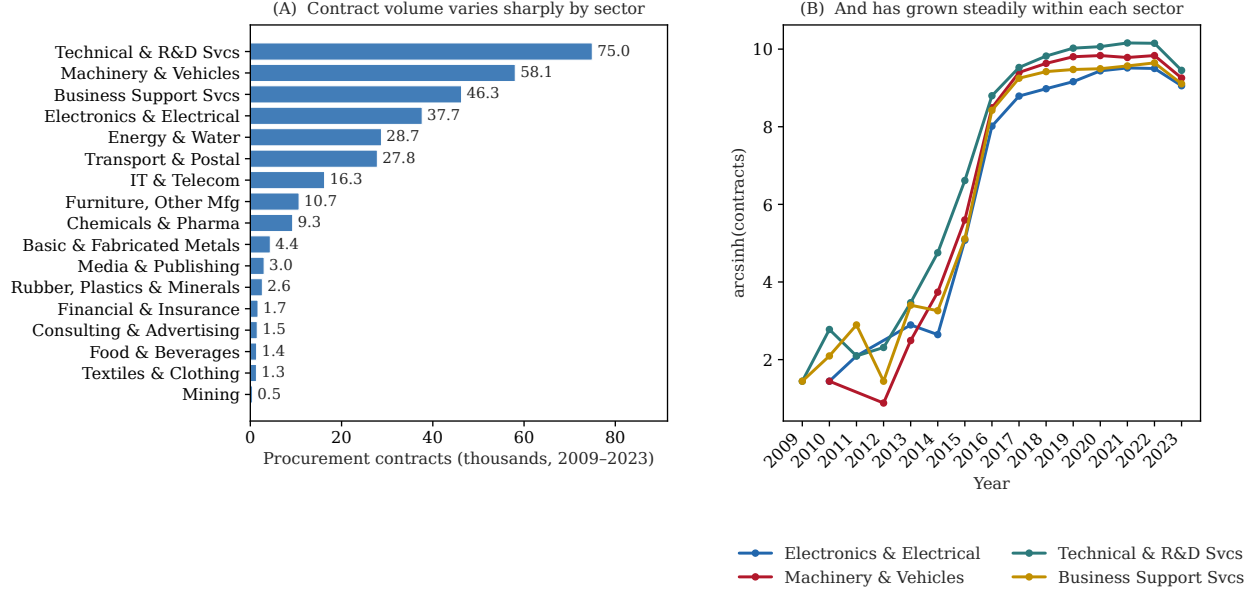


Figure 2: Procurement exposure across MIP-covered industries, 2009–2023

Notes: Panel A reports total awarded contracts, in thousands, across MIP-covered industries, aggregated over 2009–2023. Panel B plots $\text{arcsinh}(\text{contracts})$, the exposure measure, by year for the four largest sectors by total contract volume.

Source: Authors' calculations using TED, 2009–2023.

The baseline specification is

$$Y_{ist} = \alpha_i + \delta_t + \beta \text{arcsinh}(\text{contracts}_{s,t}) + \varepsilon_{ist}, \quad (1)$$

where Y_{ist} denotes a firm-level innovation outcome for firm i in industry s and year t : binary indicators for product or process innovation, or continuous measures such as revenue shares; α_i is a firm fixed effect and δ_t a year fixed effect.⁶ The coefficient β has percentage-point units for binary outcomes and natural units for continuous outcomes.

Firm fixed effects absorb time-invariant firm characteristics, including the permanent propensity to serve public buyers; year effects absorb aggregate shocks common to all firms. We cluster standard errors at the industry level, yielding twenty-one clusters. Since exposure varies at the sector-year level, β measures the average within-firm association between reported industry procurement exposure and innovation outcomes across all firms in a sector (contract winners, unsuccessful bidders, and non-participants alike). Identifying variation comes from changes within sectors over time: within a given year, sectors with above-average procurement exposure have above-average market novelty (cross-sectional selection), yet the within-firm estimate is negative.

⁶Since exposure varies at the sector-year level, the industry subscript s denotes the sector of firm i .

2.3 Main Results

Table 1 reports the baseline specification across three innovation outcomes.

Table 1: Procurement Exposure and Innovation Outcomes

	(1) Market novelty (pp)	(2) Unchanged-product revenue (%)	(3) Cost-reduction revenue (%)
$\text{arcsinh}(\text{contracts}_{st})$	-0.199* (0.101)	0.379** (0.137)	-0.047* (0.024)
Effect of +1 SD (pp)	-0.709	1.352	-0.162
Effect relative to outcome mean (%)	-5.8	1.5	-8.8
Observations	66,980	25,962	40,635
Industries	21	21	21
Firm FE	✓	✓	✓
Year FE	✓	✓	✓

Notes: Each column reports a within-firm OLS regression with firm and year fixed effects. Standard errors, in parentheses, are clustered by MIP sector. The treatment is the inverse hyperbolic sine of awarded contract counts in the firm's MIP sector and year, 2009–2023. The standardized row multiplies the coefficient by the estimation-sample standard deviation of treatment; the following row divides that effect by the estimation-sample outcome mean. Binary outcomes are multiplied by 100. Column (2) is the share of turnover from unchanged products; column (3) is the share from innovations that reduce unit production costs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Authors' calculations using the Mannheim Innovation Panel and TED.

Column (1) shows that a one-unit increase in $\text{arcsinh}(\text{contracts})$ corresponds to a 0.199 percentage-point lower probability of introducing a market-novel product; a one-standard-deviation increase in exposure (3.5 units) corresponds to a 0.7 percentage-point decline. Column (2) shows the complementary revenue pattern: the share of turnover from existing products is 0.379 percentage points higher per unit of exposure; a one-standard-deviation increase corresponds to a 1.3 percentage-point difference. Together, columns (1) and (2) document a shift in the product mix toward existing products and away from frontier novelty.

The composition shift is concentrated at the extensive margin: firms in more-exposed sectors are less likely to introduce any market-novel product, while the market-novel share within new-product revenue and process innovation show no significant association with exposure.

Column (3) traces the cost-reduction margin: the share of turnover from cost-reducing innovations is 0.047 percentage points lower per unit of exposure; a one-standard-deviation increase corresponds to a 0.16 percentage-point decline. Appendix Table A3 shows that the broader product-innovation indicator of Krieger et al. (2024) (any new or improved product) is unrelated to procurement exposure, while market novelty falls. The pattern concerns what firms innovate, not if they innovate.

2.4 Heterogeneity Across Procurement Environments

The baseline estimate pools all contract types and buyer categories. We now ask whether the composition shift concentrates in specific procurement environments. Figure 3 describes the procurement distribution in MIP-covered industries: services and supplies together account for 93 percent of contracts by count (services 51 percent, supplies 43 percent), with works contracts representing the remaining 7 percent. On the buyer side, regional and local government agencies issue 47 percent of contracts, other public bodies 45 percent, and central-government agencies 8 percent.

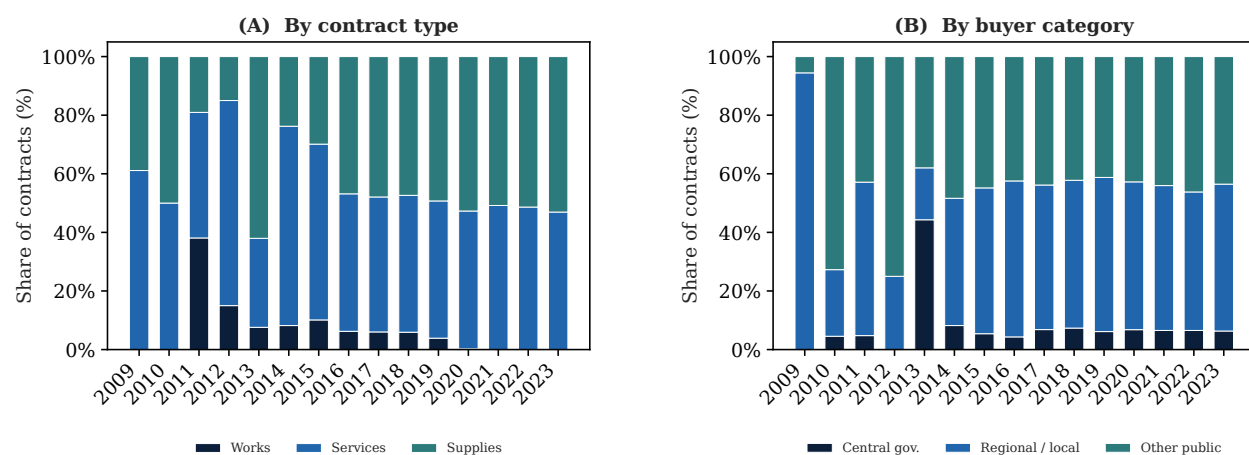


Figure 3: Procurement composition in MIP-covered industries, 2009–2023

Notes: Share of above-threshold contracts by count; Panel A by contract type (works, services, supplies), Panel B by buyer category (central government, regional and local government, other public bodies). Restricted to WZ sections covered by the Mannheim Innovation Panel (manufacturing and business services). Appendix Figure A1 reports the same decomposition by contract value.

Source: Authors' calculations using TED, 2009–2023.

Table 2 reports the two main innovation outcomes from Table 1 separately for contract types and buyer categories. The splits are descriptive: contract type, buyer category, and contract size co-vary in the procurement data, and the patterns do not isolate a single mechanism.

Table 2: Heterogeneity Across Procurement Environments

	(1) Market novelty (pp)	(2) Unchanged-product revenue (%)	(3) Cost-reduction revenue (%)
<i>Panel A. Contract type</i>			
Supplies & services	-0.211** (0.099)	0.393*** (0.137)	-0.046* (0.023)
+1 SD effect (pp)	-0.752	1.403	-0.158
Effect / mean (%)	-6.2	1.6	-8.6
Works	-0.010 (0.142)	0.180 (0.120)	0.024 (0.022)
+1 SD effect (pp)	-0.016	0.331	0.039
Effect / mean (%)	-0.1	0.4	2.1
<i>Panel B. Buyer type</i>			
Central government	-0.291** (0.131)	0.518** (0.194)	-0.045 (0.037)
+1 SD effect (pp)	-0.739	1.329	-0.112
Effect / mean (%)	-6.1	1.5	-6.1
Regional / local	-0.043 (0.137)	0.232 (0.148)	-0.028 (0.034)
+1 SD effect (pp)	-0.154	0.810	-0.096
Effect / mean (%)	-1.3	0.9	-5.2
Utilities	-0.086 (0.146)	0.313 (0.195)	-0.050 (0.033)
+1 SD effect (pp)	-0.222	0.788	-0.125
Effect / mean (%)	-1.8	0.9	-6.8
Observations	66,980	25,962	40,635
Firm and year FE	Yes	Yes	Yes
MIP-sector clusters	21	21	21

Notes: Each cell reports the coefficient from a separate within-firm OLS regression with firm and year fixed effects. Standard errors are in parentheses. Standardized rows multiply each coefficient by the estimation-sample standard deviation of its treatment; relative effects divide that change by the estimation-sample outcome mean. In Panel A, the treatment is $\text{arcsinh}(\text{sector-year lot count})$ for the indicated contract type. In Panel B, the treatment is $\text{arcsinh}(\text{sector-year lot count})$ for the indicated buyer category. All estimates use the final 21-sector, 2009–2023 panel. Standard errors are clustered by MIP sector. In joint regressions including all three buyer-category exposures, the p-values for equality of all three coefficients are 0.094 for market novelty, 0.180 for unchanged-product revenue, and 0.672 for cost-reduction revenue. These are descriptive splits; they should not be read as identifying a single mechanism. Supplies and services account for 97.8 % of contracts in the MIP-covered sample, so the supply-type contrast has limited discriminating power. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Authors' calculations using the Mannheim Innovation Panel and OpenTender DE.

Panel A splits exposure by contract type. The supplies-and-services coefficient reproduces the aggregate pattern: market novelty is 0.21 percentage points lower and existing-product revenue is 0.39 percentage points higher. The works estimate is near zero and imprecise, as expected: construction and infrastructure contracts fall outside the MIP coverage frame and contribute little identifying variation.

Panel B splits exposure by buyer category. Despite representing only 8 percent of contracts, central-government procurement drives the aggregate pattern: market novelty falls by 0.29 percentage points and existing-product revenue rises by 0.52 percentage points, both significant at the 5 percent level. Regional and local government procurement (47 percent of contracts) and utilities show no significant effect on either margin. Central-government awards cluster in specification-intensive supply-and-service categories; the composition association reflects their procurement design standards rather than their volume.

Appendix C.2 reports a related cut using TED records alone: the composition shift is stronger in sector-years where contract awards concentrate among repeat winners, an independent source of variation from Panels A–B.

2.5 Winner Outcomes: Scale and Productivity

The sector-level evidence averages over winners, unsuccessful bidders, and non-participants alike. ORBIS, matched to procurement award records for 41,801 winner firms, allows a separate look at winners directly: do firms that secure contracts expand, and is that expansion accompanied by productivity or profitability gains? Table 3 reports within-firm estimates for the matched ORBIS panel over 2009–2023.

Table 3: Winner Outcomes

	(1) Employment (log)	(2) Turnover (log)	(3) Labor productivity (log)	(4) Return on capital
arcsinh(contracts won)	0.025*** (0.002)	0.029*** (0.007)	0.007 (0.007)	0.001 (0.001)
Effect of +1 SD	0.023	0.027	0.007	0.001
Observations	88,706	38,690	33,994	28,584
NACE clusters	20	19	19	19
Firm and year FE	Yes	Yes	Yes	Yes

Notes: Sample is the ORBIS winner panel, 2009–2023, comprising 15,463 firms matched to procurement award records. Each column reports a within-firm OLS regression of the indicated outcome on arcsinh(contracts won), with firm and year fixed effects. Standard errors, in parentheses, are clustered by NACE section. The standardized row multiplies each coefficient by the estimation-sample standard deviation of treatment. Employment, turnover, and labor productivity are in natural logarithms; labor productivity is turnover divided by the number of employees. Return on capital is pretax profit divided by total assets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Authors' calculations using TED matched to ORBIS.

Columns (1) and (2) show that a one-unit increase in arcsinh(contracts won) corresponds to 2.5 percent higher employment and 2.9 percent higher turnover. Columns (3) and (4) show no statistically

significant change in revenue per employee or return on capital. Winners expand in scale without a corresponding efficiency gain. The estimates capture the intensive margin: the response to additional contracts among firms already in the winner pool.

Figure 4 traces employment, turnover, revenue per worker, and return on capital around the year of the first observed contract win. Employment and turnover rise at $t = 0$ and reach 7.5 and 19.5 percentage points by $t = 2$. Revenue per worker increases as well, but the gain is mechanical rather than a genuine efficiency improvement. Return on capital is statistically indistinguishable from zero throughout the window.

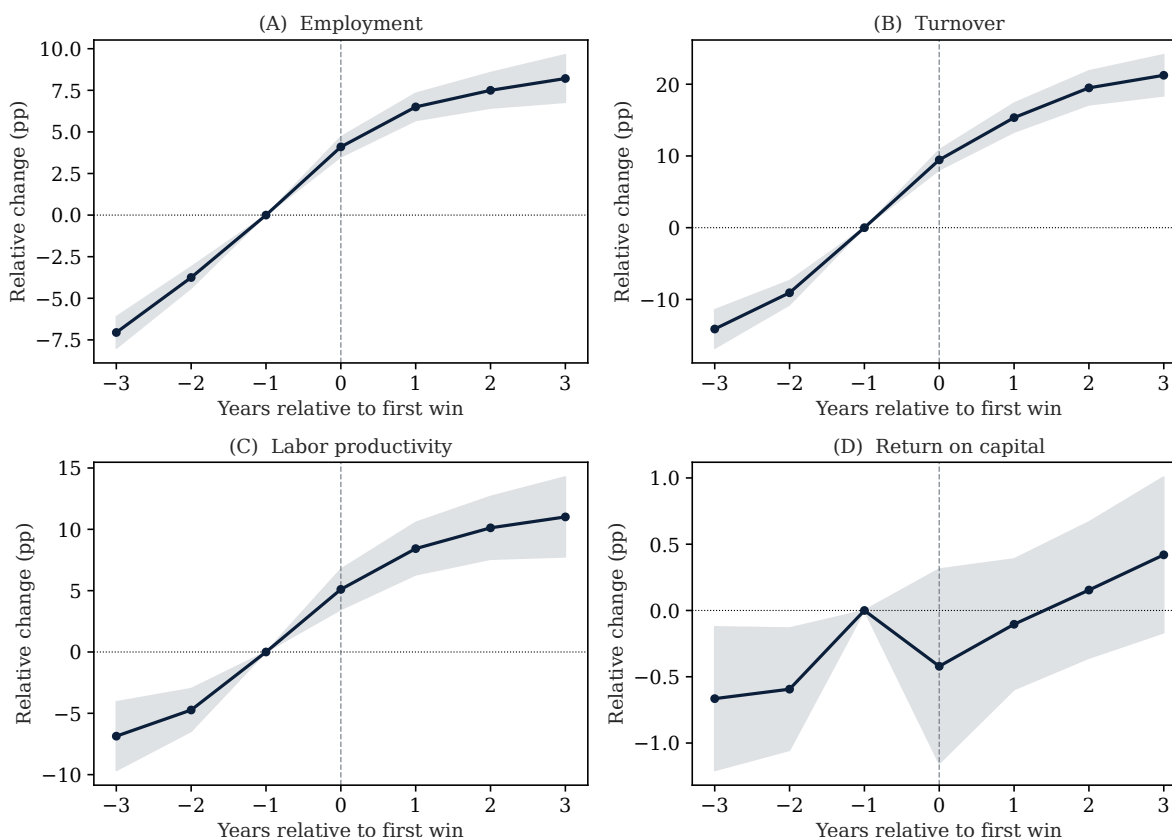


Figure 4: Scale and productivity around the first procurement win

Notes: Each panel shows the outcome demeaned relative to $t = -1$ (year before first win), expressed as percentage-point changes (log outcomes multiplied by 100). Labor productivity is revenue per employee. Return on capital is profit before tax divided by total assets. Shaded bands are 90% confidence intervals. The profile pools all matched winners and does not identify the causal effect of an award. Sample: ORBIS-matched winner panel, TED 2009–2023.

Source: Authors' calculations using TED matched to ORBIS.

Additional winner-profile checks in Appendix B support the same interpretation: procurement winners expand in scale, while profitability effects are small and imprecise. Procurement also operates through a selection margin: the firms that enter the winner pool are mostly oriented toward

scale and delivery rather than frontier innovation, compounding the composition shift measured at the sector level.

Appendix B reports a further non-survey check using PATSTAT patent filings matched to the same winner panel.

Together, the sector-level and winner-level evidence document a consistent pattern: procurement exposure shifts firm activity toward existing products, expands scale, and leaves productivity unchanged. This panel evidence identifies the sign and magnitude of the relationship but not the aggregate growth and welfare consequences of a sector facing a persistently different level of exposure. Section 3 builds a quantitative model that formalizes this mechanism and evaluates the aggregate implications of alternative procurement rules by comparing steady states at different, permanent levels of government demand rather than tracing the transition between them.

3 A Quality-Ladder Model with Procurement

We develop a quality-ladder model to organize the empirical pattern: procurement moves innovation away from frontier advances and toward incremental improvements. The framework builds on [Klette and Kortum \(2004\)](#) in which firms operate portfolios of product lines subject to creative destruction, and the quality-ladder tradition of [Aghion and Howitt \(1992\)](#); [Aghion et al. \(2023\)](#); the key extension is the composition choice μ , which lets incumbent firms select not only how much but also what type of innovation to pursue. The choice between radical (frontier) and incremental innovation follows [Acemoglu et al. \(2022\)](#), applied to a demand-composition setting. Government procurement enters through two channels: it raises the return to innovation effort, and it narrows the frontier premium because incremental lines serve both private and government demand while frontier lines serve private demand only.

3.1 Environment

Time is continuous, $t \in [0, \infty)$. A unit mass of product lines exists, and a single incumbent firm operates each line.

Households. A representative household supplies labor inelastically, owns all firms, and has preferences

$$\int_0^\infty e^{-\rho t} \log C_t dt.$$

We choose aggregate private expenditure as the numeraire; labor market clearing determines the wage w . On the balanced-growth path the household budget constraint is

$$C_0 = w(L_P + \ell_R) + \Pi_{\text{net}} - T, \quad (2)$$

where L_P and ℓ_R are aggregate production and R&D labor, defined below, Π_{net} is aggregate profit income net of innovation costs, and T is the procurement tax defined below. Entry labor $h_E(\tau - x)$ is excluded: entrants earn this wage but spend it acquiring new lines under free entry, so it adds nothing to current consumption.

Production and firm state. A competitive final-good sector aggregates active product lines with a log aggregator over the unit-mass continuum:

$$Y = \exp\left(\int_0^1 \log y_j dj\right), \quad (3)$$

where the unit mass of lines is fixed. Bertrand competition at the quality lead determines each firm's effective markup. Firms produce line-specific output with linear technology:

$$y_j = q_j \ell_j^P.$$

The economy has a fixed labor endowment $\bar{L}$. A firm is described by its portfolio (n_I, n_F) , where $n_I \geq 0$ is the number of incremental lines and $n_F \geq 0$ is the number of frontier lines. The two types differ in their quality lead over the next-best competitor:

$$\text{incremental line: } \lambda_I, \quad \text{frontier line: } \lambda_F > \lambda_I > 1.$$

Government demand and fiscal balance. Government procurement raises demand for incremental product lines. The government spends a fraction $G \geq 0$ of private-sector output per incremental line, going exclusively to specification-compliant suppliers rather than frontier innovators; Appendix E.3.3 microfound the asymmetry. The direct effect is to raise incremental-line profits by a factor $(1 + G)$, compressing the frontier premium.

The government produces output with the same technology as the private sector: one unit of labor at wage w yields one unit of quality-adjusted output, so procurement is real spending on real goods rather than waste. Firms fill government and private orders from the same workforce; procurement raises the demand for production labor without directly distorting R&D incentives.

The government finances procurement through a lump-sum tax on households. The incremental segment has measure $(1 - \mu^*)$, so total government expenditure is

$$T \equiv (1 - \mu^*)G,$$

measured as a fraction of aggregate private output. T is the procurement tax in the household budget constraint (2).

Innovation. The innovation effort rate $x \geq 0$ is the Poisson arrival rate of innovation attempts per product line. The frontier share $\mu \in [0, 1]$ is the probability that a successful innovation creates a frontier line rather than an incremental line. A firm with $N = n_I + n_F$ lines generates new lines at rate xN . Incremental and frontier lines arrive at rates

$$x(1 - \mu)N \quad \text{and} \quad x\mu N,$$

respectively. Creative destruction displaces each existing line at rate $\tau > 0$, determined in equilibrium by the aggregate innovation activity of all firms.

Innovation costs. Per-line innovation costs are

$$\Phi(x) = c_x x^\zeta w, \quad c_x > 0, \zeta > 1, \quad (4)$$

$$\Psi(\mu) = \frac{\kappa}{\eta} \mu^\eta, \quad \kappa > 0, \eta > 1. \quad (5)$$

$\Phi(x)$ is the cost of innovation effort and $\Psi(\mu)$ is the cost of frontier orientation. The model has no knowledge spillovers, which rules out composition distortions from uninternalized knowledge benefits; procurement supplies the demand-side wedge studied here.

Entry. Outside entrants acquire an existing product line by using h_E units of setup labor, with monetary entry cost wh_E . Entry is untargeted: entrants arrive randomly and take ownership of an incumbent line with equal probability, independently of its type or quality level. Entry governs firm turnover and the scale of creative destruction; quality growth comes exclusively from incumbent innovation.

Labor allocation. Firms allocate labor across three activities: production, incumbent R&D, and entry. Production labor per line satisfies $\ell_j^P = y_j/q_j$. Incumbent R&D labor per line is

$$\ell_R(x, \mu, w) \equiv \frac{\Phi(x) + \Psi(\mu)}{w}.$$

Entry labor is h_E per entrant.

3.2 Agent Problems

Pricing. Each quality leader sets the Bertrand limit price against the next-best rival; the quality lead λ_k , rather than demand elasticity, determines the markup. A line with quality lead λ_k earns the per-line profit share

$$\bar{\pi}_k = \frac{\lambda_k - 1}{\lambda_k}, \quad k \in \{I, F\}, \quad (6)$$

with $\bar{\pi}_F > \bar{\pi}_I > 0$.

Private buyers reward quality; both line types serve private demand. Government buyers reward specification compliance: incremental lines serve both private and government demand, while frontier lines serve private demand only. Per-line flow profits are

$$\pi_I(G) = \bar{\pi}_I(1 + G), \quad (7)$$

$$\pi_F(G) = \bar{\pi}_F. \quad (8)$$

The frontier premium equals

$$\pi_F(G) - \pi_I(G) = \bar{\pi}_F - \bar{\pi}_I(1 + G). \quad (9)$$

Higher government demand increases incremental profits and leaves frontier profits unchanged. The frontier premium remains positive for $G < G^* \equiv \bar{\pi}_F/\bar{\pi}_I - 1$.

Firm's Problem. A line's continuation value depends on its type and on G . The firm's portfolio value combines operating profits from each line, net of innovation costs, with the option to expand the portfolio through incremental or frontier advances: both the total number of lines and their composition matter for value. Appendix D.1 establishes that the firm value function is additive:

$$V(n_I, n_F; G) = n_I v_I(G) + n_F v_F(G). \quad (10)$$

Separability follows from the absence of cross-line interactions: profits, innovation arrivals, and destruction risks operate independently across lines, and the model has no firm-level resource constraint. Portfolio composition affects firm value through (n_I, n_F) because $v_F > v_I$.

Let

$$\bar{v}(\mu; G) \equiv (1 - \mu)v_I(G) + \mu v_F(G) \quad (11)$$

denote the expected value of a new line.

A firm with state (n_I, n_F) earns flow profits

$$n_I \pi_I(G) + n_F \pi_F(G) - (n_I + n_F) [\Phi(x) + \Psi(\mu)].$$

Each existing line generates innovation opportunities at rate x . Conditional on arrival, the firm creates an incremental line with probability $1 - \mu$ and a frontier line with probability μ . Rivals displace each existing line at rate τ .

The portfolio HJB is

$$\begin{aligned} rV(n_I, n_F; G) = \max_{x, \mu} & \left\{ \underbrace{n_I \pi_I(G) + n_F \pi_F(G)}_{\text{flow operating profits}} - \underbrace{(n_I + n_F) [\Phi(x) + \Psi(\mu)]}_{\text{innovation and frontier-orientation costs}} \right. \\ & + (n_I + n_F)x \left[(1 - \mu) \underbrace{(V(n_I + 1, n_F; G) - V(n_I, n_F; G))}_{\text{arrival of a new incremental line}} \right. \\ & \quad \left. + \mu \underbrace{(V(n_I, n_F + 1; G) - V(n_I, n_F; G))}_{\text{arrival of a new frontier line}} \right] \\ & + \tau n_I \underbrace{(V(n_I - 1, n_F; G) - V(n_I, n_F; G))}_{\text{destruction of an incremental line}} \\ & \left. + \tau n_F \underbrace{(V(n_I, n_F - 1; G) - V(n_I, n_F; G))}_{\text{destruction of a frontier line}} \right\}. \end{aligned} \quad (12)$$

Substituting (10) into the portfolio HJB reduces the problem to two value equations:

$$(r + \tau)v_I(G) = \pi_I(G) - \Phi(x) - \Psi(\mu) + x\bar{v}(\mu; G), \quad (13)$$

$$(r + \tau)v_F(G) = \pi_F(G) - \Phi(x) - \Psi(\mu) + x\bar{v}(\mu; G). \quad (14)$$

The equations differ only in flow profits. The firm chooses (x, μ) given (v_I, v_F) . Appendix D.1 derives the system.

Optimality. The effort first-order condition is

$$c_x \zeta x^{\zeta-1} w = \bar{v}(\mu; G). \quad (15)$$

The unconstrained frontier-choice condition is

$$\kappa\mu^{\eta-1} = x(v_F - v_I). \quad (16)$$

Equation (15) says that firms raise innovation effort until the marginal cost of another arrival equals the expected value of the new line. Equation (16) requires the marginal cost of frontier orientation to equal the value premium from holding a frontier rather than an incremental line. The condition (16) holds on the interior region $\mu^* \in (0, 1)$; otherwise $\mu^* \in \{0, 1\}$.

At corner solutions $\mu^* \in \{0, 1\}$, the KKT conditions replace the interior equality; Appendix D.1 states the full characterization.

3.3 General Equilibrium

The balanced-growth-path equilibrium requires household utility maximization, firm-level effort and innovation-type optimization, free entry, and labor-market clearing.

Log utility gives the balanced-growth Euler equation

$$r = \rho + g. \quad (17)$$

The unit mass of product lines is fixed. In the baseline, entrants do not create varieties or quality improvements; they use h_E units of setup labor to take ownership of existing product lines. Let τ denote the economy-wide rate at which any given line changes hands, and let e denote the rate at which outside entrants acquire lines. Since incumbents also displace rivals through quality-improving innovation at rate x per line they hold, the total ownership-change rate satisfies

$$x + e = \tau. \quad (18)$$

Hence $e = \tau - x$, aggregate entry labor is $h_E(\tau - x)$, and equilibrium requires $x < \tau$.

Potential entrants do not choose innovation effort or direction in the baseline. They draw the type of the line they acquire from the equilibrium line distribution, so the expected acquisition value is $\bar{v}(\mu^*; G)$. Free entry equates this value to the monetary setup cost:

$$\bar{v}(\mu^*; G) = wh_E. \quad (19)$$

Production labor accounts for private and government demand. Incremental lines serve private and government demand; frontier lines serve private demand only. Expected production labor per line is

$$L_P(\mu, G, w) = \frac{(1 - \mu)(1 - \bar{\pi}_I)(1 + G) + \mu(1 - \bar{\pi}_F)}{w}. \quad (20)$$

R&D labor per line is $\ell_R(x, \mu, w) = [\Phi(x) + \Psi(\mu)]/w$. Labor-market clearing requires

$$\bar{L} = L_P + \ell_R + h_E(\tau - x). \quad (21)$$

Balanced-growth output growth g is

$$g = x[(1 - \mu) \log \lambda_I + \mu \log \lambda_F]. \quad (22)$$

Only incumbent innovation appears in (22); entrant acquisitions change ownership and firm demographics but add no quality steps. Appendix E.5.2 considers the case where entrants also innovate.

Definition 3.1 (General equilibrium). Given government spending G , labor endowment $\bar{L}$, household discount rate ρ , and innovation cost scale c_x , with the interest rate $r = \rho + g$ from (17), a balanced-growth-path general equilibrium is a collection of value functions (v_I, v_F) for incremental and frontier product lines, a firm innovation effort rate x and frontier-innovation probability μ , an economy-wide wage w , and a creative destruction rate τ , with $x < \tau$, such that:

(i) *Firm value functions.* Incremental and frontier line values satisfy the Bellman equations

$$(r + \tau)v_I = \pi_I(G) - \Phi(x) - \Psi(\mu) + x\bar{v}(\mu; G), \quad (23)$$

$$(r + \tau)v_F = \pi_F(G) - \Phi(x) - \Psi(\mu) + x\bar{v}(\mu; G). \quad (24)$$

(ii) *Optimal innovation effort.* The marginal cost of an additional unit of effort equals the expected value of a new line:

$$c_x \zeta x^{\zeta-1} w = \bar{v}(\mu; G). \quad (25)$$

(iii) *Optimal innovation composition.* On the interior, the marginal cost of increasing the frontier-innovation probability equals the value premium from holding a frontier rather than an incremental line:

$$\kappa \mu^{\eta-1} = x(v_F - v_I), \quad \mu \in [0, 1]. \quad (26)$$

At $\mu \in \{0, 1\}$, the KKT conditions in Appendix D.1 replace the equality.

(iv) *Free entry*. The expected value of an acquired line equals the entrant's setup cost:

$$\bar{v}(\mu; G) = wh_E. \quad (27)$$

(v) *Labor-market clearing*. Production labor, R&D labor, and entry labor exhaust the endowment:

$$\bar{L} = L_P(\mu, G, w) + \ell_R(x, \mu, w) + h_E(\tau - x). \quad (28)$$

Computationally, the equilibrium is a system of six equations in six unknowns $(v_I, v_F, x, \mu, w, \tau)$. Given a trial wage w , the Bellman equations (23)–(24), the optimality conditions (25)–(26), and the free-entry condition (27) jointly determine $(v_I, v_F, x^*, \mu^*, \tau)$ in closed form. Bisection over w then satisfies labor-market clearing (28). Appendix D.3 describes the solution algorithm. Counterfactuals vary G and related policy parameters and re-solve the full system.

3.4 Characterization

The Compression Mechanism. The central result is that procurement compresses the frontier premium: government demand raises incremental-line profits $\pi_I(G) = \bar{\pi}_I(1 + G)$ without changing frontier-line profits $\pi_F(G) = \bar{\pi}_F$, narrowing the value gap $v_F - v_I$ that incentivizes frontier investment. Subtracting (13) from (14) makes this precise:

$$v_F(G) - v_I(G) = \frac{\pi_F(G) - \pi_I(G)}{r + \tau} = \frac{\bar{\pi}_F - \bar{\pi}_I(1 + G)}{r + \tau}. \quad (29)$$

As G rises, the numerator shrinks, so the frontier premium falls. On the interior region, substituting (29) into (16) gives

$$\mu(G) = \left[\frac{x^*(G)}{\kappa} \cdot \frac{\bar{\pi}_F - \bar{\pi}_I(1 + G)}{r + \tau} \right]^{1/(\eta-1)}. \quad (30)$$

The constrained policy is the projection of $\mu(G)$ onto $[0, 1]$.⁷

⁷Both line types generate positive flow profits for all $G \geq 0$: $\pi_I(G) = \bar{\pi}_I(1 + G) > 0$ and $\pi_F(G) = \bar{\pi}_F > 0$. Firms accept procurement demand without exit. The interior condition $G < G^*$ guarantees a positive frontier premium but does not guarantee $\mu^* < 1$; Appendix D.1 states the optimality condition used when counterfactuals reach a corner at $\mu^* = 1$.

Proposition 3.2 (Procurement compresses the frontier share). *Let $A(G) \equiv \bar{\pi}_F - \bar{\pi}_I(1+G)$. Suppose $\bar{\pi}_F > \bar{\pi}_I$, $r + \tau > 0$, $\kappa > 0$, $\eta > 1$, $G < G^* \equiv \bar{\pi}_F/\bar{\pi}_I - 1$, and $\mu^*(G) \in (0, 1)$. If*

$$\frac{1}{x^*(G)} \frac{\partial x^*(G)}{\partial G} < \frac{\bar{\pi}_I}{A(G)} + \frac{1}{r + \tau} \left(\frac{\partial r}{\partial G} + \frac{\partial \tau}{\partial G} \right),$$

then $\partial \mu^(G)/\partial G < 0$.*

Proof. Substituting (29) into (16) gives the interior solution (30). Differentiating along the equilibrium path and rearranging,

$$\frac{\partial \mu^*}{\partial G} < 0 \iff \frac{1}{x^*(G)} \frac{\partial x^*(G)}{\partial G} < \frac{\bar{\pi}_I}{A(G)} + \frac{1}{r + \tau} \left(\frac{\partial r}{\partial G} + \frac{\partial \tau}{\partial G} \right).$$

The stated condition ensures that direct frontier-premium compression and the equilibrium $r + \tau$ response dominate the effort response. $\square$

Proposition 3.2 rationalizes the pattern in Table 1.⁸ Procurement environments that place more weight on specified compliance and implementation reliability correspond to a larger effective demand wedge. As the wedge rises, the frontier premium narrows and the equilibrium frontier share falls.

Labor reallocation. When G rises, the labor market clearing condition (21) tightens against R&D: incremental lines serve both private and government demand, so production labor L_P rises with G , crowding out R&D labor ℓ_R . The compression mechanism in equation (29) captures the price channel (the frontier premium narrows) but not the quantity channel (fewer workers available for R&D). Both channels reduce frontier innovation. The policy counterfactuals reverse both: removing the demand wedge restores the frontier premium, and reducing government-driven production-labor demand frees resources for innovation. Table 6 reports ΔL_P and $\Delta \ell_R$ for each exercise, quantifying the size of the quantity channel relative to the price channel.

Directional misallocation. The composition shift is a dynamic allocation wedge. Under Bertrand pricing, frontier lines earn a higher operating-profit share than incremental lines ($\bar{\pi}_F > \bar{\pi}_I$), but procurement demand raises only the incremental payoff. The private return to frontier orientation therefore falls by more than the social growth return does: firms internalize the procurement margin

⁸Procurement affects growth through two margins. Higher G can increase innovation effort x^* (the effort channel) and lower frontier orientation μ^* (the composition channel). The marginal growth effect decomposes as $\partial g/\partial G = (\partial g/\partial x^*)(\partial x^*/\partial G) + (\partial g/\partial \mu^*)(\partial \mu^*/\partial G)$. Proposition 3.2 gives a sufficient condition under which the composition channel dominates.

on incremental lines but not the aggregate value of preserving frontier quality steps. Appendix D.2 formalizes this wedge and distinguishes it from a static markup-dispersion channel.

3.5 Stationary Distribution

The stationary distribution maps equilibrium objects into a model-implied cross section of firms. Let $N = n_I + n_F$ denote total firm scope. A firm gains lines (displacing rivals) at rate x^*N and loses lines (displaced by rivals) at rate τN . Each entrant begins with a single displaced line ($N = 1$). Since the unit mass of product lines is fixed, the entrant displacement rate satisfies $e = \tau - x^*$ and $x^* < \tau$. Define

$$q \equiv \frac{x^*}{\tau} \in (0, 1).$$

$q = x^*/\tau$ is the probability that when a product line changes ownership, it is acquired by an expanding incumbent rather than a new entrant; higher q implies larger average firm scope. Together with μ^* , it is a sufficient statistic for the full cross-sectional distribution of firm portfolios. The stationary distribution of firm scope is

$$\Pr(N = n) = \frac{q^n}{n[-\log(1 - q)]}, \quad n \geq 1. \quad (31)$$

The logarithmic form is the steady-state solution to the birth–death process in which a firm gains lines at rate x^* and loses them at rate τ ; the ratio $q = x^*/\tau$ is the single sufficient statistic for the scope distribution.⁹

The joint distribution Γ combines the scope distribution (31) and, conditional on N , a binomial over line types: each line independently becomes frontier with probability μ^* , regardless of portfolio size.

$$\Gamma(n_I, n_F; G, \psi) = \Pr(N = n_I + n_F) \binom{n_I + n_F}{n_F} (\mu^*)^{n_F} (1 - \mu^*)^{n_I}. \quad (32)$$

Conditional on $N = n$, the number of frontier lines follows:

$$\Pr(n_F = b \mid N = n) = \binom{n}{b} (\mu^*)^b (1 - \mu^*)^{n-b}. \quad (33)$$

Three cross-sectional properties follow from Γ and connect the model directly to observable moments.

⁹The normalizing constant $-\log(1 - q) = \sum_{n=1}^{\infty} q^n/n$ ensures probabilities sum to one (logarithmic series identity). The distribution is logarithmic rather than geometric because each of the firm's n lines independently generates a new line at rate x^* and loses one at rate τ : both birth and death rates scale linearly with n . A geometric distribution arises when the net gain rate is constant, independent of portfolio size; the proportional scaling here yields the logarithmic form as the unique stationary solution.

Proposition 3.3 (Cross-sectional implications of procurement). *Let $q = x^*/\tau$ and $s_F \equiv n_F/N$. Under Γ :*

(i) $\mathbb{E}[s_F] = \mu^*$ and

$$\text{Var}(s_F) = \mu^*(1 - \mu^*) \frac{\sum_{n=1}^{\infty} q^n / n^2}{-\log(1 - q)}.$$

The average frontier share across firms equals μ^ , so the composition moment used in estimation is simply the cross-sectional mean of s_F .*

(ii)

$$\Pr(n_F = 0) = \frac{-\log(1 - q(1 - \mu^*))}{-\log(1 - q)}.$$

As μ^ falls, more firms end up with no frontier lines at all: each line they hold is less likely to be frontier-oriented, so the share of purely incremental firms rises.*

(iii) $\mathbb{E}[\pi(G)] = \mu^*\bar{\pi}_F + (1 - \mu^*)\bar{\pi}_I(1 + G)$. *A lower frontier share shifts the portfolio toward incremental lines, which earn lower markups, reducing average per-line profits.*

Higher government demand lowers μ^* and, through the equilibrium entry condition, also adjusts q ; both shifts push the cross section toward incremental-only portfolios. The next section estimates $(\kappa, \lambda_I, \lambda_F, h_E)$ by matching five empirical moments that span the composition and size margins.

4 Quantitative Analysis

The quantitative analysis fits the model to observed innovation composition and firm dynamics, then compares policy reforms along the composition and effort margins. The parameterization has two blocks: externally calibrated parameters and the jointly estimated structural parameters.

4.1 Parameterization

Empirical sample. Model moments come from the 2009–2018 MIP sample (52,518 observations, 18,824 firms), when frontier-share, firm-size, and new-product variables are jointly available.

Externally calibrated objects. We set the household discount rate to $\rho = 0.04$ and use annual TFP growth $\bar{g} = 0.5\%$ as the benchmark growth normalization, consistent with Penn World Tables 10.01 and [Bergeaud et al. \(2016\)](#). The balanced-growth Euler equation implies $r = \rho + \bar{g} = 0.045$. We set the innovation-effort and frontier-orientation-cost curvatures to $\zeta = \eta = 2$.

Benchmark government demand $\bar{G} = 0.018$ implies that procurement in MIP-covered industries equals approximately 1 percent of those industries' gross output, obtained by scaling total reported German procurement (EUR 132 billion, Vergabestatistik 2022 ([BMWK, 2023](#))) by the share flowing to MIP-covered sectors (27.5 percent from TED supplier-industry composition) and dividing by Destatis Input-Output gross output for those sectors. The model uses this benchmark to translate the compositional pattern in the MIP into aggregate growth and welfare implications of a permanent change in G .

4.2 Estimation

We estimate the structural parameter vector $\psi = (\lambda_I, \lambda_F, \kappa, h_E)$ by overidentified GMM, targeting five moments with direct model counterparts: (i) the frontier innovation share, (ii) the frontier revenue share among new products, (iii) the large-firm share, (iv) the creative-destruction rate, and (v) the R&D personnel share. The GMM objective is

$$\hat{\psi} = \arg \min_{\psi} [m(\psi) - \hat{m}]^{\top} W [m(\psi) - \hat{m}],$$

where $m(\psi)$ stacks model-implied moments, $\hat{m}$ their empirical counterparts, and W is the inverse of the moment variance-covariance matrix estimated via influence functions. For each candidate ψ , the innovation cost scale $c_x(\psi)$ is adjusted to match the observed German long-run growth rate exactly, so the estimated parameters govern composition and firm dynamics independently of the aggregate growth level.

4.2.1 Identification and Model-to-Data Mapping

The system has four estimated parameters $(\kappa, \lambda_I, \lambda_F, h_E)$ and five targeted moments, yielding one overidentifying restriction. Each parameter has a primary identifying block.

The two frontier-composition moments (the frontier share among product innovators, model counterpart μ^* , and the frontier revenue share among new products, model counterpart: the frontier-weighted revenue share among new products) are most informative about the frontier-choice block $(\kappa, \lambda_I, \lambda_F)$. A higher κ increases the cost of frontier-oriented innovation, lowers the frontier share of new lines, and reduces both moments. The quality steps (λ_I, λ_F) shape the profit gap $\bar{\pi}_F - \bar{\pi}_I$ and therefore the revenue-share differential.

The three scope-and-entry moments (the large-firm share, the creative-destruction rate, and the R&D personnel share) are most informative about h_E . A higher h_E increases the labor cost of new-line creation, lowers equilibrium entry, shifts the scope distribution toward smaller firms, and increases innovation labor relative to production labor. The R&D personnel share identifies h_E along a margin

independent of firm size and turnover. The procurement coefficients are not targeted moments: their signs discipline the mechanism externally, and the model reproduces the qualitative pattern — less frontier innovation, more unchanged-product revenue, and greater scale as procurement rises.¹⁰

Table 4 presents the estimates.

Table 4: Model parameterization, estimates, and recovered objects

Parameter	Description	Estimate	Source / target
<i>Normalizations and recovered objects</i>			
w	Wage	0.953	Endogenous benchmark wage
$\bar{\pi}_F$	Frontier per-line profit	0.142	Implied by Bertrand pricing
$\bar{\pi}_I$	Incremental per-line profit	0.083	Implied by Bertrand pricing
$c_x(\psi)$	Innovation cost scale	15.446	Inner normalization: $g(\psi, c_x; \bar{G}) = \bar{g}$
<i>Externally calibrated</i>			
ρ	Household discount rate	0.040	Literature value
r	Interest rate	0.045	Euler equation: $r = \rho + g$
$\bar{g}$	TFP growth normalization	0.005	Feenstra et al. (2015); Bergeaud et al. (2016)
$\bar{G}$	Benchmark government demand	0.018	(BMWK, 2023)
ζ	Innovation-effort cost curvature	2.000	Literature value
η	Direction-cost curvature	2.000	Calibration
<i>Jointly estimated (GMM, 5 moments, 4 parameters)</i>			
λ_F	Frontier quality step	1.165 (0.003)	GMM estimate
λ_I	Incremental quality step	1.090 (0.003)	GMM estimate
κ	Direction cost	0.055 (0.004)	GMM estimate
h_E	Setup labor per entrant	1.338 (0.021)	GMM estimate

Notes: The model is overidentified, with five empirical moments and four free parameters. The targeted moments are the frontier share among product innovators, the frontier revenue share among new products, the large-firm share, the creative-destruction rate, and the R&D personnel share. Reported standard errors are GMM standard errors; the variance-covariance matrix of empirical moments is constructed using influence functions, with MIP moments clustered at the firm level and the creative-destruction moment using the cross-year standard deviation from BHP. Per-line profits follow from (λ_I, λ_F) under Bertrand pricing, with $\bar{\pi}_k = (\lambda_k - 1)/\lambda_k$. Benchmark government demand $\bar{G} = 0.018$ implies procurement in MIP-covered industries equal to $(1 - \mu^*)\bar{G} \approx 1\%$ of those industries' gross output, calibrated from Vergabestatistik 2022 and Destatis Input-Output accounts (BMWK, 2023). The cost scale $c_x(\psi)$ enters as an inner normalization rather than as an estimated parameter or target moment. *Source:* Authors' calculations using the Mannheim Innovation Panel, BHP, and Destatis.

The parameter estimates are economically sensible. The frontier quality step $\hat{\lambda}_F = 1.165$ implies a 16.5% productivity improvement per creative-destruction event; the incremental step $\hat{\lambda}_I = 1.090$ implies a 9.0% improvement per own-product advance. The ordering $\lambda_F > \lambda_I$ matches the finding of García-Macià et al. (2019) that creative-destruction events carry a larger quality improvement than own-product improvements.

¹⁰Appendix E.1.1 details the measurement equations for each targeted moment. Appendix Table E2 provides a magnitude check: under a proportional mapping from TED contract counts to G , the model implies a market-novelty coefficient of -0.101 , compared with -0.199 in the data.

Under Bertrand competition, per-line profits follow from the quality steps: $\bar{\pi}_F = 14.2\%$ and $\bar{\pi}_I = 8.3\%$, a frontier premium of 5.9 p.p. Procurement erodes the premium. Incremental lines serve both private and government demand, raising their profits by a factor of $(1 + \bar{G})$; frontier lines face private demand only. As $\bar{G}$ rises, the gap narrows and μ^* falls. The frontier-orientation cost $\hat{\kappa} = 0.055$ places adjustment frictions below the profit premium, so composition responds to the procurement-induced gap narrowing. The setup requirement $\hat{h}_E = 1.338$ labor units clears the free-entry condition near the observed creative-destruction rate.

Table 5: Model fit: empirical and model moments

Moment	Data	SE	Model	Diff.
<i>Targeted moments</i>				
Frontier share among product innovators	0.427	(0.005)	0.439	−0.012
Frontier revenue share among new products	58.249	(0.548)	56.870	1.379
Large-firm share	0.104	(0.003)	0.093	0.011
Creative destruction	0.051	(<0.001)	0.058	−0.008
R&D personnel share	0.029	(0.001)	0.035	−0.006

Notes: The table reports model fit on the 2009–2018 common MIP sample. Standard errors for MIP moments are cluster-robust at the firm level. The creative-destruction standard error is the cross-year standard deviation from BHP. The specification is overidentified, with five moments targeting four parameters. The overidentifying restriction is rejected ($J = 134.7$, one degree of freedom). The R&D personnel target measures employees formally classified as R&D personnel, whereas the model counterpart measures incumbent innovation labor; this mapping is approximate.

Source: Authors’ calculations using the calibrated model, the Mannheim Innovation Panel, and BHP.

Table 5 reports the model’s fit. The model fits the frontier-composition moments closely and leaves somewhat larger gaps on the firm-dynamics moments. At the estimated parameters, 44% of product-innovating firms pursue market-novel advances and 56% pursue incremental improvements. The model-implied creative-destruction rate is 5.8% per year, above the 5.1% data moment but still below comparable US estimates (Akçigit and Ates, 2023), consistent with Germany’s lower business dynamism. The model overpredicts the R&D personnel share (0.035 versus 0.029); part-time and multitasking R&D roles, excluded from the model, likely account for part of the gap.

4.2.2 Comparative Statics

Figure 5 traces the model’s endogenous variables as government demand rises. The pattern is a composition shift: the frontier share and frontier revenue share fall, firm size rises, and creative destruction slows.

Higher government demand increases per-line profits for incremental lines, which serve both private and government buyers, while leaving frontier-line profits unchanged. The frontier premium

narrows, reducing the return to novel innovation. Incumbents lower μ^* , the share of new lines targeted at the frontier; the fall in the frontier revenue share mirrors the reallocation within the product portfolio.

Firm dynamics reinforce the composition shift. As incumbents move toward incremental lines, the value of an existing portfolio rises relative to creative destruction; average firm scope expands and the creative-destruction rate slows, consistent with the ORBIS evidence on employment and turnover among winners. Under free entry, aggregate innovation effort adjusts to clear the entry-cost condition; procurement changes *what* firms innovate rather than how much. Because frontier innovations carry a larger quality step $\lambda_F > \lambda_I$, the shift toward incremental lines lowers the average quality improvement per innovation event and long-run growth g .

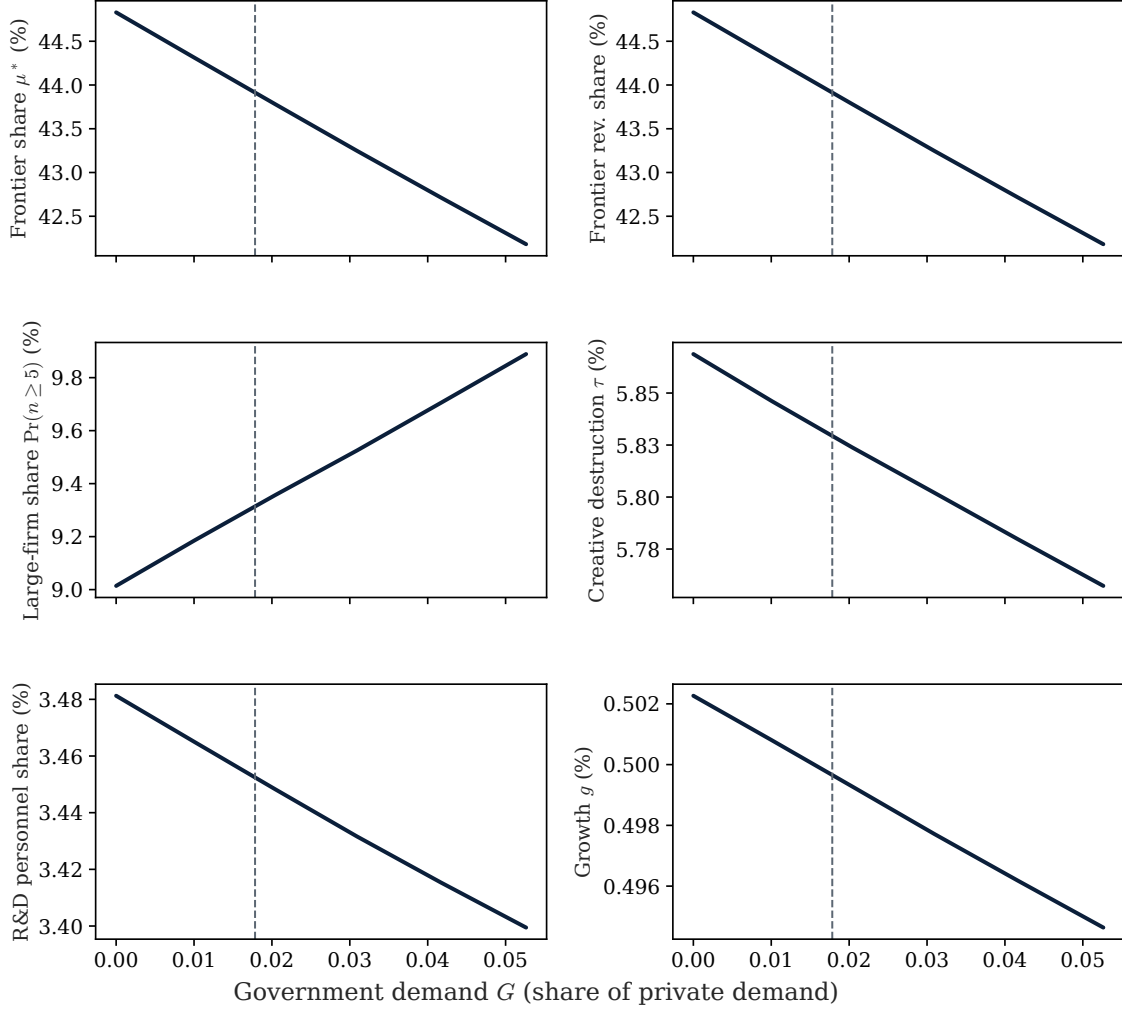


Figure 5: Comparative statics over government demand

Notes: The figure reports the five targeted GMM moments and long-run growth as functions of government demand G . The dashed vertical line marks the benchmark $\bar{G} = 0.018$. Panels report the frontier share μ^* , frontier revenue share, large-firm share $\Pr(n \geq 5)$, creative destruction τ , R&D personnel share, and growth g . The horizontal axis reports government demand as a share of private demand.

Source: Authors' calculations from the calibrated model.

5 Policy Counterfactuals

We examine three counterfactual experiments from the estimated model, each varying a single policy instrument while holding calibrated parameters at their benchmark values.

Each counterfactual uses the same fiscal accounting: procurement is financed by a lump-sum household tax $T = (1 - \mu^*)G$, equal to $T_{bm} \approx 0.010$ at the calibrated benchmark. Because a lump-sum tax is neutral in isolation, the welfare effect of each experiment depends on how the freed fiscal resources are redeployed.

Table 6 summarizes three exercises. The primary counterfactual removes procurement and redirects the freed funds to an R&D tax credit, motivated by Germany’s *Forschungszulage* (Dechezleprêtre et al., 2023). The exercise asks how outcomes change when the same fiscal resources support innovation effort directly rather than incremental demand. A design reform keeps total procurement spending fixed but changes how awards are allocated: we realign contract specifications toward open awards. A third exercise reduces G to a historically scaled estimate of its 2008 level.

Welfare metric. We report a permanent-consumption equivalent (CEV). On a balanced-growth path,

$$W = \frac{\log C_0}{\rho} + \frac{g}{\rho^2},$$

and the permanent-consumption equivalent between policy B and benchmark A is

$$\text{CEV}_{B,A} = 100 \left\{ \exp \left[\Delta \log C_0 + \frac{g_B - g_A}{\rho} \right] - 1 \right\}, \quad (34)$$

with C_0 from the household budget (2) and $\rho = 0.04$. In all scenarios, the household tax adjusts with G . The first term captures the consumption gain or loss as G and μ^* adjust; the second captures the growth gain. Both components are expressed in percent of benchmark consumption.

5.1 Main Counterfactual: Proportional R&D Tax Credit

The main counterfactual redirects the full procurement-tax envelope to a proportional R&D tax credit. R&D tax credits are a standard policy instrument for supporting private innovation effort (Hall and Van Reenen, 2000; Bloom et al., 2002). The government removes procurement ($G = 0$) and introduces a tax credit at rate s , reducing the effective innovation cost to $(1 - s)c_x$. The incumbent Bellman equations become

$$(r + \tau)v_k = \pi_k(0) - (1 - s)\Phi(x) - \Psi(\mu) + x\bar{v}(\mu; 0), \quad k \in \{I, F\}, \quad (35)$$

where the cost reduction increases equilibrium innovation effort and, through the frontier-choice condition (16), moves the innovation toward frontier lines. The tax-credit fiscal bill is $T_s = s \cdot w\ell_R$.

Germany’s *Forschungszulage*, introduced in January 2020, provides a 25 percent credit on eligible R&D labor costs (wages, employer social security contributions, and 60 percent of contracted R&D) and motivates this instrument. The counterfactual redirects the procurement tax $T_{bm} = 0.010$ to a proportional R&D credit at rate $s = 0.141$. The eligible base extends beyond R&D wages to entry costs and a production-implementation component equal to 2.78% of production labor (covering prototyping, process redesign, and adoption costs that credit designs routinely include), making total

credit expenditure equal to T_{bm} .¹¹ As a lower bound, removing procurement without an R&D credit yields a growth gain of 0.27 basis points and CEV of 0.06%. Table 6 reports the main exercise.

Under the full-budget implementation, the frontier share increases from 0.439 to 0.498 ($\Delta\mu^* = 0.059$), and innovation effort is 16.4% higher. R&D labor rises by 18.6% while production labor falls by only 0.3%: because R&D labor is a small share of the labor endowment at the benchmark, the credit draws workers overwhelmingly from a mix of production labor and entry labor rather than displacing production one-for-one, so both the price channel (frontier premium) and the quantity channel (labor available for R&D) move in the same direction. Growth rises by 10.18 basis points per year. [García-Macià et al. \(2019\)](#) estimate that the US composition drift away from creative destruction accounts for 16 of the 34 basis points of TFP growth lost between the 1980s and the 2000s; the growth change here is of the same order and operates through the same composition margin. The net CEV reflects a negative direct consumption effect: the household tax finances the credit rather than falls away, offset by a dominant long-run growth gain.

5.2 Alternative Instruments and Design Reforms

The remaining exercises separate two margins. The first asks how much the composition shifts when procurement design is altered but total spending is held fixed. The second scales aggregate demand to its 2008 level, returning the fiscal savings to households.

Specification realignment. Specification-intensive procurement concentrates government demand on incremental lines by requiring suppliers to meet product requirements rather than performance objectives. Realigned specifications shift toward open, functional awards ([Howell et al., 2025](#)): in a US Air Force R&D program that ran open and conventional competitions simultaneously, “open” topics inviting any useful new technology raised commercial innovation and military adoption, while “conventional” topics that specified the desired technology produced weaker frontier commercialization. Open awards are associated with higher frontier innovation rates, and functional specifications, rather than product-specific requirements, account for the pattern ([Uyarra et al., 2014](#); [Edquist and Zabala-Iturriagoitia, 2020](#)).

Let government demand decompose into a frontier component G_F and an incremental component G_I . In the benchmark, $G_F = 0$ and all demand flows to incremental lines; realigned specifications allow the government to place orders on both line types. Hence, the budget constraint is $(1 - \mu^*)G_I + \mu^*G_F = T_{bm}$. Setting $G_I = G_F$ and imposing the budget constraint yields $G_I = G_F = T_{bm}$: both

¹¹[Guceri and Liu \(2019\)](#) show the UK Enhanced R&D Credit explicitly covers production-adjacent labor; [Chen et al. \(2021\)](#) estimate that 24.2% of reported R&D growth under China’s super-deduction reflects relabeling. The 2.78% component is the model-implied residual, not a separate data target; Appendix E.3.1 derives how the entry-eligibility weight is fixed and the implementation share is solved as the budget-balancing residual.

line types face the same per-line government demand and profits satisfy $\pi_k = \bar{\pi}_k(1 + T_{bm})$ for $k \in \{I, F\}$. The incremental demand advantage disappears, widening the net frontier premium and raising equilibrium frontier orientation. The frontier share rises by $\Delta\mu^* = 0.013$, growth by 0.38 basis points, and CEV by 0.09%. Firms do not attempt more innovations under this reform ($\Delta x^* = 0.00\%$): the arrival rate of new lines is unchanged. What changes is what those attempts aim for. More of the same fixed number of attempts now target frontier lines, and frontier-oriented projects cost more labor to direct than incremental ones. That alone raises R&D labor by 1.0% and frees up production labor by 0.1%: a labor-market response to procurement reform that runs entirely through the type of innovation firms pursue, not through how often they try. Appendix E.3.2 formally derives the equilibrium response.

Historical rollback. This exercise sets G to its 2008 level, using the OECD national-accounts procurement share to calibrate $G_0 = 0.013$ against the benchmark $\bar{G} = 0.018$. The rollback also reduces the household tax $T = (1 - \mu^*)G$: households gain from both improved composition and a smaller fiscal burden. Because Germany's broad procurement-to-GDP ratio was relatively stable over this horizon, the absolute reduction in $\bar{G}$ is modest; the frontier share rises by $\Delta\mu^* = 0.002$, growth by 0.07 basis points per year, and CEV by 0.015%. R&D labor rises by 0.2% and production labor falls by 0.02%, an order of magnitude smaller than under either of the other two exercises, consistent with the modest scale of this reform. The direction is consistent with the empirical estimate, but the historical scaling is not a calibration target ¹²¹³.

The comparison with the design reform is instructive: specification realignment, which holds total spending fixed, generates more than five times the frontier-share gain ($\Delta\mu^* = 0.013$ versus 0.002) at the same fiscal cost. The composition distortion thus reflects how procurement resources are allocated across line types, not the aggregate scale of government demand. All three exercises compare initial and final steady states; Appendix D.5 characterizes the transition path and shows that accounting for gradual composition adjustment leaves the welfare ranking unchanged.

¹²TED contract counts grew far faster than the OECD GDP share over this period, partly reflecting improved reporting compliance rather than real procurement growth.

¹³Appendix E.3.3 quantifies how much of the aggregate decline in Figure E1 this recovers.

Table 6: Policy counterfactuals from the GMM benchmark

Counterfactual	Δx^* (%)	$\Delta \mu^*$	ΔL_P (%)	$\Delta \ell_R$ (%)	Δn_F	Δn_I	Δg (bp/yr)	CEV (%)
R&D tax credit	16.4	0.059	-0.29	18.64	0.322	0.072	10.18	1.55
Specification realignment	0.00	0.013	-0.11	0.99	0.013	-0.046	0.38	0.09
Historical rollback	0.00	0.002	-0.02	0.21	0.002	-0.008	0.07	0.015

Notes: Columns report changes relative to the calibrated benchmark. Δx^* : equilibrium innovation effort (%). $\Delta \mu^*$: frontier share of new lines (level change). ΔL_P , $\Delta \ell_R$: aggregate production and incumbent R&D labor (%), holding the fixed labor endowment $\bar{L}$ fixed; the entry-labor residual absorbs the remainder. Δn_F , Δn_I : expected frontier and incremental product lines per firm (level change), from the portfolio-scope moment used to calibrate firm size (Section 3). Δg : annual growth (basis points). CEV: permanent-consumption equivalent, $100\{\exp[\Delta \log C_0 + \Delta g/\rho] - 1\}$ with $\rho = 0.04$. *Row 1 (R&D tax credit)* removes procurement and redeploys the fixed tax $T_{bm} = 0.01001$ to a broad-base R&D credit at rate $s = 14.1\%$, with entry-eligibility weight $\omega_E = 0.5$ and production-implementation weight $\omega_P = 0.0278$. *Row 2 (specification realignment)* holds total procurement spending fixed, $(1 - \mu^*)G_I + \mu^*G_F = T_{bm}$, and reallocates it across frontier and incremental lines. *Row 3 (historical rollback)* scales G to Germany’s 2008 OECD procurement-to-GDP ratio relative to the calibrated benchmark and returns the resulting tax reduction to households. *Source:* Authors’ calculations from the calibrated model.

The welfare effects are comparable in scale to other quantitative benchmarks. The main R&D-credit counterfactual (1.55% CEV) exceeds the inflation-cost estimate of [Lucas \(2000\)](#) and the short-termism distortion in [Terry \(2023\)](#), falls below estimates for R&D misallocation and monopsony costs ([Lehr, 2024, 2025](#)), and lies at the lower end of trade-liberalization gains ([Arkolakis et al., 2012](#); [Caliendo and Parro, 2015](#); [Melitz and Redding, 2015](#); [Costinot and Rodríguez-Clare, 2015](#)). It is also below business-cycle costs of around 2% ([Krusell et al., 2009](#)). Budget-neutral design reform is smaller because it operates through composition alone; the gap between the two types of reform reflects the fiscal channel: removing procurement eliminates both the composition wedge and the household tax, while design reforms hold spending fixed.

These gains translate into long-run TFP and GDP per capita growth: in quality-ladder models, output expansion comes entirely from quality improvements. Against a German baseline of $\bar{g} \approx 50$ basis points per year ([Bergeaud et al., 2016](#)), the R&D-credit counterfactual adds 10.18 basis points and specification realignment adds 0.38 basis points. Both run through composition: procurement reform moves the innovation mix toward creative destruction, the highest-step margin. The 10.18 basis-point gain is of the same order as the composition-driven component of the US productivity slowdown: [García-Macià et al. \(2019\)](#) attribute 16 of the 34 basis points of TFP growth lost between the 1980s and 2000s to a shift away from creative destruction; against Germany’s $\bar{g} \approx 50$ basis-point baseline, the procurement composition wedge accounts for a comparable share of forgone growth. By contrast, [Atkeson and Burstein \(2019\)](#) find limited productivity gains from scale-neutral innovation subsidies, reinforcing the importance of composition over scale.

5.3 Robustness Checks

We conduct four robustness exercises. The first explores an alternative policy scenario: a procurement officer risk-preference reform that shifts demand toward frontier lines through reduced career risk, microfounded by a mean-variance government objective. The second evaluates the calibrated model at sector-specific procurement wedges, testing whether the aggregate composition pattern holds within industries. The third checks structural robustness through parameter perturbations and an extended model variant in which entrants also innovate. The fourth checks whether the welfare ranking survives once government purchases are credited with their own value rather than treated as pure transfer.

Procurement officer risk preferences. Procurement rules create career incentives that push buyers toward low-risk, specification-compliant suppliers. When transparency requirements expose shortfalls and career penalties fall on contracting officers who oversee failed frontier projects, officers rationally favor incremental lines whose delivery risk is lower and easier to document (Prendergast, 2007; Bandiera et al., 2009). Rigid award criteria that reward compliance over performance compound the bias. Procurement reforms that reduce career penalties or tighten technical evaluation lower the effective risk premium on frontier awards and shift demand toward frontier lines at zero additional fiscal cost (Uyarra et al., 2014; OECD, 2017; Decarolis et al., 2021; Bosio et al., 2022).

Let the government hold mean-variance preferences over delivered quality with risk-aversion parameter $\gamma_G \geq 0$. Frontier lines offer higher expected quality but also higher delivery variance than incremental lines. For sufficiently risk-averse buyers, the lower delivery variance outweighs the frontier quality advantage, rationalizing the baseline demand asymmetry $G_F = 0$; Appendix E.3.3 derives the preference threshold and its comparative statics. Redirecting one-quarter of the procurement budget toward frontier lines raises the frontier share by $\Delta\mu^* = 0.008$, growth by 0.21 basis points, and CEV by 0.05%.

Sector-specific wedges. We evaluate the calibrated model at sector-specific procurement wedges G_s instead of re-estimating it separately by sector. We map mean sector-year reported contract counts proportionally into model units, normalizing their mean to the covered-sector benchmark $\bar{G} = 0.018$. Among sectors with positive exposure, the resulting wedges range from $G_s = 0.008$ to $G_s = 0.025$. The frontier share in each sector is inversely related to its procurement exposure, matching the cross-sectoral pattern: the largest declines in μ_s^* occur in the most exposed sectors. Removing the sector wedge raises the frontier share by 0.48–1.44 percentage points and annual growth by 0.13–0.39 basis points. Because the count-to- G conversion is imposed rather than estimated, we treat these results as illustrative rather than as sector-specific calibration. Appendix E.4.2 reports the full mapping.

Model variants. We assess robustness to two alternative specifications. In the first, we perturb each structural parameter by one standard error around the calibrated baseline; annual growth moves by at most 0.72 basis points in either direction and the welfare change ranges from -0.313 to 0.312 percent, with κ generating the largest movement in the frontier-innovation share. In the second, we re-estimate an extended specification in which entrants also raise line quality and the policy ranking survives. Appendices [E.5](#) and [E.5.2](#) report the full results.

Government goods in welfare. The baseline CEV treats procurement spending as a pure transfer: households pay the tax that finances it, but the model assigns no utility value to what the government buys, even though procurement in practice pays for infrastructure, IT, medical supplies, and other outputs the government needs to operate. Appendix [E.5.3](#) extends household utility to include a log-weighted index of government-purchased output, weighted by $\alpha \in [0, 1]$ relative to private consumption, and recomputes CEV for all three main counterfactuals under this alternative aggregator. The design reforms are essentially unaffected (government-goods value rises slightly as μ^* increases, since more government demand reaches frontier lines). The R&D credit, which eliminates procurement entirely, loses the full benchmark government-goods term; even so, the break-even weight is $\alpha^* \approx 1.55$, above the maximum reasonable weight of $\alpha = 1$ (equal value to private and public consumption), so the credit remains welfare-improving even when government purchases are valued at par with private consumption. Only the historical rollback, whose baseline gain is small, turns welfare-reducing once α is close to one. Table [E10](#) reports the full sensitivity.

6 Conclusion

German R&D spending holds steady while the share of firms introducing market-novel products falls from 26 percent in the early 2000s to 8 percent by 2023. This paper documents public procurement as a channel associated with the observed composition shift. Government contracts specify technical standards, certification requirements, and delivery schedules that favor existing products over frontier novelty; sectors with higher procurement exposure introduce market-novel products less often and generate more revenue from existing products.

Three patterns characterize the evidence. On innovation composition, a one-unit increase in sector procurement exposure is associated with a 0.199 percentage-point lower market-novelty rate and a 0.379 percentage-point higher existing-product revenue share. On firm scale, matched procurement winners expand in scale, with higher employment and turnover, without a corresponding gain in productivity or profitability. On heterogeneity, the composition association concentrates in supplies and services contracts and is larger for central-government buyers; works contracts show no

significant association on either margin. The evidence is consistent with a shift away from frontier novelty and toward scale-oriented delivery rather than a reduction in innovation effort overall.

We extend [Klette and Kortum \(2004\)](#) by adding a direction-choice margin: each period, firms allocate innovative effort between frontier and incremental lines, facing separate quality steps and a reallocation cost. Procurement enters as a demand wedge that compresses the frontier premium, shifting the equilibrium mix toward incremental delivery. We estimate four structural parameters by GMM, targeting five empirical moments from the microdata; the fitted model attributes the growth cost to composition rather than to the level of public spending.

The primary counterfactual redirects procurement funds to an R&D tax credit modeled on Germany's *Forschungszulage*. Growth rises by 10.18 basis points per year and welfare by 1.55 percent of permanent consumption; about one-fifth of Germany's trend TFP growth and above standard inflation-cost estimates.

A budget-neutral specification reform raises growth by 0.38 basis points; a scaled volume reduction adds 0.07 basis points. Across counterfactuals, the destination of public funds matters more for growth than their volume or contractual form: redirecting procurement to an innovation subsidy yields a gain an order of magnitude larger than reforming contract design, and design reforms in turn dominate volume reduction.

The results highlight a distinction between the private and aggregate consequences of procurement. For individual firms, contracts are associated with higher scale and revenue stability, consistent with the employment and turnover patterns in the data. In the model, the composition shift toward specification-compliant delivery lowers growth because the social return to frontier innovation exceeds its private return in the presence of a procurement wedge. The model therefore gives quantitative grounding to long-standing practitioner concerns that specification-heavy procurement creates barriers to frontier innovation ([Uyarra et al., 2014](#); [OECD, 2017](#); [Edquist and Zabala-Iturriagagoitia, 2020](#)). Our mechanism may not be specific to Germany: similar channels appear active in US defense contracting ([Howell et al., 2025](#)), historical procurement ([Clemens and Rogers, 2026](#)), and government R&D programs ([Moretti et al., 2025](#)). Future work may exploit these diverse contexts to test when specification-based systems erode the frontier premium and when functional criteria preserve it; a complementary direction would decompose the composition margin into vertical quality upgrading and horizontal product-line expansion using establishment-level records, following [García-Macià et al. \(2019\)](#).

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Appendix

The appendix has five parts. Appendix A documents the merged MIP–procurement panel and exposure measures. Appendix B reports ORBIS validation exercises on winner scale, profitability, and selection. Appendix C presents additional empirical robustness checks, including public R&D subsidies, alternative exposure definitions, and common-sample specifications. Appendix D contains model derivations and numerical solution details. Appendix E presents counterfactual diagnostics, robustness exercises, and sector-level direction-cost results.

A Additional Procurement Descriptives

A.1 Variable Construction

Innovation outcomes (MIP). The primary outcome is market novelty: a binary indicator equal to one if a firm introduced at least one product new to the market over the preceding three-year reference period (the MIP questionnaire’s product-innovation module asks, e.g., "During the period $[t - 2] - [t]$, did your enterprise introduce into the market any new or significantly improved products/services that were market novelties"). Because this is a self-report resting on the firm’s own judgment of what counts as new to its market rather than new only to the firm, we treat it as a subjective measure and rely on the within-firm panel structure, rather than the cross-sectional level, to interpret it. The market-novelty measure is distinct from the broader product-innovation indicator used by Krieger et al. (2024), which equals one if a firm introduced any new or improved product (including improvements to existing product lines that are not new to the market). Appendix Table A3 reports both measures side by side.

The complementary revenue outcome is the share of total turnover from existing (unchanged) products, defined as one hundred minus the MIP share of turnover from products new or improved over the same three-year reference period. We use the complement because our hypothesis concerns the reallocation of commercial activity toward existing lines. The process-innovation outcome is likewise defined over the three-year reference period rather than the survey year alone; year fixed effects absorb the 2019 definitional expansion that raised the reported rate from 28 to 53 percent.

Additional outcomes (the market-novel revenue share, cost-reducing process innovation, geographic novelty, and R&D inputs) follow their standard MIP questionnaire definitions and are described where first used in the main text.

Procurement exposure (TED). The treatment variable is $\text{arcsinh}(\text{contracts}_{s,t})$, where $\text{contracts}_{s,t}$ is the count of contract award notices in TED for sector s in year t . Contracts are assigned to

MIP sectors by mapping each contract’s EU product code to the corresponding two-digit German industry classification, aggregated to the 21-sector industry scheme of [Czarnitzki et al. \(2020\)](#), the same classification used by [Krieger et al. \(2024\)](#).

A.2 Merged Panel and Exposure Measures

The appendix reports the merged MIP–procurement panel and the cross-sectional exposure contrast used in the main text.

Table A1: Summary Statistics: MIP–Procurement Merged Panel, 2000–2023

Variable	N	Mean	SD	Min	Max
<i>Panel A. Innovation outcomes</i>					
Market-novel product (=1)	132,311	16.19	36.83	0.000	100.0
World-first product (=1)	78,652	19.97	39.98	0.000	100.0
Product innovator (=1)	144,286	41.84	49.33	0.000	100.0
Process innovator (=1)	144,286	38.71	48.71	0.000	100.0
R&D active (=1)	140,780	30.64	46.10	0.000	100.0
New-product revenue (% of sales)	108,632	9.839	20.77	0.000	100.0
Unchanged-product revenue (% of sales)	72,999	84.48	27.77	0.000	100.0
Cost-reduction revenue (% of sales)	88,431	2.503	6.642	0.000	100.0
Market-novel share within new products (%)	33,925	25.77	37.05	0.000	100.0
<i>Panel B. Procurement and firm controls</i>					
$\text{arcsinh}(\text{Contracts, sector} \times \text{year})$	144,287	3.041	3.975	0.000	10.63
Exporter (=1)	144,287	38.09	48.56	0.000	100.0
Export intensity (% of turnover)	112,858	14.48	24.16	0.000	100.0

Sample: Full MIP panel 2000–2023 merged with OpenTender sector-year procurement exposure (procurement coded as zero for 2000–2008). Binary variables reported as proportions ($\times 100$, in percentage points). Market-novel share capped at 100 (values > 100 reflect MIP coding artefacts where reported market-novel revenue exceeds reported new-product revenue).

A.3 Comparison with Krieger, Pruefer, and Strecke (2024)

[Krieger et al.](#) study price-criterion tenders using a binary for *any* new or improved product, a measure that pools frontier novelty and incremental improvements to existing lines. We focus on *market novelty* alone, precisely because the composition mechanism predicts that procurement reallocates activity between these two types rather than suppressing innovation overall.

The composition mechanism predicts precisely this pattern. If procurement shifts firms from frontier novelty toward market-known improvements (leaving total product-innovation activity approximately unchanged), then a measure that aggregates both types should be near zero. Table [A3](#) confirms the prediction: market novelty falls with procurement exposure, while the [Krieger et al.](#)

Table A2: Innovation Outcomes by Procurement Exposure: High vs. Low Exposure Sector-Years

	Low exposure	High exposure	Difference
<i>Innovation outcomes</i>			
Market novelty (binary, %)	14.17	10.70	−3.47***
World-novel product (binary, %)	18.54	15.63	−2.90***
Any product innovator (%)	35.02	34.41	−0.61*
Any process innovator (%)	29.50	42.64	13.14***
New-product revenue (% turnover)	8.93	8.07	−0.86***
Unchanged-product revenue (% turnover)	90.00	89.01	−0.99***
Cost-reduction revenue (% turnover)	1.79	2.12	0.33***
Market-novel / new-prod revenue (%)	25.10	21.20	−3.90***
R&D active (%)	28.56	25.09	−3.47***
R&D intensity (% turnover)	1.05	1.99	0.94***
Observations	40,029	39,999	

Notes: Unit of observation is a firm-year. Low-exposure sector-years have $\text{arcsinh}(\text{contracts}_{s,t})$ at or below the sample median; high-exposure sector-years are above the median. Difference is the high-minus-low mean, with significance from a Welch two-sample t -test. Binary variables multiplied by 100 (%). R&D intensity is in percent of turnover, conditional on positive R&D activity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

aggregate indicator is near zero. The near-zero aggregate is a consequence of the composition shift, not evidence against it.

A.4 Procurement Composition by Contract Value

The main text's contract-type and buyer-category decomposition (Figure 3) is built on contract counts; Figure A1 reports the same two breakdowns by contract value instead. Works contracts account for a larger value share (around 30–50 percent in most years) than count share (7 percent), reflecting the high unit value of construction and infrastructure awards, while the composition by buyer category is similar across both measures. The count-based decomposition in the main text remains more directly relevant to the treatment variable, which is the arcsinh of contract counts.

Table A3: Variable Definition Comparison: This Paper vs. Krieger, Pruefer, and Strecke (2024)

	This paper		KPS (2024) comparable	
	(1) Mkt. novelty	(2) Existing-prod. rev. (%)	(3) Prod. innov.	(4) New-prod. rev. (%)
arcsinh(proc. lots, MIP sector)	-0.199* (0.101)	0.379** (0.137)	-0.009 (0.279)	-0.122 (0.102)
Observations	66,980	25,962	70,462	66,367
MIP-sector clusters	21	21	21	21

Firm and year FEs. SE clustered by MIP sector (21 clusters). Cols. (1)–(2): this paper’s market-novelty outcomes. Market novelty = products new to the market; existing-product rev. = share of turnover from unchanged products. Cols. (3)–(4): variable definitions of [Krieger et al. \(2024\)](#). Product innovation = new or improved products (includes market-novel and market-known improvements); new-product rev. = share of turnover from new or improved products. The market-novelty estimate (col. 1) is larger in magnitude than the product-innovation estimate (col. 3), consistent with procurement shifting composition within product innovation rather than reducing its total incidence. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

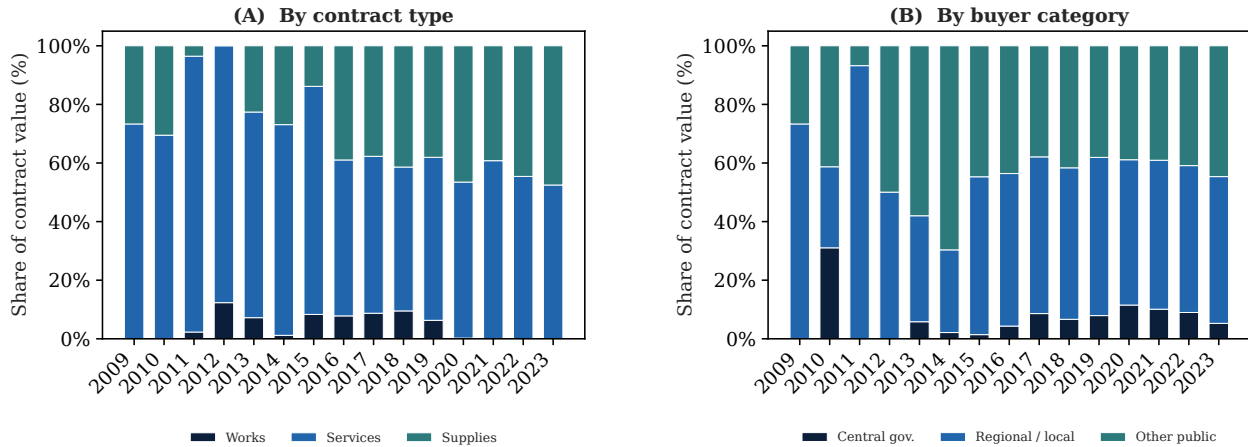


Figure A1: Procurement composition in MIP-covered industries by contract value, 2009–2023

Notes: Share of total above-threshold contract value by type (Panel A: works, services, supplies) and by buyer category (Panel B: central government, regional and local government, other public bodies). Restricted to German industry sectors covered by the Mannheim Innovation Panel. Excludes the top 0.1 percent of contracts by value to remove data-entry outliers.

Source: Authors’ calculations using TED, 2009–2023.

A second cut of the same above-threshold contracts groups them by task complexity rather than by legal type or buyer. Figure [A2](#) decomposes contracts into project/investment versus staple/routine categories for each year 2009–2023. Project and investment contracts account for 69 percent of contracts and 80 percent of contract value over the post-2016 window; staple and routine contracts account for 24 percent of contracts and 16 percent of value. The composition is stable across the post-2016 years, and category shares are unaffected by the April 2016 reporting reform.

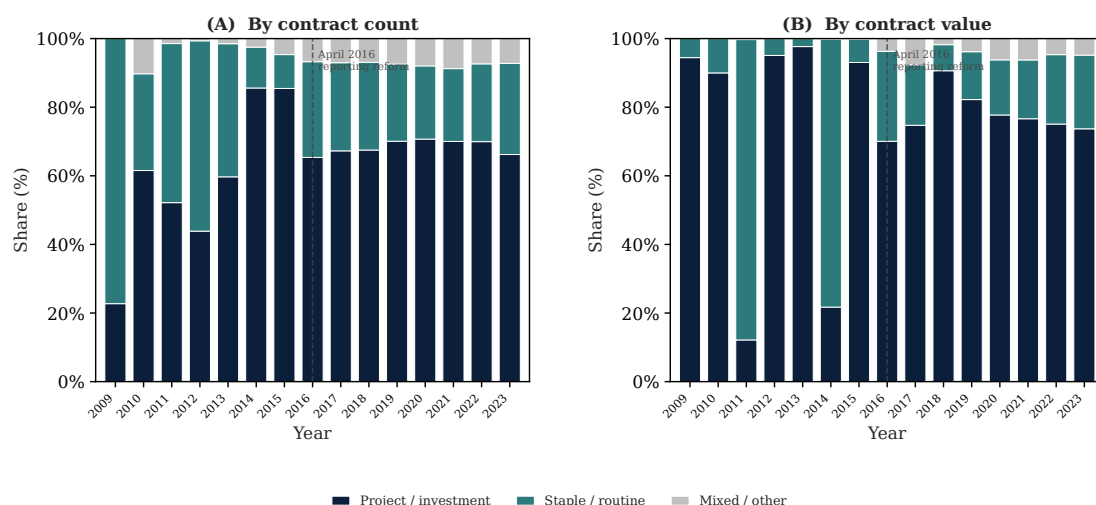


Figure A2: Contract category composition of above-threshold German procurement, 2009–2023
Notes: Share of contracts by count (A) and by value (B), grouped by contract category. Project / investment: construction, engineering, specialized equipment, IT, and R&D services. Staple / routine: food supply, printing, maintenance, financial, and social services. Dashed line: April 2016 EU-mandated reporting reform.
Source: Authors’ calculations using TED, 2009–2023.

These aggregate shares mask substantial dispersion in individual contract size within each sector. Figure A3 reports the distribution of individual contract values by sector: median lot values range from roughly EUR 12,600 in transport to EUR 201,500 in construction.

Contract counts follow a different ranking. Figure A4 shows that the same six sectors range from 17,144 lots in ICT to 142,136 in construction, so the sector with the fewest contracts (ICT) is not the sector with the cheapest ones (transport): a sector’s share of aggregate contract value need not track its share of aggregate contract activity.

A.5 Bidder Eligibility

EU Directive 2014/24/EU governs above-threshold contracts (European Union, 2014); German law implements it through the federal competition act (*GWB*, §§97ff.) (Federal Republic of Germany, 2021) and the procurement regulation (*VgV*) (Federal Republic of Germany, 2016). Any firm established in the EU or European Economic Area may bid, provided it meets the contracting authority’s technical and financial qualification criteria; the law imposes no nationality restriction beyond EU/EEA establishment, though non-EU firms can face added conditions under reciprocity rules. Contracting authorities must reject bidders with final convictions for corruption, fraud, money laundering, or participation in a criminal organization, and may reject bidders facing insolvency, grave professional misconduct, or substantial unpaid taxes or social-security contributions. Contracts below the EU thresholds follow a separate national regulation (*UVgO*) (Federal Ministry for

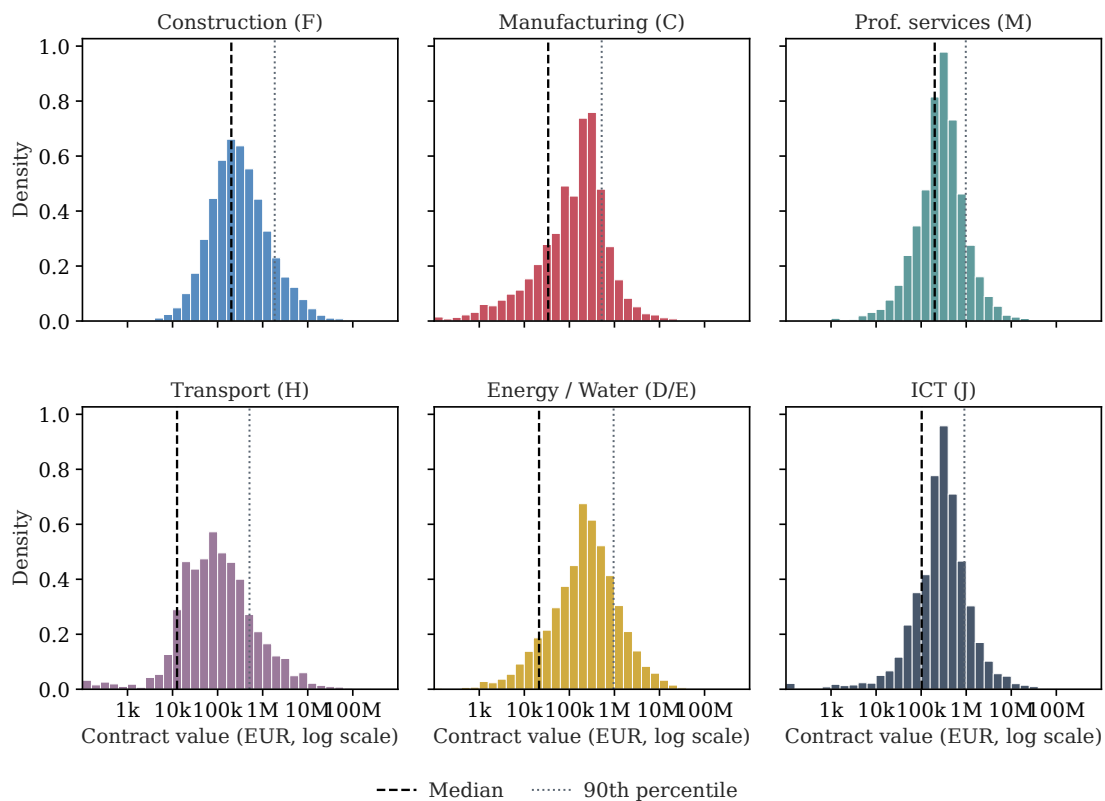


Figure A3: Distribution of contract values by sector, 2009–2023

Notes: Log-scale density of contract value in EUR, one panel per MIP-covered WZ section. Vertical lines mark the median and 90th percentile within each sector.

Source: Authors' calculations using TED, 2009–2023.

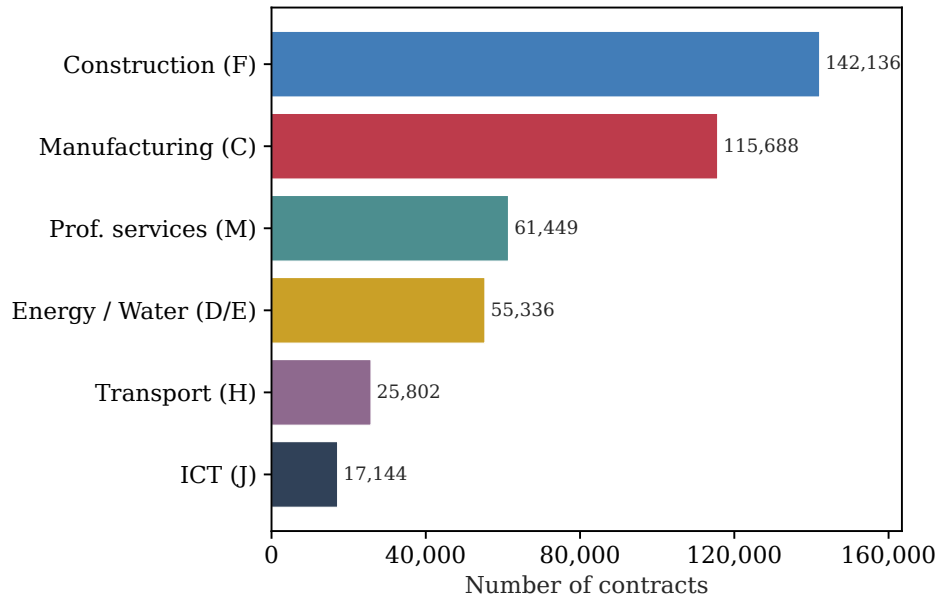


Figure A4: Number of contracts by sector, 2009–2023

Notes: Total count of above-threshold contract awards by sector, same six MIP-covered WZ sections as Figure A3.

Source: Authors' calculations using TED, 2009–2023.

[Economic Affairs and Climate Action, 2017](#)) that gives contracting authorities more discretion over the award procedure and documentation, and that carries no TED reporting obligation, one reason the coverage figures in the next subsection understate true procurement activity (Section A.6). The eligibility rules do not enter the identification strategy; we report them to establish that above-threshold competition is open to essentially the full population of qualified EU/EEA firms, not restricted by nationality or firm size.

A.6 Procurement Data Coverage, Sector Classification, and Reporting Robustness

A.6.1 Contract-Design Variables in the Rebuilt Procurement Panel

Table A4 reports coverage for the contract-design variables recovered from the raw OpenTender files and the normalized OCDS records. The award-rule and specification measures are broadly observed: the MEAT/rated-criteria indicator is available for 94.3 percent of contracts and the prescriptiveness score for 99.7 percent. Process-timing measures based on the advertisement period and the publication-to-award window are also widely observed. By contrast, explicit contract duration is too sparse to use as a main empirical variable: the year/month/day fields cover 1.2 percent of contracts and are essentially absent after 2018, while start/end-date duration covers only 2.7 percent. We therefore use award rules, specification/prescriptiveness, timing, and bid-composition variables in design robustness checks, and treat duration only as a data-coverage caveat.

Table A4: Coverage of recovered contract-design variables

Variable	Coverage	Mean	Coverage 2019–2023
MEAT / rated-criteria	94.3%	0.329	94.5%
Prescriptive score	99.7%	0.253	99.7%
Advertisement period	85.8%	45.504	88.9%
Publication-to-award period	85.8%	143.880	89.0%
Decision period	0.0%	–	0.0%
Signature period	0.0%	–	0.0%
EU-funded flag	1.4%	0.036	0.0%
Funding programme flag	100.0%	0.043	100.0%
Electronic bid count	63.7%	4.282	71.7%
Foreign EU bid count	49.0%	0.150	58.2%
Foreign non-EU bid count	48.6%	0.076	57.9%
Duration from Y/M/D fields	1.2%	2.653	0.0%
Duration from start/end dates	2.7%	1.244	2.3%

Notes: Coverage is the nonmissing share in the raw-rebuilt German award panel, 2009–2023. MEAT/rated-criteria is recovered from the normalized OCDS award-criteria field; timing, funding, bid-composition, and duration diagnostics are recovered from the flat OpenTender files.

A.6.2 TED Reporting Coverage and the 2016 Reform

German procurement agencies publish only a fraction of their contracts on TED. Published contract value amounts to approximately 1.8 percent of GDP, against an EU average of 5.8 percent ([European Commission, 2024](#)), with the gap attributed to fragmented state-level procurement systems and uneven compliance with above-threshold reporting requirements ([European Commission, 2019](#)). Our own sample gives a consistent figure: published contract value in the OpenTender microdata is below 0.05 percent of German nominal GDP before 2015, rises to 1.5–2.7 percent in the post-reform 2017–2019 window, and settles around 1.1–1.2 percent in 2021–2022. In April 2016, Germany tightened its award-notice rules and expanded electronic reporting as part of a broader EU directive transposition; TED contract counts more than doubled in the years that followed, a jump far larger than the moderate rise in public spending recorded in national accounts. Pre-2016 exposure is therefore a lower bound on true sector procurement activity rather than a representative measure.

A.6.3 Robustness to the Coverage Break

Table A5 tests whether the jump in TED contract counts after April 2016 reflects reporting compliance rather than actual procurement growth. Adding sector-specific linear time trends leaves both the market-novelty and unchanged-revenue estimates close to their baseline values. Restricting to 2017–2023, where compliance is more uniform, produces wider confidence intervals rather than

sign reversals; the shorter window reduces identifying variation rather than uncovering a different pattern. The main results are not an artifact of the reporting shift.

Table A5: Robustness to TED Coverage Break

	(A) Post-2016 only		(B) Full + sector trends		(C) Post-2016 + sector trends	
	(1) Mkt. nov.	(2) Unchanged rev.	(3) Mkt. nov.	(4) Unchanged rev.	(5) Mkt. nov.	(6) Unchanged rev.
arcsinh(proc. lots, MIP sector)	0.037 (0.207)	0.313 (0.340)	-0.299* (0.144)	0.366** (0.153)	-0.369 (0.989)	1.114 (0.993)
Observations	27,772	8,118	66,980	25,962	27,772	8,118
Clusters	21	21	21	21	21	21
Sample	2017–2023	2017–2023	2009–2023	2009–2023	2017–2023	2017–2023
Sector linear trends	No	No	Yes	Yes	Yes	Yes

Firm and year FEs throughout. Specifications (B)–(C) additionally absorb sector-specific linear time trends (sector $\times$ year). SE clustered by MIP sector (21 clusters). Outcome (1): market novelty (binary $\times 100 =$ pp). Outcome (2): share of turnover from unchanged products (percent). Post-2016 window = 2017–2023, the period of reliable TED reporting compliance. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

A.6.4 Validating the CPV-to-WZ Sector Crosswalk

The treatment variable assigns each procurement contract to an industry sector by mapping its EU product code to the corresponding German industry sector. A natural validation asks whether firms that win contracts in a given product category actually operate in the industry sector to which the mapping assigns them. We assess this by linking each contract winner in the TED records to the firm’s primary industry classification in ORBIS, using the matching described in Appendix B. Because German and European industry classification codes share the same letter-based sector scheme, the comparison is direct.

The overall count-weighted agreement rate is 54.4 percent and the value-weighted rate is 66.0 percent. Agreement is highest for product categories that map closely to a single industry: petroleum and gas services (87.8%), transport services (86.1%), and architectural and engineering services (84.3%). Agreement is lowest for product categories whose contracts attract buyers and winners from many industries: defense and public-administration services, IT office equipment (won by hardware resellers whose primary industry is often wholesale trade rather than manufacturing), and landscape and environmental services. These low-agreement categories are precisely those for which the sector assignment is most approximate; the treatment-robustness table excludes them as a check (Table C2).

Within the post-2016 window the agreement rate is stable at 53–57 percent per year, with higher agreement by contract value (60–83 percent) because large contracts tend to go to firms in the sector

the contract was written for. Table A6 summarizes the agreement rates, and Figure A5 shows the year-by-year pattern.

Table A6: CPV-to-WZ Crosswalk Validation

	Agreement rate	Observations
<i>Panel A: Overall</i>		
Count-weighted	54.4%	146,250
Value-weighted	66.8%	133,073
<i>Panel B: By supply type</i>		
Works	61.0%	49,384
Services	52.1%	49,380
Supplies	49.7%	46,788
<i>Panel C: Pre- vs. post-2016 reform</i>		
2009–2015	59.4%	633
2016–2023	54.4%	145,617
Difference ($p = 0.011$)	-5.0%	

Notes: Agreement is the share of contracts where the CPV-mapped WZ section of the contract matches the NACE section of the winning firm as recorded in ORBIS. The sample includes contracts with at least one winning bidder matched to ORBIS. WZ section codes correspond directly to NACE Rev. 2 section letters. Pre-2016 agreement is lower because compliance with TED electronic reporting was incomplete; the post-2016 subsample has both higher coverage and higher within-sector agreement. Sources: TED award notices and ORBIS financial data.

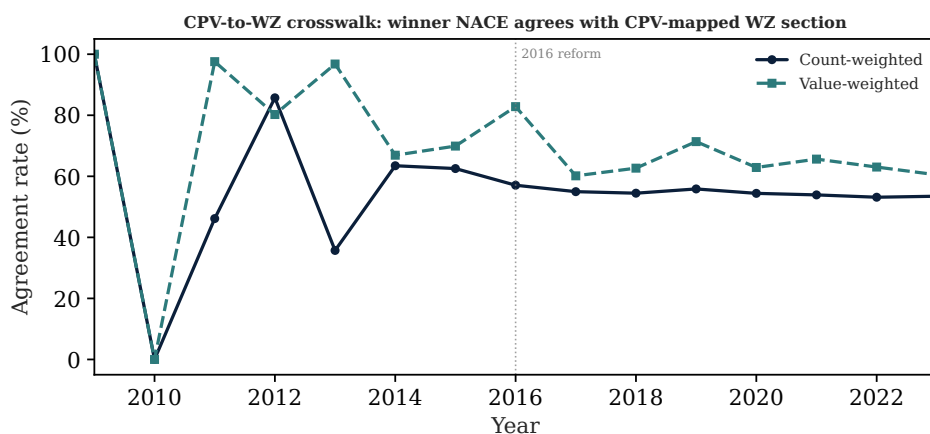


Figure A5: Sector assignment accuracy by year

Notes: Agreement is the share of matched contract rows where the industry sector assigned by the EU product-code mapping matches the primary industry classification of the winning firm as recorded in ORBIS. Count-weighted (solid) and value-weighted (dashed) rates are shown. The vertical dotted line marks April 2016 (EU Procurement Directive transposition). Pre-2016 observations are fewer than 500 rows due to low TED reporting compliance before the reform; the stable post-2016 series is the informative window. Sample: TED 2009–2023, ORBIS-matched winning bidders. *Source:* Authors' calculations using TED and ORBIS.

B ORBIS Validation

The MIP results describe the innovation-composition margin. ORBIS provides evidence on the scale margin among matched procurement winners. Figure B1 describes the profile of winners; the regression tables that follow show that procurement exposure is associated with higher employment and turnover, while profitability effects remain small. The full-sample estimates appear in Table 3 in the main text.

B.1 Winner Profile and Scale Dynamics

Figure B1 describes the size distribution and sector composition of procurement winners in the matched ORBIS panel. Winners skew toward small and medium-sized enterprises: 46 percent have 10–49 employees and 16 percent are micro firms; large firms (250 or more employees) account for only 10 percent of winners. Construction and manufacturing together account for 44 percent of contracts won; professional services, wholesale trade, and administrative-support sectors account for a further 35 percent.

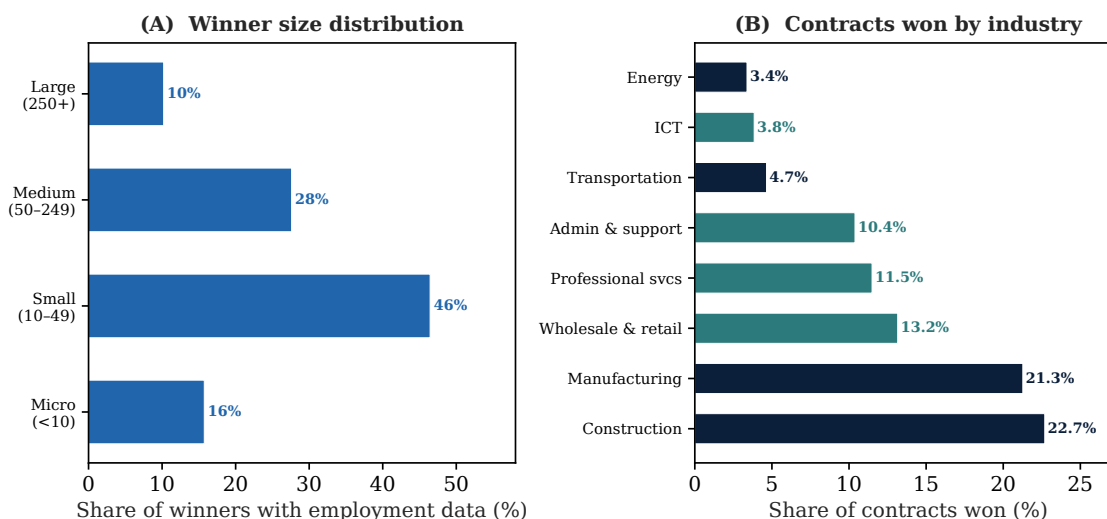


Figure B1: Size distribution and sector composition of procurement winners

Notes: Panel A reports the size distribution of winning firms by employment class among firms with employment data at first win; the median winner has 33 employees. Panel B reports the share of contracts won by the eight largest industry sectors (accounting for 91 percent of all contracts); navy bars indicate production-intensive sectors (construction, manufacturing, energy, transportation), teal bars indicate knowledge and service sectors. Sample: ORBIS-matched winner panel, TED 2009–2023.

Source: Authors' calculations using TED matched to ORBIS.

Figure B2 reports the share of turnover accounted for by awarded contracts: total contract value divided by same-year turnover. Dependency falls sharply with firm size: the median micro firm (fewer than 10 employees) derives 6.1 percent of turnover from public contracts in a win year,

against 1.3 percent for small firms, and effectively zero for medium and large firms. Consistent with Figure B1, public contracts are economically significant primarily for the smallest winners.

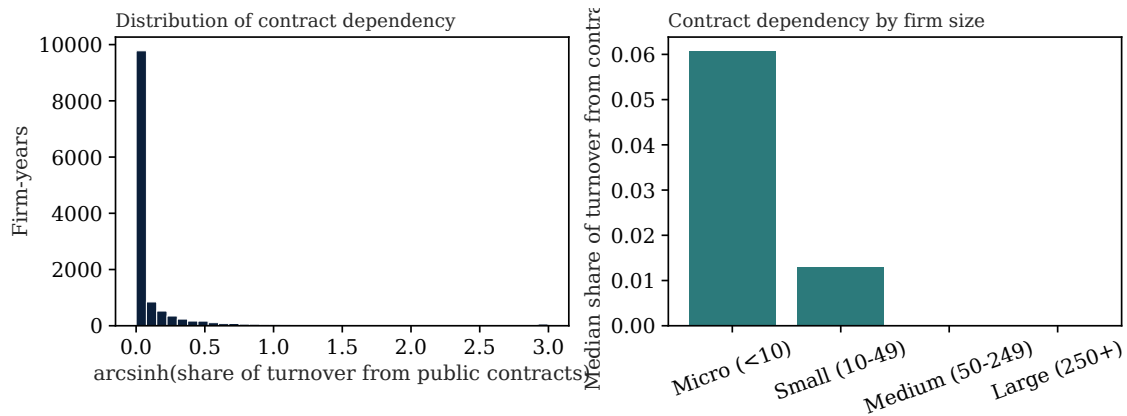


Figure B2: Contract value as a share of winner turnover

Notes: Panel A: arcsinh-transformed distribution of the share of turnover from awarded public contracts, across firm-years with matched turnover data, capped at a ratio of 10. Panel B: median share of turnover from public contracts, by employment class at first win. Sample: ORBIS-matched winner panel with non-missing turnover, TED 2009–2023. *Source:* Authors’ calculations using TED matched to ORBIS.

Additional ORBIS heterogeneity splits by winner age, size, ownership, and repeat-contractor status are available in the replication files. We do not report them here because they do not change the core validation result: matched winners expand in scale, while profitability effects are small and imprecise.

B.2 A Staggered Event Study with a Not-Yet-Treated Control Group

The winner-panel results above and the descriptive event study in the main text (Figure 4) share a limitation: every firm in the ORBIS-matched panel is, by construction, a procurement winner, since firms enter the match precisely because they won a contract. There is no never-treated comparison group, so the main-text figure plots each firm’s own deviation from its own year before first win, averaged across firms; it is descriptive of winners’ trajectories, not a causal comparison.

This appendix reports a version with a genuine control group: firms that win their first contract *later* serve as controls for firms that win earlier, following the not-yet-treated logic of Callaway and Sant’Anna (2021) and de Chaisemartin and D’Haultfœuille (2020).¹⁴ For each cohort of firms winning their first contract in year c and each event time k , we run a clean two-period difference-in-differences between $t = -1$ and $t = c + k$, using as controls only firms whose own first win

¹⁴A stacked, clean-control version following Cengiz et al. (2019), requiring each control firm to remain untreated throughout the full event window, gives qualitatively the same conclusion but with less power: requiring seven consecutive years of ORBIS coverage per firm shrinks the usable sample sharply, since ORBIS coverage of German firms expanded over calendar time and firms newly entering coverage mid-window have no true pre-period baseline.

has not yet occurred as of year $c + k$, the maximal valid not-yet-treated comparison group for that specific comparison, which can differ across k . Cohort-level estimates are then combined across cohorts for each k by inverse-variance weighting. Between six and ten cohorts (of the ten first-win years 2014–2023 with enough treated firms; earlier cohorts exist but are too small, a consequence of the pre-2016 TED reporting gap documented in Appendix A.6) contribute a valid estimate at each k , depending on how many cohorts have enough not-yet-treated firms remaining at that specific horizon.

Table B1 and Figure B3 report the result across all six outcomes. No outcome shows a statistically significant change at any *post*-treatment event time ($k = 0, +1, +2, +3$): point estimates are small relative to their standard errors throughout, in contrast to the clear positive pattern in the main-text descriptive figure for employment and turnover. What is significant is concentrated entirely in the *pre*-period: turnover at $k = -2$, return on capital at both $k = -3$ and $k = -2$, and the any-patent indicator at $k = -2$ all show significant differences between eventual winners and their not-yet-treated peers before any contract is won. This is the same pattern documented for the MIP sector-level analysis in Appendix C.3: evidence of selection into procurement timing that predates treatment, not a designed experiment. Firms are not randomly assigned an award year; whatever makes a firm win in 2016 rather than 2019 is correlated with its turnover, profitability, and patenting trajectory beforehand.

The turnover pattern specifically is consistent with reversion to the mean rather than a treatment effect: winners' turnover is 2.0 percentage points lower than not-yet-treated peers two periods before the award ($k = -2$, significant at the 10 percent level) and 3.4 percentage points higher two periods after ($k = +2$, not significant), a pre-award dip and a same-order post-award recovery that a temporary downturn followed by mechanical rebound would produce whether or not the contract caused any of it.¹⁵

¹⁵Splitting eventual winners by their own pre-award trajectory supports this reading directionally: among the 795 firms whose own turnover fell in the year before winning, post-award point estimates against the same not-yet-treated controls are positive and rising ($k = +2$: 1.0; $k = +3$: 4.4; log points $\times 100$, neither significant); among firms without a pre-award dip, post-award estimates show no such pattern (one point, $k = +1$, is significantly negative). The split is underpowered on its own (both subsamples are smaller than the pooled cohort), but the direction is what reversion, not a broad treatment effect, would predict.

Table B1: Callaway-Sant'Anna-style event study: selected outcomes

	Employment (log $\times 100$)	Turnover (log $\times 100$)	Return on capital (pp)	arcsinh(patent applications)	Any patent (pp)
<i>Pre-period</i>					
$k = -3$	-0.610 (0.666)	0.194 (1.339)	-0.649* (0.356)	-0.000 (0.003)	-0.076 (0.166)
$k = -2$	0.099 (0.477)	-2.035* (1.065)	-0.580** (0.292)	0.005 (0.003)	0.315** (0.148)
<i>Post-period</i>					
$k = +0$	0.207 (0.433)	0.951 (1.011)	-0.498 (0.339)	0.001 (0.003)	-0.006 (0.149)
$k = +1$	-0.198 (0.644)	1.220 (1.549)	-0.091 (0.343)	-0.004 (0.004)	-0.114 (0.188)
$k = +2$	-0.260 (0.949)	3.354 (2.078)	-0.612 (0.426)	-0.004 (0.005)	0.035 (0.219)
$k = +3$	-1.051 (1.374)	3.140 (2.405)	-0.218 (0.523)	0.000 (0.006)	0.098 (0.275)

Notes: Callaway-Sant'Anna-style not-yet-treated event study. Each cohort-event-time cell is a clean 2-period DiD (firm and year FE, clustered by firm) using firms not yet treated as of that specific calendar year as controls; cohort estimates combined via inverse-variance weighting. Standard errors in parentheses. $k = -1$ is the omitted reference period. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Authors' calculations using TED matched to ORBIS and PATSTAT.

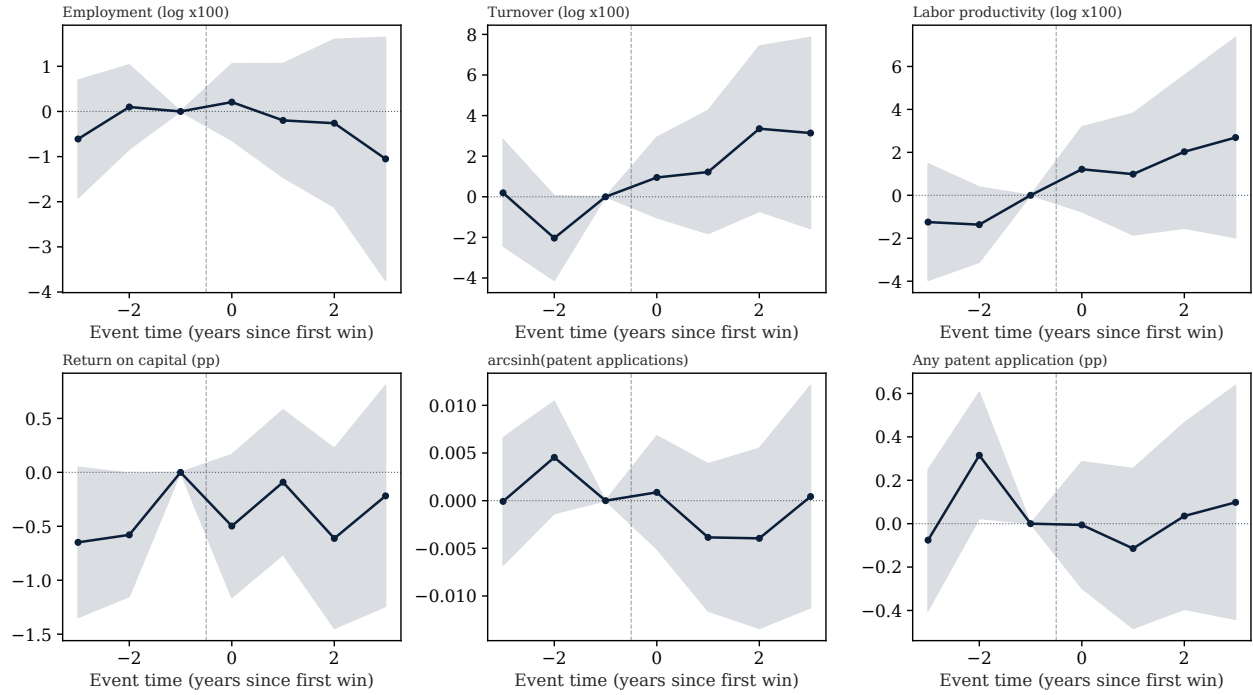


Figure B3: Not-yet-treated event study: ORBIS winners vs. firms that have not yet won

Notes: Each event-time coefficient combines cohort-specific two-period DiD estimates (firm and year FE, clustered by firm) via inverse-variance weighting; the control group for each cohort-event-time cell is firms not yet treated as of that specific calendar year. Shaded bands are 95% confidence intervals. $k = -1$ is the omitted reference period.

Source: Authors' calculations using TED matched to ORBIS and PATSTAT.

C Mechanism Robustness

This appendix presents four supplementary analyses. The first compares procurement exposure to public R&D subsidies, showing that the composition pattern is specific to procurement rather than common to government funding broadly. The second rules out a denominator artifact and shows the pattern is robust to alternative exposure definitions and common-sample restrictions. The third checks pre-treatment parallel trends with a dose-response event study. The fourth grounds the specification mechanism with individual contract records drawn from the TED microdata.

C.1 Procurement versus Public R&D Subsidies

Table C1 compares two public-funding instruments on three composition outcomes: market novelty (binary, pp), market-novel share (fraction of new-product revenue that is new to the market), and existing-product revenue share. Procurement exposure is associated with higher existing-product revenue and lower market novelty, consistent with a within-firm composition shift. Public R&D subsidies show the opposite pattern: they are associated with higher market novelty and lower existing-product revenue, consistent with an overall innovation stimulus. The two instruments are associated with opposite patterns of innovation composition.

Table C1: Procurement vs. Public Subsidies: Two Channels

	(1) Market novelty (pp)	(2) Market-novel share (% of new products)	(3) Existing-product revenue (pp)
<i>Panel A. Procurement exposure, 2009–2023</i>			
<i>arcsinh</i> (contracts)	−0.199* (0.101)	−0.099 (0.198)	0.379** (0.137)
<i>N</i>	66,980	15,830	25,962
<i>Panel B. Public R&D subsidy (subsidy-wave years)</i>			
Public R&D subsidy	5.133*** (1.258)	1.152 (1.801)	−3.819*** (1.103)
<i>N</i>	26,936	5,396	14,112
Firm & year FE	Yes	Yes	Yes
Industries	21	21	21

Notes: Each channel estimated separately on its available sample; firm and year fixed effects throughout; standard errors clustered by industry (21 industries). The procurement rows use the full 2009–2023 panel (same specification as Table 1); the subsidy rows restrict to waves in which is asked. Market novelty: binary $\times 100$ (pp); market-novel share: share of new-product revenue that is new to the market (%); existing-product revenue: share of total turnover from unchanged products (pp).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Sample and Exposure Robustness

Ruling out a denominator artifact. One concern is that procurement exposure coincides with higher total sales, which would mechanically raise the existing-product revenue share even if the level of new-product activity were unchanged. Two additional revenue-share outcomes speak against this interpretation. The share of sales from market-novel products falls with exposure, and the share from any newly introduced product also falls directionally. If the pattern were a pure denominator artifact, these new-product shares would remain flat; the fact that all three revenue-mix margins move in the same direction is consistent with a genuine reallocation.

Alternative exposure definitions. Table C2 checks whether the results change when procurement exposure is measured differently. Replacing contract counts with total awarded value in euros produces smaller point estimates for both outcomes, and neither is statistically significant, consistent with contract value being measured with more error in TED records than counts. Scaling exposure by the number of MIP-surveyed firms in each sector-year (normalizing contract counts by how many firms could plausibly compete for them in that sector-year) makes the market-novelty point estimate more negative but imprecise, no longer significant on its own; the unchanged-revenue estimate remains significant and larger than the baseline. Restricting to project and investment contract categories, which account for 69 percent of contracts and 80 percent of contract value post-2016, is the only alternative scaling where the pattern strengthens across every outcome: market novelty is more negative (-0.335 versus -0.199 at baseline), unchanged-product revenue more positive (0.456 versus 0.379), cost-reduction revenue more negative (-0.060 versus -0.047), and process innovation turns significantly positive (0.275 , statistically indistinguishable from zero at baseline). Only the market-novel share of new-product revenue stays insignificant, as at baseline. The consistency across outcomes suggests the baseline understates the association in the most economically meaningful contract types.

Table C2: Robustness: Alternative Treatment Variable Scalings

	(1) arcsinh(contracts) [baseline]					(2) arcsinh(€ value)					(3) arcsinh(contracts / MIP firms)					(4) arcsinh(investment contracts)				
	(1) Mkt. nov.	(2) Unchanged rev.	(3) Mkt-nov. shr.	(4) Proc. inn.	(5) Cost-red.	(6) Mkt. nov.	(7) Unchanged rev.	(8) Mkt-nov. shr.	(9) Proc. inn.	(10) Cost-red.	(11) Mkt. nov.	(12) Unchanged rev.	(13) Mkt-nov. shr.	(14) Proc. inn.	(15) Cost-red.	(16) Mkt. nov.	(17) Unchanged rev.	(18) Mkt-nov. shr.	(19) Proc. inn.	(20) Cost-red.
arcsinh(contracts)	-0.199*	0.379**	-0.099	0.101	-0.047*															
	(0.101)	(0.137)	(0.198)	(0.169)	(0.024)															
arcsinh(€ value)						-0.039	0.055	0.003	0.054	-0.012										
						(0.036)	(0.041)	(0.050)	(0.037)	(0.008)										
arcsinh(contracts / MIP firms)											-0.533	0.849**	-0.295	0.251	-0.109					
											(0.311)	(0.387)	(0.401)	(0.340)	(0.075)					
arcsinh(investment contracts)																-0.335***	0.456**	-0.099	0.275**	-0.060*
																(0.105)	(0.173)	(0.171)	(0.109)	(0.031)
Obs.	66,980	25,962	15,830	70,462	40,635	66,980	25,962	15,830	70,462	40,635	66,980	25,962	15,830	70,462	40,635	66,980	25,962	15,830	70,462	40,635
Clusters	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21	21

Firm and year FEs throughout. SE clustered by MIP sector (21 clusters). Binary outcomes $\times 100 =$ pp. Treatment (1): arcsinh of sector-year contract count (baseline spec). Treatment (2): arcsinh of total contract value in euros. Treatment (3): arcsinh of contracts divided by MIP-surveyed firms in sector-year. Treatment (4): arcsinh of contracts in project and investment categories only (construction, engineering, equipment, IT/R&D CPV divisions), restricted within each MIP sector's own CPV concordance. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Wild-cluster bootstrap inference. Table C3 reports wild-cluster bootstrap p-values for the three main-text outcomes. The bootstrap p-values are close to the conventional cluster-robust p-values, and if anything slightly stronger: market novelty remains marginal at the 10 percent level, unchanged-product revenue remains significant at the 1 percent level, and cost-reduction revenue is significant under the bootstrap ($p = 0.018$) where it was only marginal under cluster-robust inference ($p = 0.063$). The result is therefore not an artifact of relying only on asymptotic cluster inference with 21 MIP-sector clusters.

Table C3: Wild Bootstrap Robustness

Outcome	β	SE (cluster)	p (cluster)	p (wild bootstrap)
Mkt. novelty (2009–2023)	-0.199*	(0.101)	0.063	0.059
Unchanged-prod. rev. (%) (2009–2023)	0.379**	(0.137)	0.012	0.007
Cost-red. rev. (%) (2009–2023)	-0.047*	(0.024)	0.063	0.018

Notes: Firm and year FEs. SE clustered by MIP sector (21 clusters). Wild bootstrap uses boottest (Webb 2023) with 9,999 Monte Carlo Rademacher draws; full enumeration (2^{21} patterns) is infeasible for $G = 21$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (cluster SE).

Source: Authors' calculations using the Mannheim Innovation Panel and OpenTender DE.

Additional subsamples. Table C4 repeats the 21-sector specification for the three main-text outcomes in two narrower cuts of the data: manufacturing firms only, and procurement exposure restricted to supplies and services contracts. In both subsamples, procurement exposure is associated with lower market novelty, higher unchanged-product revenue, and lower cost-reduction revenue. These cuts are not new identification designs, but they show that the baseline composition pattern is not coming only from non-manufacturing sectors or public works contracts.

Table C4: Manufacturing and Supplies-and-Services Subsamples

	(1) Manufacturing	(2) Supplies & services
Market novelty (pp)	−0.439*** (0.094)	−0.211** (0.099)
Unchanged-product revenue (%)	0.478* (0.238)	0.393*** (0.137)
Cost-reduction revenue (%)	−0.090*** (0.025)	−0.046* (0.023)
<i>Observations by outcome</i>		
Market novelty	29,851	66,980
Unchanged-product revenue	12,682	25,962
Cost-reduction revenue	17,395	40,635
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes

Notes: We use the three main-text MIP outcomes from Table 1. Column (1) restricts the sample to the ten manufacturing MIP sectors and uses the baseline 21-sector procurement exposure. Column (2) uses the full 21-sector sample but restricts procurement exposure to supplies and services contracts only, excluding works entirely. Standard errors are clustered by industry. Binary outcomes are multiplied by 100. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Authors' calculations using the Mannheim Innovation Panel and TED.

Restricting to consistently reporting buyers. A different reporting-driven concern is that new buyers enter TED disclosure over time, so part of the exposure trend could reflect the extensive margin of buyer coverage rather than existing buyers procuring more. Table C5 restricts the 2017–2023 window to 1,013 buyer-postcode entities that report in every year from 2017 through 2022, accounting for 40.0 percent of lots over the period, and compares this stable-buyer subsample to the full set of reported lots over the same window. Neither restriction is significant for market novelty or unchanged-product revenue: the point estimates are close to zero and change sign across the six specifications (−0.566 to 0.233 for market novelty; −0.298 to 0.313 for unchanged revenue). The cost-reduction-revenue margin is the exception, staying negative in every specification and significant in four of six.

This lack of precision is not, by itself, evidence against the main result. The full 2009–2023 baseline sample gives −0.199* for market novelty and 0.379** for unchanged revenue (Table 1); the least restrictive row in Table C5, all reported lots over 2017–2023 with no buyer restriction at all, already loses that significance (0.037 and 0.313, both insignificant). Narrowing the sample window from sixteen years to seven costs power well before the stable-buyer restriction is even applied, consistent

with the shorter window’s reduced power documented in Appendix A.6. The stable-buyer restriction does not by itself indicate that buyer entry drives the baseline result; the two explanations remain observationally similar in this subsample.

Table C5: Exposure restricted to consistently reporting buyers

	Market novelty (pp)	Unchanged-product revenue (%)	Cost-reduction revenue (%)
All reported lots	0.037 (0.207)	0.313 (0.340)	−0.146 (0.094)
Stable-buyer lots	0.028 (0.269)	−0.193 (0.491)	−0.322*** (0.107)
Stable-buyer contract value	0.130 (0.117)	0.097 (0.293)	−0.051 (0.041)
Number of stable buyers	0.233 (0.407)	−0.240 (0.685)	−0.377** (0.150)
Lots per stable buyer	−0.566 (0.531)	−0.298 (0.904)	−0.493** (0.231)
Predetermined buyer–CPV–cell lots	0.089 (0.245)	0.018 (0.389)	−0.243** (0.094)
Observations	27,772	8,118	13,268

Notes: Each cell is a separate within-firm regression over 2017–2023 with firm and year fixed effects. Stable buyers are buyer-name–postcode entities observed in every complete year from 2017 through 2022: 1,013 buyers representing 40.0 percent of lots over 2017–2023. Predetermined buyer–CPV links are fixed using 2017–2019 activity. All treatments are inverse-hyperbolic-sine transformations. Standard errors clustered by MIP sector are in parentheses. This restriction removes variation generated by buyer entry into TED but does not address endogenous procurement demand. Adding sector-specific linear trends leaves the pattern unchanged (available on request): cost-reduction revenue remains the only outcome that stays significant across specifications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Authors’ calculations using the Mannheim Innovation Panel and TED.

Award-rule design: MEAT versus price-only. The rebuilt procurement panel also separates contracts whose award rule uses rated criteria (MEAT/scoring-rule procurement) from contracts awarded on price only. Table C6 re-estimates the three main MIP outcomes with separate sector-year exposure measures for MEAT/rated-criteria and price-only contracts. The split does not support a simple interpretation in which scoring-rule procurement mechanically stimulates frontier innovation: MEAT exposure is not precisely estimated, while price-only exposure is associated with higher unchanged-product revenue and lower cost-reduction revenue at the 10 percent level. The main takeaway is therefore more conservative: the baseline composition pattern is robust to reconstructing the procurement database and can be decomposed by award-rule design, but the design split is descriptive rather than a standalone causal mechanism.

Table C6: Procurement exposure by award-rule design

	(1) Market novelty	(2) Unchanged revenue	(3) Cost reduction
arcsinh(MEAT/rated-criteria lots)	-0.153 (0.263)	-0.230 (0.326)	0.079 (0.060)
arcsinh(price-only lots)	-0.079 (0.260)	0.656* (0.339)	-0.125* (0.064)
Obs.	66,980	25,962	40,635
Clusters	21	21	21

Notes: Within-firm OLS regressions with firm and year fixed effects. Standard errors are clustered by MIP sector. MEAT/rated-criteria and price-only exposures are inverse hyperbolic sine transformations of sector-year contract counts in each award-rule category, constructed from the raw-rebuilt procurement panel. Binary market novelty is multiplied by 100.

Contractor persistence. A related question is whether the composition shift is stronger where the same suppliers keep winning. For each sector-year, using TED records alone, we compute the share of contract awards going to a bidder who already won a contract in that industry before; the measure is a market-level aggregate, not a firm-level link to MIP. Table C7 interacts this share with procurement exposure: the interaction is negative and significant for market novelty and for cost-reduction revenue, positive but not significant for unchanged-product revenue. Markets where the same suppliers keep winning show a larger novelty decline, consistent with a market converging toward delivering what buyers already ask for rather than competing on new designs.

Table C7: Heterogeneity by contractor persistence

	Market novelty (pp)	Unchanged revenue (%)	Cost-reduction revenue (%)
arcsinh(sector contracts)	-0.167 (0.192)	-0.273 (0.519)	-0.107 (0.085)
High repeat-winner share (=1)	5.684*** (1.880)	-3.006 (3.677)	1.302*** (0.451)
arcsinh(sector contracts) × High repeat-winner share	-0.808*** (0.267)	0.399 (0.446)	-0.184*** (0.058)
Obs.	56,884	21,735	33,456
Clusters	19	19	19
Firm and year FE	Yes / Yes	Yes / Yes	Yes / Yes

Notes: Firm and year fixed effects; standard errors clustered by MIP sector in parentheses. Repeat-winner share is the sector-year share of contract awards going to a bidder that already won at least one contract in the same industry in an earlier year; the indicator equals one above the sample median. Restricted to sector-years with positive procurement activity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: Authors' calculations using TED, 2009–2023.

C.3 Pre-Trends and Identification

The baseline specification’s identifying assumption is that, absent procurement exposure, sectors would have followed parallel trends in market novelty. We check that assumption with a pre-period event study below; Appendix A.6 reports a complementary sector-specific-trend robustness check.

Do sectors already differ before treatment? Table C8 reports a dose-response event study using each sector’s average exposure over 2009–2011 as a fixed dose, interacted with year indicators, relative to a 2007 baseline. The regression sample is firm-year, but the dose varies only across 23 MIP sectors, so the identifying power behind each event-study coefficient is bounded by a 23-cluster design, not by the firm count. Judged against that benchmark, the genuinely pre-treatment years show no deviation: 2003 ($p = 0.715$), 2004 ($p = 0.579$), and 2005 ($p = 0.607$) are all statistically indistinguishable from the 2007 reference. The joint pre-period F-test is significant only because of 2006 (-20.57 pp, $p = 0.012$), a year in which TED above-threshold reporting had already begun to phase in; 2006 is therefore a boundary year, not a clean pre-treatment placebo. We do not read the 2006 coefficient as evidence that sectors with higher subsequent exposure already had diverging market novelty before the analysis window begins: the three unambiguously pre-treatment years show no such pattern, and a single sector-year cell can move a 23-cluster joint test.

C.4 Illustrative Contract Records

The heterogeneity splits in Section 2.4 are descriptive regressions; this subsection grounds the specification mechanism with individual contract records drawn directly from the TED microdata. Table C9 summarizes three records side by side before the prose discussion below.

Table C8: Pre-trends Test: Event Study around Procurement Exposure Onset

Year	$\hat{\beta}$ (pp)	SE	p-value
<i>Pre-treatment period (relative to 2007)</i>			
2003	−4.603	(12.450)	0.715
2004	−6.929	(12.305)	0.579
2005	−6.315	(12.091)	0.607
2006	−20.571**	(7.513)	0.012
2007	0.000	—	—
<i>Joint F-test: $H_0: \beta_{2003} = \dots = \beta_{2006} = 0$</i>			
<i>F(4, 22)</i>	3.084**	(<i>p</i> = 0.037)	
<i>Post-reference (selected years)</i>			
2008 (<i>crisis</i>)	−17.719***	(4.991)	0.002
2009	7.490	(6.676)	0.274
2012	7.039	(6.840)	0.315
2015	6.374	(8.047)	0.437
2018	14.787*	(7.211)	0.052
2021	26.776**	(10.169)	0.015
<i>N</i>	91,516 firm-year obs		
Sector clusters	23		

Notes: Dose variable S_s = sector s 's average $\text{arcsinh}(\text{contracts}_{s,t})$ over 2009–2011 (early-treatment intensity at MIP-sector level), normalized to sum to one. Outcome: market novelty $\times 100$ (pp). Firm and year fixed effects. Standard errors clustered by MIP sector (23 clusters). Reference year: 2007 (last pre-OpenTender year; avoids the 2008 financial-crisis trough, which disproportionately depressed innovation in high-procurement sectors). The pre-period F-test covers 2003–2006; 2006 overlaps with TED above-threshold procurement, so the truly pre-treatment window is 2003–2005. Post-reference years 2009–2023 correspond to the OpenTender sample period. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C9: Illustrative Contract Records

Buyer	Year	Value (EUR)	Requirement	Bidders	Reading
Paul-Ehrlich-Institut (federal vaccine regulator)	2020	539,236	Cryo-CLEM and STED super-resolution microscope retrofit; named a patented imaging technology sold by a single manufacturer	1	Requirement rules out every other supplier by design; won by Leica
Julius Kühn-Institut, agricultural research (award administered by the Bundesanstalt für Immobilienaufgaben)	2020	1,065,183	New laboratory building; fire-safety and utilities engineering restricted to licensed specialists	1	Narrow licensing requirement; won by Planungsgruppe M+M
Bundesamt für Sicherheit in der Informationstechnik (federal cybersecurity agency)	2021	216,056	AI system for industrial-control-network security; 5,000-word technical requirement	31	Detailed but functional: universities, security firms, and research labs all qualified; won by Fraunhofer

Notes: Individual TED award records, verified directly against the underlying microdata (OpenTender DE). All three buyers are federal-level agencies. The first two records name a specific technology or a licensed credential that only one supplier meets; the third writes a long but functional requirement that leaves room for a wide field. Requirement length alone does not compress competition: what matters is whether the requirement names something only one supplier can provide.

Source: TED award notices, 2020–2021.

All three records are federal-level awards, consistent with Panel B's finding that the composition association concentrates in central-government procurement. Two fit the specification-compression story directly. The Paul-Ehrlich-Institut, Germany's federal vaccine and biomedicines regulator, awarded a 2020 contract for a Cryo-CLEM and STED super-resolution microscope retrofit at EUR 539,236; the requirement named a patented imaging technology sold by a single manufacturer, and only one bidder, Leica, qualified despite the procedure being open to any EU-established firm. A 2020 laboratory building for the Julius Kühn-Institut, Germany's federal research centre for cultivated plants, required licensed fire-safety and utilities engineers at EUR 1,065,183; again only one bidder, Planungsgruppe M+M, met the licensing requirement.

The third record shows the story does not simply reduce to specification length. The Bundesamt für Sicherheit in der Informationstechnik, Germany's federal cybersecurity agency, wrote a 5,000-word technical requirement for an AI system to secure industrial control networks at EUR 216,056, yet drew 31 bidders spanning universities, security firms, and research labs, and Fraunhofer won. The requirement was long but functional: it specified performance objectives rather than naming a proprietary technology or credential, leaving room for a wide field of qualified suppliers. What compresses competition is not how much a specification says, but whether it names something only one supplier can provide.

D Technical Derivations for the Klette–Kortum Model

The appendix collects the technical material behind the model: the incumbent fixed-point reduction, the numerical solution, the stationary distribution, and several extensions. It also clarifies the entry and creative-destruction accounting used in the quantitative block.

D.1 Fixed-Point Reduction

The incumbent block is governed by

$$(r + \tau)v_I = \pi_I(G) - \Phi(x) - \Psi(\mu) + x\bar{v}(\mu; G), \quad (36)$$

$$(r + \tau)v_F = \pi_F(G) - \Phi(x) - \Psi(\mu) + x\bar{v}(\mu; G), \quad (37)$$

where $\bar{v}(\mu; G) \equiv (1 - \mu)v_I + \mu v_F$ and $r + \tau - x > 0$ ensures finite values. The value gap $v_F - v_I = [\bar{\pi}_F - \bar{\pi}_I(1 + G)]/(r + \tau)$ follows from subtracting these two equations; its derivation appears in the main text. Defining

$$v_F - v_I = \frac{\bar{\pi}_F - \bar{\pi}_I(1 + G)}{r + \tau}, \quad (38)$$

premultiplying (36) by $(1 - \mu)$ and (37) by μ , then adding, gives the average line value:

$$\bar{v}(\mu; G) = \frac{\bar{\pi}(\mu; G) - \Phi(x) - \Psi(\mu)}{r + \tau - x}, \quad (39)$$

where

$$\bar{\pi}(\mu; G) \equiv (1 - \mu)\pi_I(G) + \mu\pi_F(G).$$

The effort and direction first-order conditions are

$$c_x \zeta x^{\zeta-1} w = \bar{v}(\mu; G), \quad (40)$$

$$\kappa \mu^{\eta-1} = x(v_F - v_I). \quad (41)$$

Using (38), the interior direction condition becomes

$$\mu^{\text{int}} = \left[\frac{x}{\kappa} \frac{\bar{\pi}_F - \bar{\pi}_I(1 + G)}{r + \tau} \right]^{1/(\eta-1)}. \quad (42)$$

Because $\mu \in [0, 1]$, the constrained policy is

$$\mu^* = \min \left\{ 1, \max \left\{ 0, \mu^{\text{int}} \right\} \right\}. \quad (43)$$

Equivalently, for $D(\mu) = x(v_F - v_I) - \kappa\mu^{\eta-1}$, the KKT conditions are $D(\mu^*) = 0$ when $0 < \mu^* < 1$, $D(0) \leq 0$ when $\mu^* = 0$, and $D(1) \geq 0$ when $\mu^* = 1$. For $\eta > 1$, the two boundary conditions are $x(v_F - v_I) \leq 0$ and $x(v_F - v_I) \geq \kappa$, respectively. Counterfactuals apply these inequalities through the projection in (43).

Equations (39), (40), and (43) reduce the incumbent problem to a two-dimensional fixed point in (x, μ) . The first equation gives the value of an average new line, the second determines innovation effort, and the third determines the frontier share.

D.2 Directional Reallocation and Markup Misallocation

This subsection separates two notions of misallocation. The first is the static markup-misallocation statistic in Peters (2020). The second is the dynamic directional wedge in our model, through which procurement changes the composition of innovation and the stationary allocation of product lines across firms.

Peters (2020) studies an economy in which markups are dynamically determined by the balance between firms' own-innovation and creative destruction. Conditional on the resulting markup distribution, however, the aggregate misallocation object is a static sufficient statistic. Let m denote a line-level gross markup. With log aggregation, the Peters wedge can be written as

$$M = \frac{\exp\{-\mathbb{E}[\log m]\}}{\mathbb{E}[m^{-1}]} \leq 1, \quad (44)$$

where $M = 1$ if markups are equalized. In his Pareto-markup case, the distribution is pinned down by the churning intensity ϑ and the quality step λ : $\theta = \log(1 + \vartheta)/\log \lambda$, so $M = e^{-1/\theta}(1 + \theta)/\theta$.

In our model, there is no Pareto tail of accumulated markups. Line-level markups take only two values, $m_I = \lambda_I$ and $m_F = \lambda_F$, with frontier mass μ . The two-point analogue of the Peters wedge is therefore

$$M^{\text{static}}(\mu) = \frac{\exp\{-(1 - \mu) \log \lambda_I - \mu \log \lambda_F\}}{(1 - \mu)/\lambda_I + \mu/\lambda_F}, \quad (45)$$

or, equivalently,

$$-\log M^{\text{static}}(\mu) = \log \left[\frac{1 - \mu}{\lambda_I} + \frac{\mu}{\lambda_F} \right] + (1 - \mu) \log \lambda_I + \mu \log \lambda_F \geq 0. \quad (46)$$

This expression is useful as a diagnostic, but it is not the source of the paper's welfare results. Since procurement lowers μ and $\lambda_F > \lambda_I$, the two-point markup distribution becomes slightly less dispersed; the static Peters-style loss is small and mechanically declines. This is a distinctive implication of the procurement setting. Public contracts raise the payoff to specification-compliant incremental lines rather than letting firms accumulate a long right tail of markups within a market. Incumbents gain demand on incremental lines, but precisely for that reason they have weaker incentives to redirect innovation toward frontier lines. The model's cost therefore comes from a dynamic direction wedge rather than from higher static markup dispersion.

The private frontier-orientation condition is

$$\kappa\mu^{\eta-1} = x(v_F - v_I) = \frac{x}{r + \tau} [\bar{\pi}_F - \bar{\pi}_I(1 + G)]. \quad (47)$$

Holding (x, r, τ) fixed, the no-procurement benchmark would satisfy

$$\kappa\mu_0^{\eta-1} = \frac{x}{r + \tau} (\bar{\pi}_F - \bar{\pi}_I). \quad (48)$$

Subtracting (47) from (48) gives the marginal private return removed from frontier orientation by procurement:

$$\Delta_G^{priv} = \frac{x\bar{\pi}_I G}{r + \tau}. \quad (49)$$

Thus, for an interior solution,

$$\frac{\mu(G)}{\mu(0)} = \left[\frac{\bar{\pi}_F - \bar{\pi}_I(1 + G)}{\bar{\pi}_F - \bar{\pi}_I} \right]^{1/(\eta-1)} < 1. \quad (50)$$

Equation (50) is the allocative distortion: procurement lowers the privately chosen frontier share even though frontier innovations carry the larger quality step, $\lambda_F > \lambda_I$.

At fixed effort, the implied growth loss from this directional wedge is

$$g(0) - g(G) = x [\mu(0) - \mu(G)] (\log \lambda_F - \log \lambda_I) > 0. \quad (51)$$

The full quantitative model also lets effort, wages, entry, and creative destruction adjust, but the sign of the direct composition margin is the same. This is the margin disciplined by the market-novelty evidence: procurement does not need to reduce total innovation effort to lower growth; it is enough to tilt a given innovation arrival rate away from the higher-step frontier margin.

Table D1 reports equilibrium diagnostics from the general-equilibrium comparative-static grid. As procurement demand rises, the model reallocates activity toward larger incumbent portfolios with

fewer frontier lines. The frontier share and growth rate fall, entry labor contracts, mean firm scope rises, and the probability that a firm owns no frontier lines increases. These movements are the quantitative counterpart of the directional wedge above: procurement sustains firms and product lines that are privately valuable because of public demand even though they contribute less to aggregate quality growth. The final row reports the static Peters-style markup loss in (46); its small decline confirms that the welfare results are driven by labor and product-line reallocation toward lower-step innovation, not by static markup dispersion.

Table D1: Procurement demand and dynamic reallocation

	No procurement	Benchmark	High procurement
Procurement demand G	0.000	0.020	0.299
Frontier share μ^*	0.451	0.440	0.312
Growth g (%/yr)	0.503	0.500	0.465
Entry labor L_E	0.0216	0.0210	0.0134
Mean firm scope	2.123	2.154	2.679
Pr(no frontier lines)	0.386	0.394	0.483
Static markup loss $-\log M$ (%)	0.064	0.064	0.055

Notes: The table reports selected objects from the calibrated model's general-equilibrium comparative-static grid over procurement demand. The benchmark column uses the stored grid point closest to the calibrated procurement level. High procurement uses the largest displayed interior point in the same grid. "No frontier lines" is the stationary probability that a firm owns zero frontier product lines. The static markup loss is computed from (46) using the calibrated quality steps.

Source: Authors' calculations from the calibrated model.

D.3 Numerical Solution

Let $\mathbf{z} \equiv (x, \mu)^\top$ denote the common per-line innovation choice. Using the Bellman equations and the closed-form value gap, the incumbent block can be written as

$$F(\mathbf{z}; G, \psi) = 0, \quad (52)$$

where ψ collects the parameters held fixed in the incumbent problem. In the interior region, the two residuals are

$$F_1(\mathbf{z}; \cdot) \equiv c_x \zeta x^{\zeta-1} w - \bar{v}(\mathbf{z}; G) = 0, \quad (53)$$

$$F_2(\mathbf{z}; \cdot) \equiv \kappa \mu^{\eta-1} - \frac{x[\bar{\pi}_F - \bar{\pi}_I(1+G)]}{r+\tau} = 0. \quad (54)$$

At the boundaries $\mu = 0$ or $\mu = 1$, the algorithm applies the projection in (43).

We solve the system by damped fixed-point iteration. Given an initial guess $\mathbf{z}^{(0)}$, the algorithm updates μ from the projected direction condition and x from the effort condition until

$$\|\mathbf{z}^{(k+1)} - \mathbf{z}^{(k)}\| < 10^{-5}.$$

Once (x^*, μ^*) is obtained, (v_I, v_F, g) follow analytically.

The outer loop updates wages and entry until labor-market clearing and free entry hold jointly. Because the benchmark shuts down cross-line cost interactions, the policy functions are flat in the firm state (n_I, n_F) : all lines use the same (x^*, μ^*) in a given equilibrium.

At the benchmark, the numerical routine nests the general-equilibrium solve inside a scalar search over the cost scale c_x . For each candidate structural vector ψ , the algorithm recovers the unique $c_x(\psi)$ that delivers the benchmark growth target. Macro growth is a normalization, not a GMM target moment.

Optimization and Inference. The GMM objective is minimized by a multistart global search. At each start, we draw candidate parameter vectors uniformly from the admissible region, evaluate $Q(\psi)$ at each candidate, and retain the best. The search spans tens of thousands of function evaluations across multiple random seeds, avoiding reliance on gradient information unreliable given the discrete choice in μ^* .

Inference uses the delta method. Let $\hat{\mathbf{J}} = \partial \mathbf{m}(\psi) / \partial \psi^\top \big|_{\hat{\psi}}$ be the Jacobian of model moments at the estimates, computed by symmetric finite differences. The asymptotic covariance of the parameter vector is

$$\text{Var}(\hat{\psi}) = (\hat{\mathbf{J}}^\top W^* \hat{\mathbf{J}})^{-1} \hat{\mathbf{J}}^\top W^* \hat{\Sigma} W^* \hat{\mathbf{J}} (\hat{\mathbf{J}}^\top W^* \hat{\mathbf{J}})^{-1}, \quad (55)$$

where $\hat{\Sigma}$ is the empirical moment covariance and $W^* = \hat{\Sigma}^{-1}$ is the efficient weighting matrix estimated via influence functions, with MIP moments clustered at the firm level and the creative-destruction moment using the cross-year standard deviation from BHP. Standard errors are square roots of the diagonal elements. Appendix E.5 replaces the influence-function covariance with a firm-block bootstrap over 399 replications; parameter uncertainty changes little.

D.4 Value and Policy Functions in the Firm State

The value function varies with portfolio composition:

$$V(n_I, n_F; G) = n_I v_I(G) + n_F v_F(G).$$

Firms with the same total scope can have different values if their frontier shares differ. Innovation policies, by contrast, are flat in the firm state because the benchmark abstracts from cross-line interactions.

Figure D1 illustrates the result at $\bar{G} = 0.018$. The left panel reports $V(n_I, n_F)$. The middle and right panels report the effort and direction policies. The flat policy surfaces are the graphical counterpart of the model's linearity result.

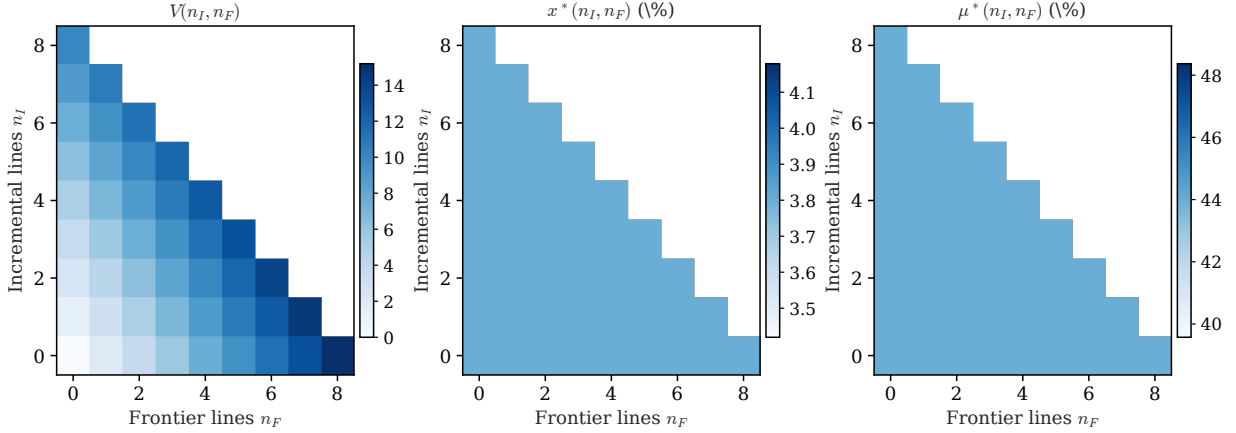


Figure D1: Benchmark value and policy functions in the firm state

Notes: The left panel reports firm value $V(n_I, n_F)$ at the benchmark $\bar{G} = 0.018$. The middle and right panels report effort and direction policies over the same state space. Policy surfaces are flat across firm states because the benchmark shuts down cross-line interactions.

Source: Authors' calculations from the calibrated model.

D.5 Transition to a New Steady State

The counterfactual comparisons in Section 5 compare initial and final steady states. This subsection characterizes the path connecting them under perfect foresight, following an unanticipated, permanent change in policy at $t = 0$ (a “MIT shock,” Boppart et al., 2018).

State versus jump variables. The firm-level state is the portfolio (n_I, n_F) , and firm value is additive in that state, $V(n_I, n_F; G) = n_I v_I(G) + n_F v_F(G)$ (eq. (10)). Every incremental line is worth v_I and every frontier line v_F regardless of portfolio size, so the aggregate transition does not require tracking the full cross-sectional distribution $\Gamma(n_I, n_F)$: one scalar aggregate state suffices. Let $\mathcal{M}_t \in [0, 1]$ denote the population share of *existing* lines that are frontier type at time t . This differs from μ_t , the frontier probability among *newly created* lines, which is pinned each instant by the static optimality system given $(v_{I,t}, v_{F,t})$ and is therefore a jump variable, not a state. The two coincide only on a balanced growth path, by Proposition 3.3(i). Off it, production labor, profit income, and the procurement tax base are functions of the actual line stock $\mathcal{M}_t$, not the flow policy μ_t ; equations (20) and (67) use μ in place of $\mathcal{M}$, which is exact only at a single steady state.

Each incumbent innovation event (aggregate rate x_t , since the unit mass of lines each attempt at rate x_t) creates one new line of the chosen type and displaces one uniformly random existing line, drawn with probability $\mathcal{M}_t$ of being frontier type, since displacement risk is common across types (eq. (12)). Entrant takeovers change ownership, not type. The state therefore follows

$$\dot{\mathcal{M}}_t = x_t(\mu_t - \mathcal{M}_t), \quad \mathcal{M}_0 = \mu_{\text{old}}^*, \quad (56)$$

with $\mathcal{M}_0$ the old steady state's stationary frontier share.

Non-stationary values. Adding a time-derivative term to the per-line Bellman equations gives, for $k \in \{I, F\}$,

$$\dot{v}_{k,t} = (r_t + \tau_t)v_{k,t} - \pi_k(G) + \Phi(x_t) + \Psi(\mu_t) - x_t \bar{v}_t, \quad (57)$$

with terminal condition $v_{k,t} \rightarrow v_k^{\text{new}}$ as $t \rightarrow \infty$. Subtracting the two equations gives a value-gap ODE, $\dot{D}_t = (r_t + \tau_t)D_t - [\pi_F(G) - \pi_I(G)]$: linear in D_t alone, but not autonomous, since τ_t inherits $\mathcal{M}_t$ -dependence through labor clearing. This is the only channel through which the state affects values: profits and the Bellman equations reference $\mathcal{M}_t$ nowhere directly, only τ_t and w_t do, both general-equilibrium objects.

Proposition D.1 (Effort invariance). *Combining the effort first-order condition (25) with free entry (27) gives a closed form for equilibrium effort,*

$$x^* = \left(\frac{h_E}{c_x \zeta} \right)^{1/(\zeta-1)}, \quad (58)$$

independent of G , μ , and w . Effort therefore does not transition: it is constant throughout the path except for a discrete jump at $t = 0$ when c_x itself changes.

Proof. Free entry requires $\bar{v}(\mu; G) = wh_E$ at every instant that entrants can costlessly enter, which the effort condition equates to $c_x \zeta x^{\zeta-1} w$. Setting the two equal and dividing by $w > 0$ gives $c_x \zeta x^{\zeta-1} = h_E$, independent of μ , G , and w . $\square$

At $\zeta = 2$, $x^* = h_E/(2c_x)$. This matches the counterfactual table exactly: specification realignment and historical rollback change only G , leaving $\Delta x^* = 0.00\%$ identically rather than approximately; the R&D credit changes $c_x \rightarrow (1-s)c_x$, giving $x_{\text{new}}^* = x_{\text{old}}^*/(1-s)$, a 16.4% rise at $s = 0.141$, matching Table 6 to the reported precision.

Given constant effort, the direction condition collapses to a static, invertible map,

$$\mu_t = \min\left\{1, \max\{0, [x^* D_t / \kappa]^{1/(\eta-1)}\}\right\}, \quad (59)$$

so μ_t needs no law of motion of its own, only D_t 's.

Corrected static block. At each instant, given $(\mathcal{M}_t, v_{I,t}, v_{F,t})$: μ_t follows from the direction map; $w_t = \bar{v}_t/h_E$ from free entry; τ_t from labor clearing with $\mathcal{M}_t$ replacing μ in L_P ,

$$\tau_t = x^* + \frac{1}{h_E} \left[\bar{L} - L_P(\mathcal{M}_t, G, w_t) - \ell_R(x^*, \mu_t, w_t) \right];$$

and consumption from the household budget with $\mathcal{M}_t$, not μ_t , in the profit and tax terms,

$$\begin{aligned} C_t &= w_t \bar{L} + [\mathcal{M}_t \pi_F(G_F) + (1 - \mathcal{M}_t) \pi_I(G_I) - \Phi(x^*) - \Psi(\mu_t)] - T_t - w_t L_{E,t}, \\ T_t^G &= (1 - \mathcal{M}_t) G_I + \mathcal{M}_t G_F. \end{aligned} \quad (60)$$

For the R&D-credit counterfactual, $G_I = G_F = 0$ and the balanced-budget tax instead equals the credit's contemporaneous fiscal cost, $T_t^s = s w_t (L_{R,t} + \omega_E L_{E,t} + \omega_P L_{P,t})$, using the same eligible-base shares as in Table 6. Let $\hat{C}_t$ denote the detrended consumption object in the budget equation above. Since level consumption is $C_t^{\text{level}} = \hat{C}_t Q_t$ and $\dot{Q}_t/Q_t = g_t$, with $g_t = x^*[(1 - \mu_t) \log \lambda_I + \mu_t \log \lambda_F]$, the household Euler equation is

$$r_t = \rho + g_t + \frac{d \log \hat{C}_t}{dt}. \quad (61)$$

This nests the balanced-growth restriction $r = \rho + g$ when detrended consumption is constant.

Proposition D.2 (Local convergence rate). *Suppose $(r_t + \tau_t)$ does not vary with $\mathcal{M}_t$. Then $(v_{I,t}, v_{F,t})$ jump to their new steady-state levels immediately, and $\mathcal{M}_t$ converges to μ_{new}^* exponentially at rate x^* .*

Proof. With $(r + \tau)$ fixed at its new-steady-state value $(r + \tau)_{\text{new}}$ and $\mu_t \equiv \mu_{\text{new}}^*$, the linearized (v_I, v_F) system has Jacobian

$$J = \begin{pmatrix} (r + \tau)_{\text{new}} - x^*(1 - \mu_{\text{new}}^*) & -x^* \mu_{\text{new}}^* \\ -x^*(1 - \mu_{\text{new}}^*) & (r + \tau)_{\text{new}} - x^* \mu_{\text{new}}^* \end{pmatrix},$$

with trace $2(r + \tau)_{\text{new}} - x^*$ and determinant $(r + \tau)_{\text{new}}[(r + \tau)_{\text{new}} - x^*]$. The finite-value condition $r + \tau > x$, footnoted at eq. (36), makes both positive, so both eigenvalues have positive real part:

(v_I, v_F) is fully unstable and must jump to remain bounded. The $\mathcal{M}_t$ equation is then the scalar $\dot{\mathcal{M}}_t = -x^*(\mathcal{M}_t - \mu_{\text{new}}^*)$, one stable eigenvalue at $-x^*$, matching the count required for determinacy given one predetermined state and two jump variables. $\square$

This benchmark switches off the channel through which $\mathcal{M}_t$ actually affects τ_t and w_t through labor clearing, so it provides a transparent reference rate rather than the full general-equilibrium rate; Table D2 reports both.

Numerical solution. The full system, with the wage-market feedback restored, is solved by forward-backward relaxation on a discretized time grid ($\Delta t = 0.05$ years, $T = 150$ years): march $(v_{I,t}, v_{F,t})$ backward from the terminal condition given guesses for the $\mathcal{M}_t$ and r_t paths, march $\mathcal{M}_t$ forward from $\mathcal{M}_0$ given the resulting μ_t path, update r_t from the Euler equation (61), and iterate to convergence. As a validation check, the solver's own effort path comes out numerically constant and equal to eq. (58) at every point on the path, to five significant digits, independently of having assumed the result.

Table D2 reports the fitted convergence rate, from a log-linear fit of $|\mathcal{M}_t - \mu_{\text{new}}^*|$ over the back half of each path, against the closed-form benchmark, together with a transition-integrated welfare measure that replaces the two-point CEV (34) with the discounted path integral

$$\begin{aligned} \log(1 + \lambda) &= \left[\rho \int_0^\infty e^{-\rho t} \log C_t dt - \log C_0^{\text{old}} \right] + \left[\int_0^\infty e^{-\rho t} g_t dt - \frac{g_{\text{old}}}{\rho} \right], \\ \text{CEV}^{\text{trans}} &= 100 \left(e^{\log(1+\lambda)} - 1 \right), \end{aligned} \quad (62)$$

splitting into a consumption-level term and a growth term exactly as eq. (34) does at two points, here integrated along the whole path and approximated by the discretized path to T plus a tail assuming convergence beyond it.

Convergence is slow relative to the household discount rate: the composition margin closes half its gap in 14 to 29 years, an order of magnitude slower than the discount horizon implied by $\rho = 0.04$. The transition-integrated CEV nonetheless lands close to the two-point figure in Table 6 for all three exercises, running marginally above it rather than below, since the forward-looking jump variable μ_t moves most of the way to its new steady-state level immediately after the shock, so the growth benefit is realized faster than the two-point comparison assumes even though the composition stock itself adjusts slowly. Labor-market feedback through w_t also slows convergence unevenly across exercises: the fitted rate sits close to the closed-form benchmark for the R&D credit (14.6 versus 13.8 years) but is nearly double the benchmark half-life for the historical rollback (29.2

Table D2: Transition dynamics for the policy counterfactuals

Counterfactual	Half-life (years)		CEV (%)	
	Benchmark x^*	Fitted	Two-point (endpoints only)	Transition-integrated (full path)
R&D tax credit	13.75	14.60	1.55	1.567
Specification realignment	16.01	18.31	0.09	0.104
Historical rollback	16.01	29.22	0.015	0.025

Notes: Benchmark half-life is $\ln 2/x^*$, the closed-form rate from Proposition D.2 (identical across the specification-realignment and historical-rollback rows since neither changes c_x , and hence x^* , from its calibrated value). Fitted half-life is $\ln 2$ divided by the slope of a log-linear fit of $|\mathcal{M}_t - \mu_{\text{new}}^*|$ over the back half of the numerically solved path. The two CEV columns answer the same welfare question two different ways. Two-point CEV, from Table 6, assumes the economy jumps directly from the old steady state to the new one and compares only those two endpoints. Transition-integrated CEV instead values household consumption and growth along the entire solved adjustment path, respecting how slowly $\mathcal{M}_t$ actually converges, approximating the infinite horizon with the solved path to $T = 150$ years plus a tail assuming convergence beyond it. The two are close because most of the welfare-relevant jump in μ_t happens immediately even though $\mathcal{M}_t$ itself adjusts slowly, as discussed in the text.

Source: Authors' calculations from the calibrated model, forward-backward relaxation solver.

versus 16.0 years), consistent with a smaller policy change leaving relatively more room for the general-equilibrium wage channel to matter.

Figure D2 illustrates the same jump-then-drift mechanism generically, for a permanent increase in G by one standard deviation of the sector-year exposure distribution ($\Delta G = 0.0132$, computed from the same proportional mapping used for the sector-specific wedges in Section 5) rather than for a specific policy. The frontier share falls from 43.90% to a new steady state of 43.19% ($\Delta \mu^* = -0.70$ p.p.), with a fitted half-life of 9.6 years, faster than the 16.0-year closed-form benchmark since the larger shock moves the system further from the region where wage feedback is weak. R&D labor and growth trace the mirror image of the pattern in the three policy exercises: because a positive G shock pushes μ_t below its new, lower steady-state level on impact, both variables start at their post-shock trough and rise back toward the new steady state as $\mathcal{M}_t$ catches up, rather than starting high and settling down. The transition-integrated CEV for this shock is -0.043% , consistent in sign with a composition-driven welfare loss and an order of magnitude smaller than the R&D-credit gain, matching its smaller size relative to the calibrated benchmark $\bar{G}$.

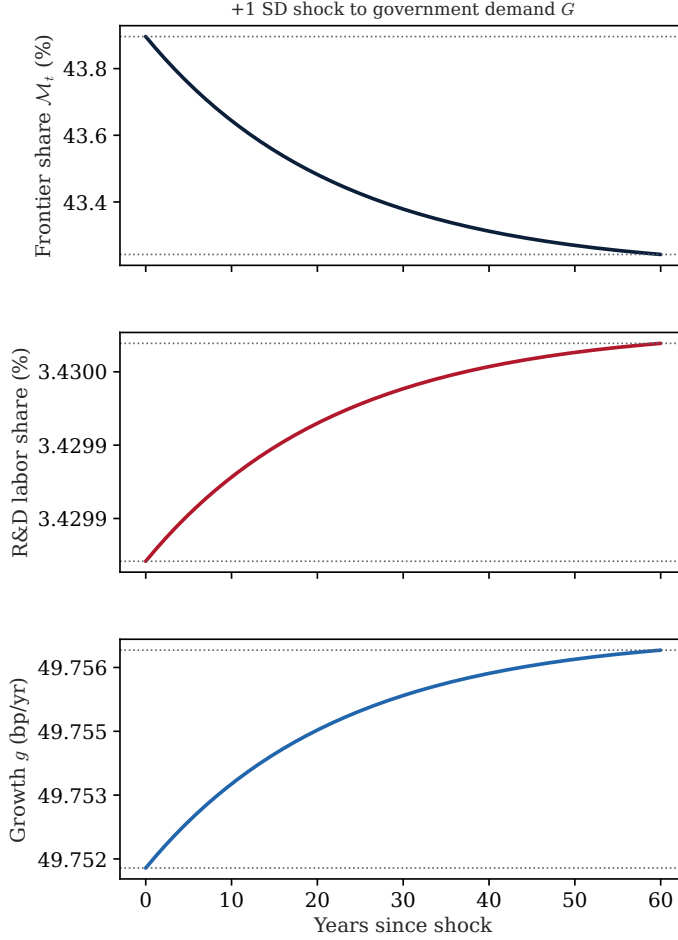


Figure D2: Transition path for a one-SD shock to government demand

Notes: Top: the state $\mathcal{M}_t$ (stock frontier share), first sixty years after a permanent increase in G of one sector-year standard deviation. Middle: the R&D labor share $L_{R,t}/\bar{L}$. Bottom: the flow quality-growth rate g_t . Dotted horizontal lines mark the first plotted point (immediately post-shock) and the converged new steady-state level in each panel.

Source: Authors' calculations from the calibrated model.

D.6 Entry, Creative Destruction, and Stationarity

The baseline has two sources of ownership changes but only one source of quality growth. Incumbent innovation improves a rival line and transfers it to the innovator at rate x^* per existing line. Entrants use setup labor to acquire existing lines and create new firms with one line, without generating an additional quality step. Existing lines change owner at total hazard τ . With a unit mass of product lines, stationarity requires

$$x^* + e = \tau, \quad (63)$$

where e is the entrant line-creation rate. Hence,

$$e = \tau - x^*. \quad (64)$$

The entry-labor requirement is

$$L_E = h_E e = h_E(\tau - x^*), \quad (65)$$

rather than $h_E \tau$, whenever τ denotes total line destruction.

Free entry determines the expected value of an entrant line:

$$\bar{v}(\mu^*; G) = w h_E. \quad (66)$$

Here h_E is the fixed setup-labor requirement per entrant and $w h_E$ is the corresponding monetary cost. Labor-market clearing uses production labor, incumbent R&D labor, and entrant labor:

$$\bar{L} = L_P(\mu, G, w) + \ell_R(x, \mu, w) + h_E(\tau - x). \quad (67)$$

D.7 Stationary Firm-Size Distribution

Let $N = n_I + n_F$ denote firm scope. Conditional on a firm having $N = n$ lines, births arrive at rate $x^* n$ and deaths arrive at rate τn . Entrants enter with one line at rate $e = \tau - x^*$ under the unit line-mass normalization. We require $x^* < \tau$ for a stationary firm-size distribution.

Let m_n denote the mass of firms with n product lines. For $n \geq 2$, stationarity implies

$$0 = x^*(n-1)m_{n-1} + \tau(n+1)m_{n+1} - (x^* + \tau)n m_n. \quad (68)$$

The solution is

$$m_n = \frac{e}{x^*} \frac{1}{n} \left(\frac{x^*}{\tau} \right)^n, \quad n \geq 1. \quad (69)$$

Let

$$q \equiv \frac{x^*}{\tau} \in (0, 1).$$

Conditional on observing an active firm, the firm-scope distribution is

$$\Pr(N = n) = \frac{q^n}{n[-\log(1-q)]}, \quad n \geq 1. \quad (70)$$

Equation (70) gives the Klette–Kortum/Yule distribution, which differs from a geometric distribution by the factor $1/n$.

Conditional on $N = n$, each line is frontier with probability μ^* :

$$\Pr(n_F = b \mid N = n) = \binom{n}{b} (\mu^*)^b (1 - \mu^*)^{n-b}. \quad (71)$$

The joint distribution is

$$\Gamma(n_I, n_F; G) = \Pr(N = n_I + n_F) \binom{n_I + n_F}{n_F} (\mu^*)^{n_F} (1 - \mu^*)^{n_I}. \quad (72)$$

The product-innovation probability over a horizon T is

$$p_{\text{innov}} = 1 - \sum_{n=1}^{\infty} \Pr(N = n) e^{-Tx^*n}. \quad (73)$$

Using (70),

$$p_{\text{innov}} = 1 - \frac{-\log(1 - qe^{-Tx^*})}{-\log(1 - q)}. \quad (74)$$

Proof of Proposition 3.3. (i) By iterated expectations $\mathbb{E}[s_F] = \mu^*$. For the variance, $\mathbb{E}[(n_F/N)^2 | N] = \mu^*(1 - \mu^*)/N + \mu^{*2}$, so $\text{Var}(s_F) = \mu^*(1 - \mu^*)\mathbb{E}[1/N]$. Under (70),

$$\mathbb{E}[1/N] = \frac{\sum_{n=1}^{\infty} q^n/n^2}{-\log(1 - q)},$$

giving the stated formula.

(ii) $\Pr(n_F = 0) = \sum_{n \geq 1} \Pr(N = n)(1 - \mu^*)^n = -\log(1 - q(1 - \mu^*)) / (-\log(1 - q))$, which follows by substituting (70) and recognizing the Taylor series of $\log(1 - z)$.

(iii) $\mathbb{E}[\pi(G)] = \mu^*\bar{\pi}_F + (1 - \mu^*)\bar{\pi}_I(1 + G)$ by linearity of expectation over the portfolio and the per-line profit expressions. $\square$

Figure D3 plots both marginal distributions at the GMM benchmark.

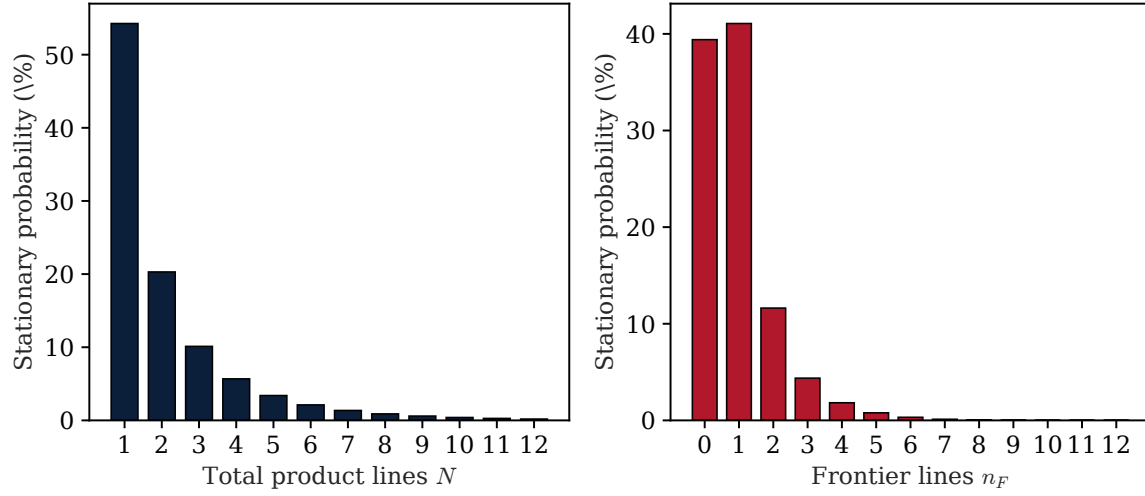


Figure D3: Model-implied stationary distributions at the GMM benchmark

Notes: Panel A reports the firm-scope distribution in (70). Panel B reports the marginal frontier-line distribution implied by binomial mixing under the benchmark equilibrium.

Source: Authors' calculations from the calibrated model.

E Quantitative Analysis

The appendix collects the quantitative exercises supporting the model and data mapping.

E.1 Identification and Model Fit

E.1.1 Detailed Measurement Mapping and External Validation

Table E1 reports the corresponding external-validation exercise. Because the empirical and model treatments are measured in different units, the table does not treat the magnitudes as a formal fit criterion. It asks whether a transparent increase in model procurement reproduces the joint empirical signs for innovation composition and incumbent scale.

Table E1: External validation: empirical and model responses to procurement

Outcome	Data effect	Model effect	Sign match	Role
Market novelty	−0.709 pp	−0.024 pp	Yes	Composition validation
Unchanged-product revenue	1.352 pp	0.139 pp	Yes	Composition validation
Employment / firm scope	2.300 percent	0.339 percent	Yes	Scale validation

Notes: Data effects correspond to a one-standard-deviation increase in the relevant procurement exposure. Model effects compare the benchmark $\bar{G} = 0.01783$ with a 25% increase, $G = 1.25\bar{G}$, holding structural parameters and the recovered innovation-cost scale fixed while re-solving general equilibrium. The two changes are not treated as equivalent units; the exercise validates signs and reports transparent magnitudes. Market novelty maps to the model probability that a firm produces at least one frontier innovation over the MIP horizon. Unchanged-product revenue is one minus the model new-product revenue share. Employment maps to mean product scope. Model magnitudes are provisional pending re-estimation after the Euler-equation correction.

To make the unit conversion explicit, Table E2 combines the model derivative of the market-novelty counterpart with a proportional mapping from sector-year lot counts to sector procurement shares.

Table E2: Model-implied procurement coefficient

Object	Value	Unit
$\partial MN / \partial G$	−5.632	pp / unit- G
$\partial G / \partial \operatorname{arcsinh}(\ell)$	0.01791	unit- G / $\operatorname{arcsinh}$ unit
$\hat{\beta}_{\text{model}}$	−0.101	pp / $\operatorname{arcsinh}$ unit
$\hat{\beta}_{\text{empirical}}$	−0.199	pp / $\operatorname{arcsinh}$ unit

Notes: $\partial MN / \partial G$ is computed by symmetric finite differences at $\bar{G} = 0.01783$, holding the recovered innovation-cost scale c_x fixed. $\partial G / \partial \operatorname{arcsinh}(\ell)$ assumes that sector-year procurement shares are proportional to sector-year lot counts, giving $\partial G / \partial \operatorname{arcsinh} = \bar{G} \times \cosh(\bar{a}) / \sinh(\bar{a}) \approx \bar{G}$ at the sample mean $\bar{a} = 3.1$. The model-implied coefficient is the product of these two components and is not targeted in estimation. Because the lot-count-to- G conversion is assumed rather than independently estimated, the comparison is a sensitivity exercise rather than a structural fit test.

Historical rollback and the aggregate trend. The historical rollback should not be read as a decomposition of Germany’s aggregate novelty decline. In relative terms, μ^* falls from 0.441 under 2008-level procurement to 0.439 at the benchmark, a 0.5 percent relative decline. Over the same window, the raw market-novelty rate in Figure E1 fell by 32 percent in relative terms, from 18.0 percent of firms in 2008 to a 2009–2023 average of 12.2 percent. The rollback therefore recovers under two percent of the aggregate decline. This gap is expected rather than a failure of fit: the counterfactual isolates the procurement channel alone, identified from cross-sectional exposure variation across sectors and years, while the aggregate time-series trend plausibly reflects globalization of R&D, catch-up growth, and sectoral reallocation forces the model does not speak to.

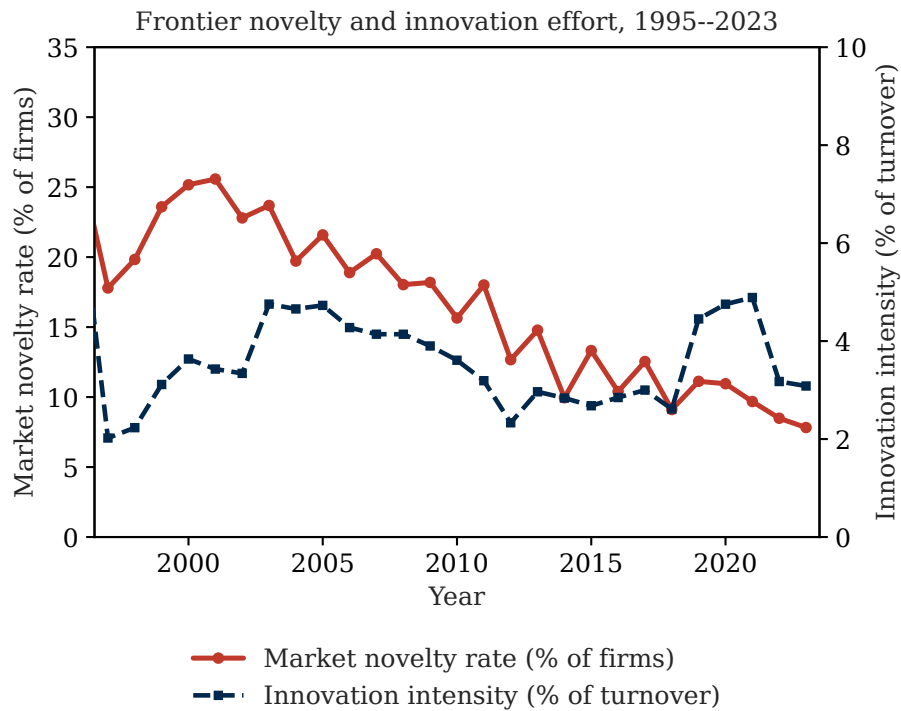


Figure E1: Novelty fell despite steady innovation spending

Notes: The red line plots the market-novelty rate, defined as the share of firms introducing products new to the market, on the left axis. The blue dashed line plots innovation intensity, defined as innovation expenditure divided by turnover, on the right axis. The aggregate decline is not a story of reduced innovation effort: spending stays broadly stable while frontier novelty falls.

Source: Mannheim Innovation Panel, 1995–2023.

E.2 Moment Sensitivity

Figure E2 shows how each targeted moment responds to individual parameter variation. Each parameter has a distinct signature: κ and λ_I strongly rotate the frontier-share moments (rows 1–2); h_E primarily moves the large-firm share and creative-destruction rate (rows 4–5); λ_F shifts frontier

revenue while leaving other moments relatively flat. Government demand G (rightmost column) reduces frontier composition and creative destruction monotonically, motivating the composition mechanism. The parameter effects are separable enough to identify all four structural parameters, though the quality steps share some influence over rows 1–2.

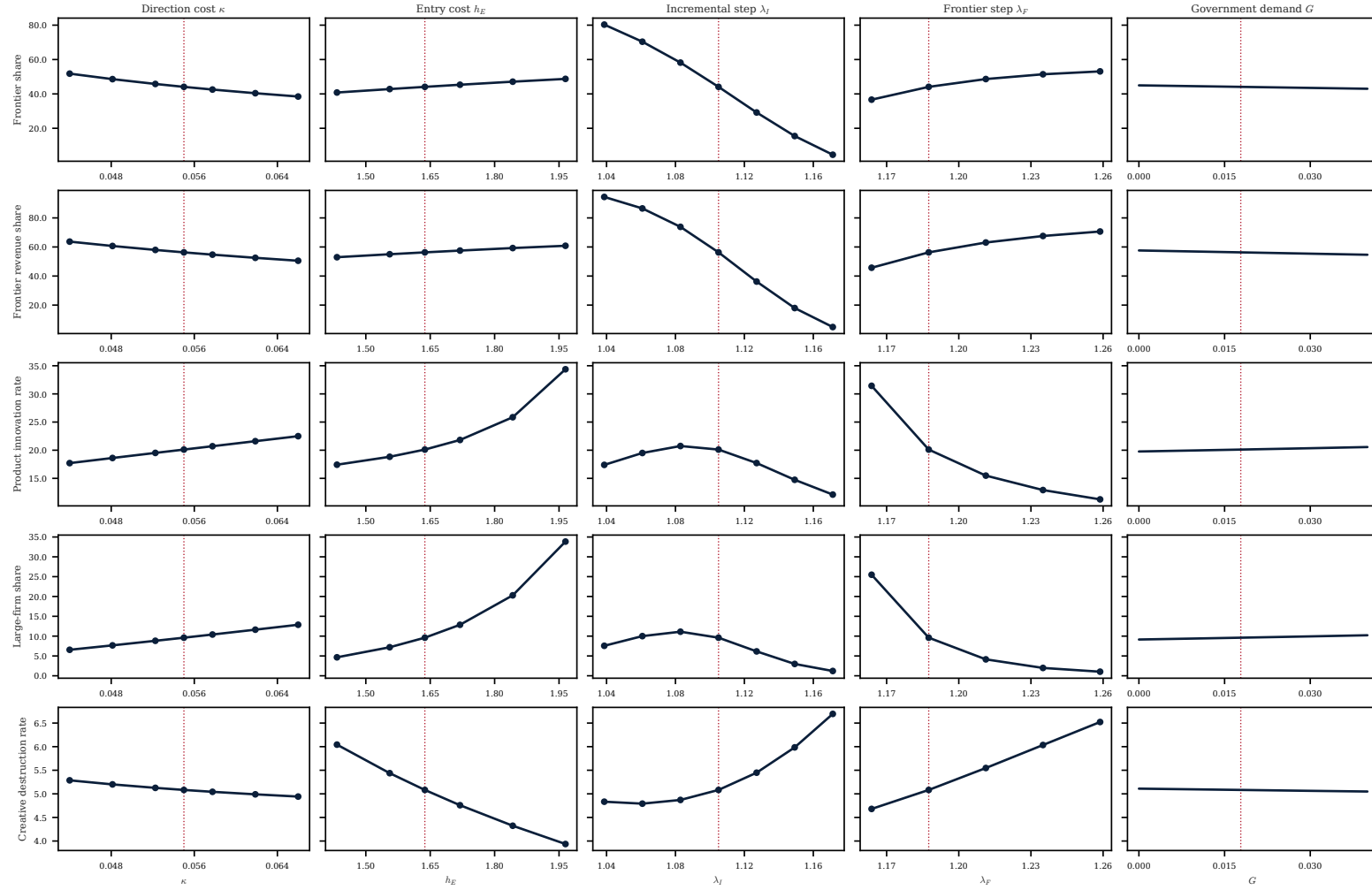


Figure E2: Moment comparative statics: identification

Notes: Rows are targeted moments; columns are parameters. For each structural parameter (κ , h_E , λ_I , λ_F), the curve evaluates up to seven values spanning $\pm 20\%$ of the calibrated estimate ($\pm 6\%$ for the quality steps), with all other parameters held fixed at their calibrated values and $G = \bar{G} = 0.018$; dots mark converged equilibria. The G column (right) traces $G \in [0, 0.04]$ at benchmark structural parameters. The dotted red vertical line marks the calibrated benchmark of each parameter.

Source: Authors' calculations from the calibrated model.

E.3 Counterfactual Design Mechanics

E.3.1 R&D Tax Credit Design

The R&D-credit counterfactual sets $G = 0$ and finances a proportional subsidy to innovation cost, $\Phi(x) \rightarrow (1 - s)c_x x^\zeta w$, at a rate s chosen so that the credit's fiscal cost exactly replaces the procurement tax it displaces. The eligible base covers incumbent R&D labor, a share ω_E of entry labor, and a share ω_P of production labor, mirroring how the *Forschungszulage* and comparable credits subsidize adoption and implementation costs alongside R&D wages (main text, Section 5). The credit rate solves the fixed point

$$s \cdot w(s) \left[\ell_R(x^*(s), \mu^*(s), w(s)) + \omega_E h_E(\tau(s) - x^*(s)) + \omega_P L_P(s) \right] = T_{bm}, \quad (75)$$

where $w(s)$, $x^*(s)$, $\mu^*(s)$, $\tau(s)$, and $L_P(s)$ are the equilibrium wage, effort, frontier share, creative-destruction rate, and aggregate production labor under credit rate s , and $T_{bm} = (1 - \mu_{bm}^*)\bar{G}$ is the benchmark procurement tax it replaces.

Equation (75) has no closed form. Effort itself is pinned down exactly, $x^*(s) = x_{bm}^*/(1 - s)$ by Proposition D.1 applied to the credit-adjusted cost $(1 - s)c_x$, but the tax base on the left side depends on the full general-equilibrium labor allocation (w, μ^*, τ, L_P) , which itself responds to s through the innovation-cost subsidy. We solve (75) numerically: re-solve the GE fixed point at a trial s , evaluate the resulting credit expenditure, and adjust s by bisection until expenditure matches T_{bm} to numerical tolerance. The entry-eligibility weight $\omega_E = 0.5$ is fixed ex ante at half the entry-labor cost; the production-implementation weight ω_P is then the residual value that clears (75) exactly at the resulting s . This delivers $s = 14.1\%$ and $\omega_P = 2.78\%$ (Table 6, Row 1).

E.3.2 Specification Design Reform

The specification design reform reallocates a fixed procurement budget T_{bm} across line types. Let $\alpha \in [0, 1]$ denote the share of the budget assigned to frontier lines. Given the equilibrium frontier mass μ , the per-line demand intensities are

$$G_I(\mu, \alpha) = \frac{(1 - \alpha)T_{bm}}{1 - \mu}, \quad G_F(\mu, \alpha) = \frac{\alpha T_{bm}}{\mu}, \quad (76)$$

so $(1 - \mu)G_I + \mu G_F = T_{bm}$ in every counterfactual equilibrium. The modified profit expressions are

$$\pi_I(\alpha) = \bar{\pi}_I[1 + G_I(\mu, \alpha)], \quad (77)$$

$$\pi_F(\alpha) = \bar{\pi}_F[1 + G_F(\mu, \alpha)]. \quad (78)$$

At $\alpha = 0$ all demand flows to incremental lines. At the benchmark, $T_{bm} = (1 - \mu_{bm})\bar{G}$, so $G_I = \bar{G}$ and $G_F = 0$. The value gap under the reform is

$$v_F(\alpha) - v_I(\alpha) = \frac{\bar{\pi}_F[1 + G_F(\mu, \alpha)] - \bar{\pi}_I[1 + G_I(\mu, \alpha)]}{r + \tau}. \quad (79)$$

The direction condition gives

$$\mu^*(\alpha) = \frac{x^*}{\kappa} \cdot \frac{\bar{\pi}_F[1 + G_F(\mu^*, \alpha)] - \bar{\pi}_I[1 + G_I(\mu^*, \alpha)]}{r + \tau}. \quad (80)$$

Proposition E.1 (Budget-neutral specification realignment). *Suppose the equilibrium is interior. Reallocating a marginal unit of the fixed procurement budget from incremental to frontier lines raises the current profit gap by*

$$\frac{\partial(\pi_F - \pi_I)}{\partial \alpha} = T_{bm} \left(\frac{\bar{\pi}_F}{\mu} + \frac{\bar{\pi}_I}{1 - \mu} \right) > 0.$$

Holding general-equilibrium prices fixed, this raises frontier orientation. Quantitatively, the model jointly resolves the induced change in μ , wages, and creative destruction while imposing equation (76).

Proof. Raising α by a unit shifts government demand away from incremental lines and toward frontier lines. From (77)–(78), frontier profits rise by $\bar{\pi}_F G$ and incremental profits fall by $\bar{\pi}_I G$. Both changes widen the frontier value gap $(v_F - v_I)$ in (79), so the direction first-order condition moves μ^* toward frontier orientation. Formally, differentiating (80) with respect to α :

$$\frac{\partial \mu^*}{\partial \alpha} = \frac{x^*}{\kappa(r + \tau)} \cdot \frac{\partial}{\partial \alpha} [\bar{\pi}_F(1 + \alpha G) - \bar{\pi}_I(1 + (1 - \alpha)G)] = \frac{x^* G (\bar{\pi}_F + \bar{\pi}_I)}{\kappa(r + \tau)}.$$

Since $G, \bar{\pi}_F, \bar{\pi}_I, \kappa, r + \tau, x^* > 0$, this is strictly positive. The factor G enters multiplicatively because specification reform can only redirect existing procurement demand: the larger the procurement footprint, the more demand is available to realign toward frontier lines. $\square$

E.3.3 Government Risk Aversion and the Demand Wedge

The active model treats procurement demand as reduced form. The extension microfound the demand wedge through a government objective over expected delivered quality and implementation risk.

Suppose the government procures one line at a time. An incremental line delivers

$$q_I \sim \mathcal{N}(\lambda_I, \sigma_I^2),$$

while a frontier line delivers

$$q_F \sim \mathcal{N}(\lambda_F, \sigma_F^2), \quad \lambda_F > \lambda_I, \quad \sigma_F^2 > \sigma_I^2.$$

The frontier line has higher expected quality but also higher delivery risk: frontier projects are harder to specify, verify, and monitor ex ante.

The government values expected quality and dislikes implementation risk:

$$U_G(k) = \mathbb{E}[q_k] - \frac{\gamma_G}{2} \text{Var}(q_k), \quad k \in \{I, F\},$$

where $\gamma_G \geq 0$ is the government's risk-aversion parameter. Under the Gaussian specification,

$$U_G(I) = \lambda_I - \frac{\gamma_G}{2} \sigma_I^2, \quad U_G(F) = \lambda_F - \frac{\gamma_G}{2} \sigma_F^2.$$

Proposition E.2 (Government preference threshold). *Suppose $\lambda_F > \lambda_I$ and $\sigma_F^2 > \sigma_I^2$. The government prefers incremental procurement to frontier procurement if and only if*

$$\gamma_G > \gamma_G^* \equiv \frac{2(\lambda_F - \lambda_I)}{\sigma_F^2 - \sigma_I^2}.$$

Proof. The government prefers incremental procurement when $U_G(I) \geq U_G(F)$. Using the expressions above,

$$\lambda_I - \frac{\gamma_G}{2} \sigma_I^2 \geq \lambda_F - \frac{\gamma_G}{2} \sigma_F^2$$

if and only if

$$\gamma_G(\sigma_F^2 - \sigma_I^2) \geq 2(\lambda_F - \lambda_I).$$

Since $\sigma_F^2 > \sigma_I^2$, division gives the threshold. □

Proposition E.2 gives the buyer-side logic behind the demand wedge. When the government is sufficiently risk averse, it forgoes the expected quality advantage of frontier lines and steers procurement toward specification-compliant incremental supply. Higher γ_G corresponds to a larger effective incremental-demand wedge in the active model.

The extension maps to the contract-design interpretation in a limited way. The main text reports descriptive splits by contract type and buyer category. Those splits are consistent with the idea that some procurement environments place more weight on compliance and implementation reliability than on frontier upside, but they do not isolate the buyer-side preference parameters in the model.

Remark E.3 (Monitoring capacity as a procurement-design reform). The threshold γ_G^* rises when monitoring or verification technology reduces σ_F^2 . For a fixed γ_G , better monitoring makes the government less likely to prefer incremental procurement over frontier procurement. In the active model, lower frontier-delivery uncertainty works like a reduction in the effective demand wedge G .

Table E3 evaluates the threshold using the GMM estimates $\lambda_F = 1.165$ and $\lambda_I = 1.090$. Setting $\sigma_I = 0$ and varying σ_F gives the following values:

Table E3: Government preference threshold at the GMM benchmark

σ_F	γ_G^*
0.15	6.67
0.20	3.75
0.25	2.40
0.30	1.67
0.35	1.22

Notes: Computed from $\gamma_G^* = 2(\lambda_F - \lambda_I)/\sigma_F^2$ with $\lambda_F - \lambda_I = 0.075$ and $\sigma_I = 0$. *Source:* Authors' calculations from the calibrated model.

The incremental-demand regime holds when frontier projects carry sufficient delivery uncertainty relative to the quality-step gap. Table E3 shows the frontier delivery standard deviation σ_F required for the regime to hold at a given risk-aversion level. We do not estimate σ_F or γ_G separately; the table provides a reference for how delivery uncertainty maps to the preference threshold given the calibrated quality steps $\lambda_F = 1.165$ and $\lambda_I = 1.090$.

Identifying the two preference parameters would require contract-level ex post performance data (completion rates, cost overruns, or independent quality assessments) not available in the current ad-

ministrative records. The extension serves as a microfoundation for the baseline demand asymmetry rather than as an independently estimated buyer-side model.

E.4 Heterogeneity

E.4.1 Heterogeneous Direction Costs

The benchmark estimates a single direction-cost parameter κ common to all firms and sectors. Allowing sector-specific direction costs $\kappa_s > 0$ leaves the *proportional* compression of the frontier share unchanged, separating the demand-composition mechanism from the supply-side forces documented by Bloom et al. (2020).

Let sector s have direction cost $\Psi_s(\mu) = (\kappa_s/2)\mu^2$, with $\kappa_s > 0$, and let all other parameters be common across sectors. The direction first-order condition in sector s is

$$\kappa_s \mu_s^* = x^*(v_F - v_I) = x^* \frac{\bar{\pi}_F - \bar{\pi}_I(1 + G)}{r + \tau}, \quad (81)$$

so the equilibrium frontier share is

$$\mu_s^* = \frac{x^*}{\kappa_s} \cdot \frac{\bar{\pi}_F - \bar{\pi}_I(1 + G)}{r + \tau}. \quad (82)$$

Sectors with larger κ_s (frontier innovations costlier to orient toward) have lower frontier shares at any given G , while their innovation behavior is otherwise governed by the same demand-payoff margin.

Proposition E.4 (Conditional invariance of proportional compression). *Let direction costs be sector-specific, $\kappa_s > 0$, and suppose sectors take the common equilibrium responses $(x(G), r(G), \tau(G))$ as given. For $\eta = 2$, proportional compression satisfies*

$$\frac{\partial \ln \mu_s^*}{\partial G} = \frac{1}{x} \frac{\partial x}{\partial G} - \frac{\bar{\pi}_I}{\bar{\pi}_F - \bar{\pi}_I(1 + G)} - \frac{1}{r + \tau} \left(\frac{\partial r}{\partial G} + \frac{\partial \tau}{\partial G} \right),$$

which is independent of κ_s . This invariance is conditional: it need not survive if sectoral direction costs change aggregate equilibrium responses or if sectors face different wages or destruction hazards.

Proof. Taking logs of (82):

$$\ln \mu_s^* = \ln x^* - \ln \kappa_s + \ln [\bar{\pi}_F - \bar{\pi}_I(1 + G)] - \ln(r + \tau).$$

Differentiating with respect to G along the common equilibrium path gives

$$\frac{\partial \ln \mu_s^*}{\partial G} = \frac{1}{x} \frac{\partial x}{\partial G} - \frac{\bar{\pi}_I}{\bar{\pi}_F - \bar{\pi}_I(1 + G)} - \frac{1}{r + \tau} \left(\frac{\partial r}{\partial G} + \frac{\partial \tau}{\partial G} \right),$$

which does not depend directly on κ_s under the maintained common-equilibrium-response assumption. $\square$

Proposition E.4 isolates a partial-equilibrium comparison across sectors. Conditional on common responses of effort, growth, and creative destruction, sectors with high κ_s have lower frontier shares and therefore smaller absolute changes, while their proportional response is the same. We do not use the conditional result to claim invariance in a multisector general equilibrium.

Empirical range of direction costs. Table E4 recovers $\hat{\kappa}_s$ for each sector and for the 21 MIP industries by inverting (82) at the GMM benchmark, using the sector-level mean frontier share among product innovators as the empirical counterpart of μ_s^* .

The recovered values range from 0.047 (Manufacturing) to 0.099 (Transport). Manufacturing subsectors (particularly chemicals, electrical equipment, and machinery) sit below the benchmark, reflecting higher baseline frontier shares; service sectors and transport sit above it. Figure E3 plots the full subsector distribution.

Table E4: Sector-level direction costs

WZ	Sector	$\bar{\mu}_s$	$\hat{\kappa}_s$ (SE)
G	Trade/Retail	0.513	0.047 (0.001)
C	Manufacturing	0.497	0.049 (0.000)
M	Prof. Services	0.440	0.055 (0.001)
K	Finance	0.374	0.065 (0.002)
D	Energy/Water	0.367	0.066 (0.003)
J	ICT/Media	0.334	0.073 (0.002)
N	Other Services	0.286	0.085 (0.005)
B	Mining	0.274	0.088 (0.008)
H	Transport	0.244	0.099 (0.007)
<i>Homogeneous GMM benchmark</i>			0.070 (0.004)

Notes: $\bar{\mu}_s$ is the mean market-novelty share among product innovators in sector s , MIP 2003–2018. $\hat{\kappa}_s$ is recovered from the direction first-order condition $\kappa_s \bar{\mu}_s = x^* D_{bm} / (r + \tau)$ using GMM benchmark parameters; standard errors propagate sampling uncertainty in $\bar{\mu}_s$ via the delta method. Proposition E.4 establishes that the proportional compression of the frontier share by procurement demand is the same across all sectors regardless of sector-level direction costs. *Source:* Authors' calculations using the Mannheim Innovation Panel and OpenTender procurement records.

Table E5: Direction costs at the 21-subsector level

Subsector	$\bar{\mu}_s$	N	$\hat{\kappa}_s$ (SE)
<i>Mining (WZ B)</i>			
Mining	0.274	339	0.088 (0.008)
<i>Manufacturing (WZ C)</i>			
Chemicals & Pharma	0.572	1637	0.042 (0.001)
Glass & Ceramics	0.545	721	0.044 (0.002)
Machinery & Vehicles	0.539	2291	0.045 (0.001)
Electrical Equip.	0.531	3077	0.046 (0.001)
Textiles	0.498	1000	0.049 (0.002)
Plastics	0.494	1103	0.049 (0.001)
Metals	0.472	1862	0.051 (0.001)
Furniture & Med. Tech.	0.460	1401	0.053 (0.002)
Wood & Paper	0.399	1141	0.061 (0.002)
Food & Tobacco	0.381	1202	0.064 (0.002)
<i>Energy/Water (WZ D)</i>			
Energy/Water	0.367	632	0.066 (0.003)
<i>Trade/Retail (WZ G)</i>			
Retail	0.576	1772	0.042 (0.001)
Wholesale	0.323	588	0.075 (0.004)
<i>Transport (WZ H)</i>			
Transport	0.244	704	0.099 (0.007)
<i>ICT/Media (WZ J)</i>			
IT/Telecom	0.348	1570	0.070 (0.002)
Media Services	0.313	1068	0.078 (0.004)
<i>Finance (WZ K)</i>			
Finance	0.374	1413	0.065 (0.002)
<i>Prof. Services (WZ M)</i>			
Prof. Services I	0.480	2044	0.051 (0.001)
Prof. Services II	0.375	1230	0.065 (0.002)
<i>Other Services (WZ N)</i>			
Other Services	0.286	766	0.085 (0.005)
<i>Homogeneous GMM benchmark</i>			0.070 (0.004)

Notes: $\bar{\mu}_s$ is the mean market-novelty share among product innovators in the subsector, MIP 2003–2018. $\hat{\kappa}_s$ is recovered from the direction first-order condition; SE propagates sampling error in $\bar{\mu}_s$ via the delta method. Procurement exposure in the regression analysis varies at the broader WZ section level; $\hat{\kappa}_s$ values within a section share the same underlying data variation. Source: Authors' calculations using the Mannheim Innovation Panel.

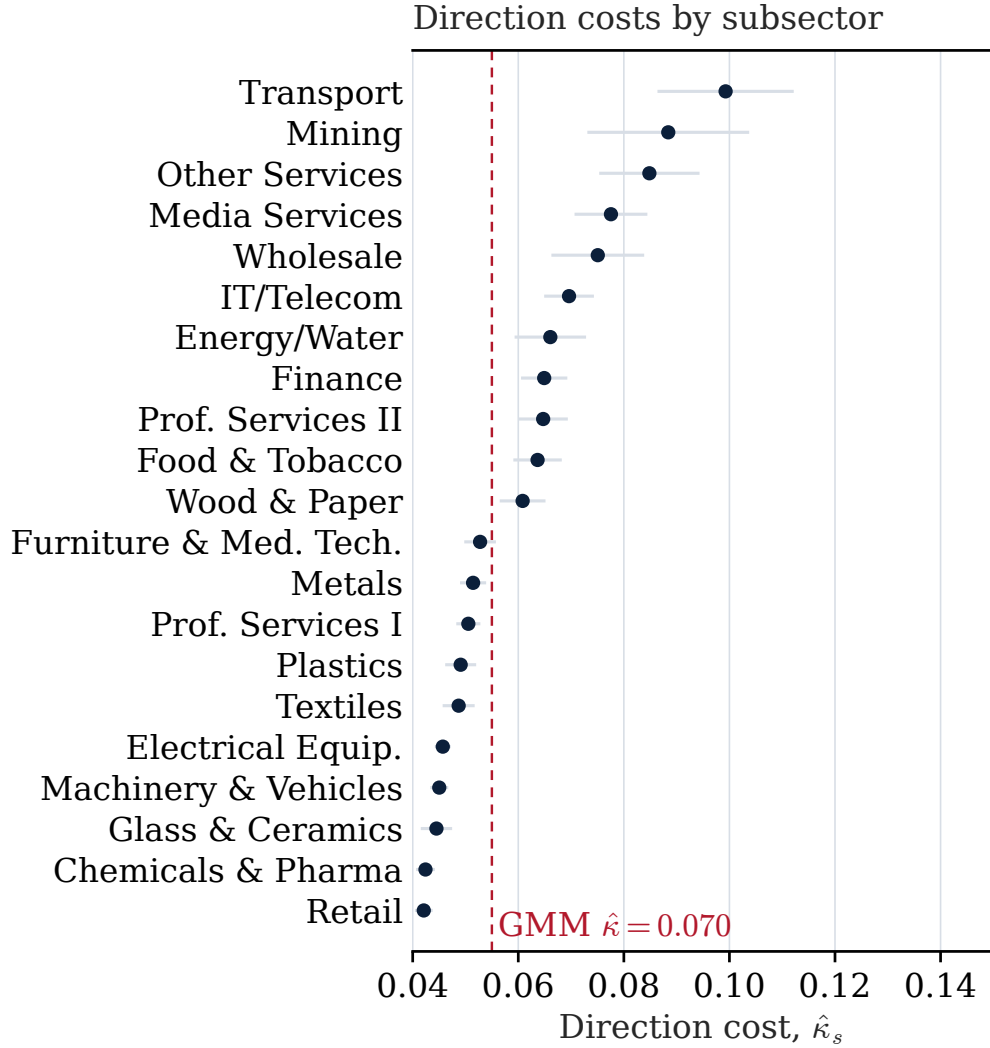


Figure E3: Sector-level direction costs

Notes: The figure plots $\hat{\kappa}_s$ for all 21 MIP subsectors, with 95% confidence intervals (delta-method SE). The dashed vertical line marks the homogeneous GMM benchmark $\hat{\kappa} = 0.055$. Proposition E.4 establishes conditional proportional invariance when sectors share the same equilibrium responses of effort, interest rates, and creative destruction.

Source: Authors' calculations using the Mannheim Innovation Panel, 2003–2018.

E.4.2 Sector Exposure Heterogeneity

The composition wedge scales with the government demand parameter G . We map mean sector-year reported contract counts to model units proportionally, normalizing the mean across sectors with positive exposure to the benchmark $\bar{G} = 0.018$. Manufacturing and professional services receive the largest mapped wedges: without procurement, their frontier-innovation shares are 1.44 and 1.35 percentage points higher, respectively. Transport and IT services lie in the middle range (1.16 and 1.11 pp), while finance and mining have smaller gains (0.67 and 0.48 pp). The corresponding

annual growth gains range from 0.13 to 0.39 basis points. The pattern is the composition-channel counterpart to Proposition E.1: sectors assigned larger G_s face larger composition wedges.

Table E6: Frontier-Innovation Distortion by Sector

Sector	$\bar{G}_s$	$\hat{\mu}_s^*$	$\Delta\mu^*$ (pp)	Δg (bp)
Manufacturing	0.025	0.435	+1.3	+0.4
Professional services	0.023	0.436	+1.2	+0.4
Transport	0.020	0.438	+1.0	+0.3
IT and information	0.019	0.438	+1.0	+0.3
Administrative services	0.017	0.439	+0.9	+0.3
Energy	0.017	0.439	+0.9	+0.3
Finance	0.012	0.442	+0.6	+0.2
Mining	0.008	0.444	+0.4	+0.1
Wholesale and retail	0	0.448	0	0
Economy-wide benchmark	0.018	0.439	+0.9	+0.3

Notes: $\bar{G}_s$ is the sector G equivalent, computed by mapping mean sector-year arcsinh contract counts to model units proportionally: the economy-wide mean corresponds to $\bar{G} = 0.018$. This proportional mapping is a descriptive incidence exercise rather than a sector-specific calibration. $\hat{\mu}_s^*$ is the equilibrium frontier-innovation share at that G , read from the estimated model's CF1 G-sweep. $\Delta\mu^*$ and Δg report the gain in frontier share and annual growth if procurement were removed ($G_s \rightarrow 0$), holding the R&D tax credit fixed. The R&D credit contributes ≈ 10 additional basis points economy-wide in the main counterfactual, independent of sector exposure.

E.5 Robustness Exercises

Three exercises assess robustness of the quantitative results. We perturb structural parameters locally, allow entrants to raise quality directly, and add government goods to the welfare aggregator. None of these exercises removes the model-specification rejection; they instead characterize numerical reliability, alternative entry timing, and welfare sensitivity.

E.5.1 Parameter Sensitivity

Table E7 perturbs each structural parameter by one conventional GMM standard error, holding the remaining parameters at their estimates, and re-solves the equilibrium at the benchmark procurement level. The exercise holds the unperturbed benchmark R&D cost scale fixed, so each entry is a local deviation from the calibrated baseline rather than a new counterfactual rollback. Across all eight perturbations, annual growth moves by at most 0.72 basis points in either direction (Figure E4) and the full CEV ranges from -0.313 to 0.312 percent. The frontier quality step λ_F and the frontier-

orientation cost κ generate comparable welfare movements; κ generates the largest movement in the frontier-innovation share.

Table E7: Baseline Equilibrium Sensitivity to Parameter Perturbations

Parameter	−1 SE shock			+1 SE shock		
	$\Delta\mu^*$ (pp)	Δg (bp)	CEV (%)	$\Delta\mu^*$ (pp)	Δg (bp)	CEV (%)
λ_I	1.26	-0.01	-0.041	-1.26	0.03	0.047
λ_F	-1.11	-0.71	-0.313	1.08	0.72	0.310
κ	2.60	0.71	0.312	-2.32	-0.64	-0.285
h_E	-0.35	-0.39	-0.041	0.35	0.38	0.041

Notes: Entries are deviations from the calibrated baseline. Each row perturbs one parameter by one posterior SE in either direction, holds all other parameters at their GMM estimates, and re-solves the equilibrium at the benchmark procurement level with the unperturbed R&D cost scale fixed. $\Delta\mu^*$ is in percentage points; Δg is in basis points per year; CEV is the full permanent-consumption equivalent, $\Delta \log C_0 + \Delta g/\rho$. Standard errors: $SE(\lambda_I) = 0.0032$, $SE(\lambda_F) = 0.0027$, $SE(\kappa) = 0.0038$, $SE(h_E) = 0.0214$.

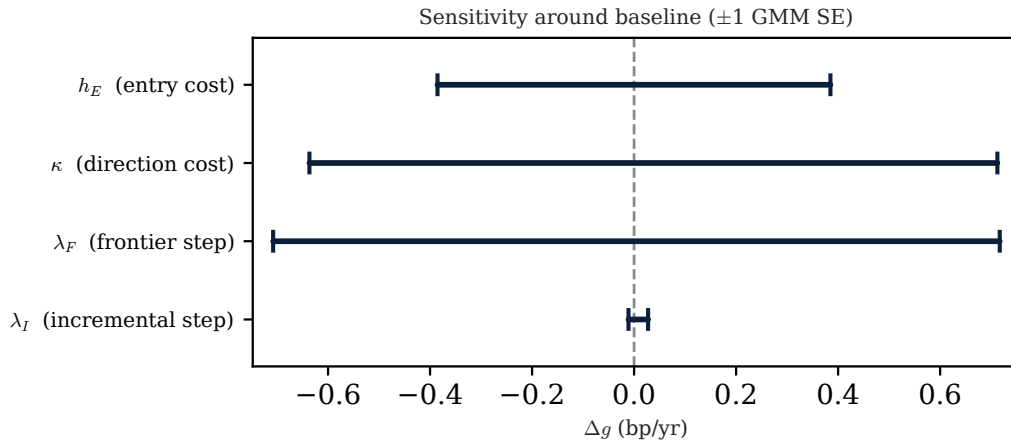


Figure E4: Baseline sensitivity (± 1 GMM SE per parameter)

Notes: Each row perturbs one estimated parameter by one GMM standard error in each direction, holding the remaining parameters at their benchmark values. The horizontal segment spans $[\Delta g(-1 \text{ SE}), \Delta g(+1 \text{ SE})]$ relative to the unperturbed baseline; the dashed vertical line marks zero. The unperturbed benchmark R&D cost scale $c_x(\psi)$ is held fixed.

Source: Authors' calculations from the calibrated model.

E.5.2 Entrant-Innovation Robustness

As a robustness, consider the Klette–Kortum interpretation in which a successful entrant both takes ownership and raises line quality. To avoid adding unidentified entrant-specific parameters, impose

the incumbent composition on entrants, $\mu_E = \mu$, and the same type-specific quality steps. Aggregate growth then becomes

$$g^{KK} = (x + e) \left[(1 - \mu) \log \lambda_I + \mu \log \lambda_F \right] = \tau \left[(1 - \mu) \log \lambda_I + \mu \log \lambda_F \right]. \quad (83)$$

This alternative requires re-normalizing the innovation-cost scale and re-estimating the structural parameters before recomputing the policy counterfactuals. Table E8 reports that exercise. The re-estimated specification matches the frontier share, firm-size distribution, creative destruction, and R&D personnel share closely, but overpredicts the frontier revenue share by 3.53 percentage points. Its overidentifying restriction is rejected more strongly than in the baseline. We therefore treat innovative entry as a specification stress test rather than mechanically replacing x with τ or using it as the preferred quantitative model.

Table E8: Robustness with quality-improving entry: estimates and model fit

	Data	Baseline	Innovative entry
<i>Panel A. Estimated and recovered objects</i>			
λ_I		1.0900	1.0650
λ_F		1.1650	1.1575
κ		0.0550	0.0700
h_E		1.3375	1.1875
c_x		15.446	15.415
Incumbent innovation, x		0.04330	0.03852
Innovative entry, e		0	0.01232
<i>Panel B. Targeted moments</i>			
Frontier share	0.4268	0.4390	0.4246
Frontier revenue share	58.249	56.870	61.778
Large-firm share	0.1039	0.0932	0.1026
Creative destruction	0.0507	0.0583	0.0508
R&D personnel share	0.0290	0.0345	0.0295
J statistic (one d.f.)		134.7	36,342.0

Notes: The innovative-entry specification imposes $\mu_E = \mu$ and the same type-specific quality steps for entrants and incumbents. Growth is $g = \tau[(1 - \mu) \log \lambda_I + \mu \log \lambda_F]$. Both columns re-estimate the four structural parameters against the same five moments and recover c_x to match annual growth of 0.5 percent. The innovative-entry estimates are from a two-stage local grid search. Its overidentifying restriction is strongly rejected, primarily because it overpredicts the frontier revenue share. Standard-error and J calculations use the same weighting matrix and numerical-Jacobian procedure as the baseline.

Source: Authors' calculations.

Table E9 recomputes the design-reform and historical-scaling exercises under the re-estimated specification. Specification realignment, screening, and historical scaling all rise under innovative entry, to 0.86, 0.99, and 0.14 basis points respectively, against 0.38, 0.42, and 0.07 basis points at baseline. Entry responds directly to these reforms and contributes to growth, which makes the design exercises larger even though their frontier-share changes are smaller. The R&D tax credit exercise does not converge to an interior equilibrium under the innovative-entry calibration at the baseline subsidy rate and is omitted from the comparison.

Table E9: Counterfactual robustness with quality-improving entry

Exercise	Baseline entry			Innovative entry		
	$\Delta\mu$	Δg (bp)	CEV (%)	$\Delta\mu$	Δg (bp)	CEV (%)
Specification realignment	0.0132	0.38	0.094	0.0079	0.86	0.210
Screening ($\alpha = 0.5$)	0.0145	0.42	0.097	0.0091	0.99	0.243
Historical scaling	0.0023	0.07	0.015	0.0011	0.14	0.029

Notes: Each specification is re-estimated and its innovation-cost scale is separately normalized to annual growth of 0.5 percent. All fiscal constraints and exact CEV formulas are unchanged. Under innovative entry, both incumbent innovation and entrant takeovers raise quality, so changes in τ contribute directly to growth. The R&D tax credit exercise is omitted: it does not converge to an interior equilibrium under the innovative-entry calibration at the baseline subsidy rate. The exercise is a specification stress test; the innovative-entry model fits the targeted moments substantially worse than the baseline.

Source: Authors' calculations.

E.5.3 Government Goods in the Welfare Aggregator

The baseline CEV captures gains from higher private consumption and faster growth but assigns no household utility to government-purchased output itself: infrastructure maintenance, IT systems, medical supplies, and other goods government buys. This extension adds that value back and checks whether the main welfare ranking survives. We define a utility share for government goods, α , and recompute the CEV for each counterfactual.

Environment. Extend household utility to $U = \log C^{priv} + \alpha \log Q^{gov}$, where

$$\log Q^{gov} = (1 - \mu^*) \log(1 + G_I) + \mu^* \log(1 + G_F)$$

is a log-weighted index of government demand across line types, and $\alpha \geq 0$ governs the relative value of public services. At the benchmark $G_I = \bar{G}$ and $G_F = 0$, so only incremental lines generate

public-good value. With log aggregation, total output is

$$Y^{total} = Y^{private} \cdot (1 + G)^{1-\mu^*}, \quad (84)$$

and no equilibrium condition changes. The parameter α enters only the welfare calculation, adding one term to the main-text CEV:

$$CEV_{B,A}^{gov} = 100 \left\{ \exp \left[\Delta \log C_0 + \alpha \Delta [(1 - \mu^*) \log(1 + G_I) + \mu^* \log(1 + G_F)] + \frac{\Delta g}{\rho} \right] - 1 \right\}. \quad (85)$$

Setting $\alpha = 0$ recovers the main-text formula exactly.

Mechanism. Each counterfactual type has a distinct effect on Q^{gov} . Budget-neutral reforms hold G fixed; Q^{gov} changes only because μ^* rises: more frontier lines, fewer incremental lines receiving government demand. The effect is small and positive: the public-good term adds 0.004% under specification realignment.

The historical rollback reduces G directly, partially destroying Q^{gov} and generating a large offset (−0.247 p.p. at $\alpha = 1$) that turns the net CEV negative once α is close to one. The R&D credit eliminates procurement entirely, losing the full benchmark public-good term $(1 - \mu_{bm}^*) \log(1 + \bar{G}) = 0.988\%$.

The literature does not pin down a single value of α for our setting. [Jones and Klenow \(2016\)](#) motivate reporting welfare in consumption-equivalent units rather than treating GDP as welfare, but their benchmark components are consumption, leisure, mortality, and inequality, not a calibrated public-good aggregator. [Bandiera et al. \(2009\)](#) show that procurement spending need not translate one-for-one into service value: in their Italian standardized-goods setting, passive waste accounts for 83 percent of total estimated waste. We therefore report the full range $\alpha \in [0, 1]$ rather than selecting one welfare weight. The point of the exercise is that the main-text welfare ranking is not an artifact of treating government purchases as pure waste, since the R&D-credit ranking survives even when government purchases are valued at par with private consumption.

Table E10: Welfare effects with and without government goods in the aggregator

Counterfactual	$\alpha = 0$	$\alpha = 0.5$	$\alpha = 1$	Break-even α^*
<i>Design reform (G fixed, budget neutral)</i>				
Specification realignment	0.094%	0.096%	0.098%	< 0
<i>Volume reduction</i>				
Historical rollback	0.015%	-0.108%	-0.231%	0.06
<i>Fiscal redeployment to R&D credit</i>				
R&D credit ($s = 14.1\%$) [†]	1.55%	1.05%	0.55%	1.55

Notes: Entries report exact permanent-consumption equivalents after adding government services to household utility with weight α : $100\{\exp[\Delta \log C_0 + \alpha \Delta((1 - \mu) \log(1 + G_I) + \mu \log(1 + G_F)) + \Delta g/\rho] - 1\}$, with $\rho = 0.04$. Thus $\alpha = 0$ recovers the main-text CEV, $\alpha = 1$ values the government-service index at parity with private consumption, and α^* is the value that makes the CEV equal zero. For the design reform, $\alpha^* < 0$ because the government-service term is positive. The R&D credit sets procurement demand to zero while retaining the benchmark tax to finance the credit.

Source: Authors' calculations from the calibrated model.